The 6-in-1, Quick, Daily Reference

SCIENCE, TECHNOLOGY AND MATHEMATICS DICTIONARY

A Comprehensive Approach

With Effective Knowledge Acquisition Mechanism For Every Student & For Easy Learning

COVERS IN DETAIL:

1. *Biology*
2. *Chemistry*
3. *Physics*
4. *Agricultural Science*
5. *Information & Communication Technology*
6. *Mathematics*

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(The University Child and Youth Development Centre)
P.O. Box KW536, Accra, Ghana.

E-mail: godfriedfunkor@yahoo.com
gfunkor@ucydecentre.com

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PATMOS-i (M: +233 (0)209 277 169, E: horla@patmos-i.com)

EDITOR

Godfried Funkor

CONTRIBUTORS

Abdulai, Deen Jamal (Dr.) - Computer Science: Head, Computer Science Department, University of Ghana, Legon.

Aboh, Innocent Kwame (Prof.) - Physics: Ag. Director-General, Nuclear Regulatory Authority, Ghana & Former Lecturer in Physics, University of Ghana, Legon.

Acquaah, Ebenezer (Mr.) - Physics: Former Assistant Headmaster (Administration), Achimota Senior High School, Accra and Headmaster, Ada Senior High School.

Adams, Issa (Dr.) - Biology: Lecturer, Department of Biological Sciences, Kwame Nkrumah University of Science and Technology, Kumasi.

Adane, Samuel Kwabena (Mr.) - Physics: Head of Physics Department, Achimota Senior High School, Accra.

Adu-Gyamfi, Douglas (Dr.) -Mathematics: Lecturer and former Head of Mathematics Department, University of Ghana, Legon.

Agbeko, Mawutor Kwame (Mr.) - Biology: Biology Teacher, Achimota Senior High School, Accra.

Ahmed, Khalid (Mr.) - Chemistry: Former Senior Research Scientist, Ghana Atomic Energy Commission and Lecturer in Chemistry, University of Ghana, Legon.

Alhassan, Mohammed B. (Mr.) - Agriculture: Headmaster, Ghana-Lebanon Islamic Senior Secondary School, Accra.

Anamuah-Mensah, Jophus (Prof.) - Chemistry: Former Vice Chancellor, University of Education, Winneba.

Anthony-Krueger, Christian (Prof.) - Dept. of Science and Mathematics Education, University of Cape Coast.

Anornu, Geophrey (Prof.) - Lecturer, Civil Engineering Department, Kwame Nkrumah University of Science and Technology, Kumasi.

Asiedu, Kingsley (Mr.) - Biology: Biology Teacher, Achimota Senior High School, Accra.

Asiedu, Seth (Mr.) - Mathematics: Mathematics Teacher, Achimota Senior High School, Accra.

Attuquayefio, Daniel K. (Prof.) - Biology: Former Head of Animal Biology and Conservation Sciences Department, University of Ghana, Legon.

Avi, Christian (Dr.) - Chemistry: Head of Science Laboratory Technology Department, Accra Technical University, Accra.

Bansa, David (Mr.) - Chemistry: Senior Research Scientist, Ghana Atomic Energy Commission, Accra.

Boateng, Samuel (Dr.) - Computer Science: Post Doctoral Research Fellow, Computational Science Centre, Korea Institute of Science and Technology.

Carboo, Derick (Prof.) - Chemistry: Lecturer, Chemistry Department, University of Ghana, Legon.

Danso, Kenneth (Prof.) - Agriculture: Associate Professor of Plant Biotechnology, Ghana Atomic Energy Commission and Lecturer, Graduate School of Nuclear and Allied Sciences, University of Ghana, Legon.

Doku-Amponsah, Kwabena (Dr.) - Head, Department of Statistics & Actuarial Science, University of Ghana Legon.

Dzukey, Stephanie Kafui (Ms.) - Mathematics: UCYDE Centre, Accra.

Esiah, John Wilberforce (Mr.) - Physics: Former Science Programme Officer, UNESCO, Ghana.

Gyamera, Emmanuel Atta (Mr.) - Chemistry: Former National Coordinator, Ghana Science Resource Centre Project, Ministry of Education, Accra.

Gyasi Appiah, George (Mr.) - Communication Studies: Lecturer, Institute of Commercial Management (Central University), Accra.

Harrison, Jerry Joe (Dr.) - Chemistry: Lecturer, Department of Chemistry, University of Ghana, Legon.

Katsriku, Ferdinand Apietu (Dr.) - Computer Science: Former Head, Computer Science Department, University of Ghana/ Senior Lecturer, School of Engineering and Information Sciences, Middlesex University, United Kingdom.

Kuffour, Benjamin Asiamah (Mr.) - Chemistry: Chemistry Teacher, Achimota Senior High School, Accra.

Lartey, T. E. K. (Mr.) - Agriculture: Former Headmaster, Nsawam Senior High School, Nsawam, Ghana.

Logo, Peter (Mr.) - Biology: Former President, Ghana Association of Science Teachers and Headmaster, Presbyterian Secondary Technical School, Aburi.

Martey, William (Mr.) - Physics: Former Teaching Assistant, Physics Department, University of Ghana, Legon.

Mireku, Kofi (Prof.) - Mathematics: Head of Mathematics Education Department, University of Education, Winneba.

Nyamavor, Ebenezer (Mr.) - Chemistry: Chemistry Teacher, Nsawam Senior High School, Nsawam.

Okyne, Kobina Otu (Mr.) - Biology: Former Teaching Assistant, Department of Botany, University of Ghana, Legon.

Oppong-Wusu, Nana Kwadwo (Mr.) - Chemistry: Chemistry Teacher, Achimota Senior High School, Accra.

Ossei-Anto, Theophilus A. (Prof.) - Physics: Former Lecturer, University of Cape Coast, Cape Coast.

Pekyi, Eli Davis (Mr.) - Mathematics: Mathematics Teacher, Achimota Senior High School, Accra.

Wiredu, Benjamin (Mr.) - Computer Science: Lecturer, Computer Science Department, University of Ghana, Legon.

Yamoah, Grace Gyamfuah (Mrs.) - Computer Science: Lecturer, Computer Science Department, University of Ghana, Legon.

EDITORIAL CONSULTANTS

Antwi Aning: Head of Assessment Unit, National Council for Curriculum and Assessment (NaCCA), Accra.

Agatha Akonor Mills: Expert in Front Matters, Ghana Publishing Authorities, Accra.

Ohui Agbenyega Allotey: Ghana Book Development Council, Accra.

Maxwell Agbewale Koku: Logistics and Supplies, Ghana Education Service Headquarters, Accra.

Veronica Odoom: In Charge of Textbooks and Educational Materials, National Council for Curriculum And Assessment (NaCCA), Accra.

Joana Adna Vanderpuiye: In Charge of Textbooks and Educational Materials, National Science Resources Centre, Accra.

Sampson Osenda: In Charge of Referencing, Ghana Library Board, Accra.

Peter Essien: Bureau of Ghana Languages, Accra.

Cynthia Amankwa: Head of Information and Communications Technology Department, Presbyterian Boys Senior High School, Legon, Accra.

Dr Isaac Wiafe, Lecturer: Dept. of Computer Science, University of Ghana, Legon.

Joseph K. Abusah: Mathematics Teacher, and First Runner-up for National Best Teacher Award (2017), Presbyterian Boys Senior High School, Legon, Accra.

Raymond Sedzro: Biology Teacher, Presbyterian Boys Senior High School, Legon, Accra.

Michael Osei: Chemistry Teacher, Acquinas Senior High School, Cantonments, Accra.

Emmanuel Oduro: Physics Teacher, Acquinas Senior High School, Cantonments, Accra.

Peter Adjetey: Biology and Agricultural Science Teacher, Achimota Senior High School, Accra.

David Addo: Head of Science Department, West Africa Senior High School, Accra.

Robert Obodai Torgbor: Mathematics and ICT Teacher, West Africa Senior High School, Accra.

Dominique Julius Cudjoe: Former Physics Teacher, West Africa Senior High School, Accra.

Winfred Agbeko: Former Chemistry Teacher, West Africa Senior High School, Accra.

Kwame Asante: Former Agricultural Science Teacher, Christian Methodist Senior High School, Accra.

Aggrey, Leslie (Mr.) - Graphic Designer: Managing Director, Indes Procom Limited, Accra.

Duwor-Adamadoh, Horla (Mrs.) - Graphic Designer: Patmos-i (Managing Director), Accra.

Acquah, Jeffrey (Mr.) - Graphic Designer: Patmos-i (Art Director), Accra.

Funkor, John (Mr.) - Data Entry & Editing: Former Electrical Engineer, Pacific International Lines, Singapore.

ASSESSMENT AND RECOMMENTATIONS

COUNCIL FOR SCIENTIFIC AND INDUSTRIAL RESEARCH
HEAD OFFICE

P.O. BOX M. 32
Accra
Ghana
WEST - AFRICA

TEL: 233-30-2777651-4 (4 Lines)
FAX: 233-30-2777655
E-MAIL: headoffice@csir.org.gh
WEBSITE: www.csir.org.gh

Our Ref: CSIR/

Date: *7th May, 2018*

THE HON. MINISTER
MINISTRY OF EDUCATION
ACCRA

Dear Sir,

ASSESSMENT AND RECOMMENDATION OF SCIENCE, TECHNOLOGY AND MATHEMATICS DICTIONARY.

On behalf of the Governing body of the Council for Scientific and Industrial Research (CSIR), I wish to send my compliments to the Honourable Minister for Education.

I humbly wish to submit my assessment and recommendation on a book titled "Science, Technology and Mathematics Dictionary" for further attention.

ASSESSMENT

In assessing the book I find these strong attributes:

- It is a well-researched product of academics and consultants with track-proven background in their various areas of expertise. Many of these are very conversant with the technical language that is required at the Junior and Senior High School levels of examinations.
- Presentation of material with its lucid diagrams and annotations goes beyond the dictates of an ordinary dictionary.
- With six subjects all packed into one book, it offers a one-stop, desktop and comprehensive reference material for the student and teacher.
- It seeks to help bridge the yawning gap in science subjects mastering between less endowed schools in the rural areas and urban centres, where the disparity between more well-trained and resourceful teachers and the less resourceful ones, is clearly demonstrated in the outcome of their examination results. Motivated students are able to self-learn and master technical science concepts with little help.

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- Osmosis, a game included as part of the package helps generate interest in the acquisition and mastery of scientific concepts easily in a fascinating way. Playing the Osmosis game reminds players of the SCRABBLE game that facilitates easy and faster acquisition of English vocabulary.

RECOMMENDATION AND CONCLUSION

I strongly recommend the dictionary to every student who earnestly desire to make good career in Science, Technology, Mathematics and related disciplines in future. Parents and teachers who want to see a decline in the failure rate in Science and Mathematics should own this dictionary and encourage their wards to do same.

Science and Technical Education has its unique place in equipping a society towards its drive to achieve measurable development. A country's state of technological advancement and its emphasis on Science, Technology and Mathematics Education are strongly correlated. There is the need to acknowledge the importance of these subjects, so as to place the youth in an advantageous position to participate and be beneficiaries to the ever rapidly advancing frontiers of technology and its role of poverty reduction.

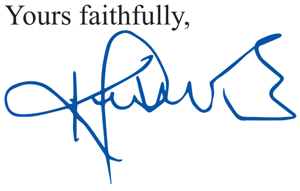
However, the teaching and learning of Science, Technology and Mathematics have their challenges in schools. Lack of teaching resources, inadequately trained teachers for these subjects in some schools and the perception among teachers and students that these are difficult subjects to tackle, especially by females, are just a few of the numerous challenges. These, in turn, further fuel the increasingly poor performance in examinations. There is, therefore, an urgent need for these to be properly addressed.

The 6-in-1 Science, Technology and Mathematics Dictionary written by the UCYDE (University Child and Youth Development Centre) in partnership with the Ghana Association of Science Teachers, is going to boost learning and easy grasping of difficult concepts as well as technical language by students at the Junior and Senior High School levels (and even beyond). The introduction of the book is meant to address some of the incipient challenges associated with science learning as enumerated above. It will help to demystify Science and Mathematics as 'no go areas' among children.

Please accept my highest appreciation.

Thank you

Yours faithfully,



PROFESSOR ROBERT KINGSFORD-ADABOH
CHAIRMAN OF COUNCIL, CSIR
& AG. DEAN OF THE SCHOOL OF PHYSICAL AND MATHEMATICAL SCIENCES OF THE UNIVERSITY OF GHANA

Cc: THE DIRECTOR
UNIVERSITY CHILD AND YOUTH DEVELOPMENT CENTER (PUBLISHERS)

2

UNIVERSITY OF CAPE COAST
COLLEGE OF EDUCATION STUDIES
FACULTY OF SCIENCE AND TECHNOLOGY EDUCATION
DEPARTMENT OF SCIENCE EDUCATION

Tel: 03320 96801/96951
Email: dse@ucc.edu.gg
Website: www.ucc.edu.gh

University Post Office
Cape Coast
Ghana

Your Ref:
Our Ref: DSE/R.2/V.1/17

19th June, 2018

ASSESSMENT AND RECOMMENDATION FOR THE USE OF THE '6-IN-1' SCIENCE, TECHNOLOGY AND MATHEMATICS DICTIONARY
(A Comprehensive Approach with "Effective" Knowledge Acquisition Game for Easy learning)
IN EDUCATIONAL INSTITUTIONS: Published by THE UCYDE CENTRE-ACCRA

From my extensive reading of this book, research and studies, I can sincerely and truthfully say that this type of project (The 6-In-1 Science, Technology and Mathematics Dictionary) is the first of its kind undertaken in Ghana in terms of the number of subject areas put together as well as the depth of knowledge organization, due to the carefully selected outstanding Science, Mathematics and ICT teachers, lecturers and consultants involved in the project. It was an ambitious project, bravely initiated, well researched, painstakingly written in detail by the contributors, with deep understanding of their subject areas of specialization and effectively accomplished. There is no Dictionary of Science, Technology and Mathematics like this in Ghana.

Our students at all levels of the educational sector, from the basic to tertiary, must take advantage of this dictionary to achieve their individual goals by using this resource to build their vocabulary in their chosen fields.

I, therefore, strongly recommend this book to Senior High as well as Basic School students who desire to pursue Science to a high level or make a career out of the Science programme. Again, this book is recommended to Undergraduate Students as well as all teachers (from the tertiary to the basic level) for personal use. Finally, I recommend this book to all groups of learners/educated persons; for their personal and family use, as well as for libraries in their offices/organisations.

Though, I got involved at the tail end, I am pleased that I saw and studied key aspects of this book (mainly my speciality - Physics) before its final launch; very outstanding and excellent work done.

I would like to congratulate the vision bearers of this Book Project and I dare say that this will not just be used in Ghana alone in the near future, but other African countries.

It must be emphasized that Science/Technology is the only doorway to sustained prosperity for any nation in the world; hence, the need to promote it at all cost in Ghana and, therefore, Africa that is lagging behind in technological advancement (and as a result relatively poor).

As a nation, if we are able to encourage the interest and creativity of our students from the Basic level in the area of Science and Technology, we will be on the path to producing many citizens in the various fields of Science and Technology, including the following: Scientists for knowledge production; and this will assist Technologists to discover and invent things. In fact, we need another "Silicon Valley" in Ghana, where inventions are given birth to. This is very achievable if we start our pupils and students early, hence, the usefulness of this Book Project - **Science, Technology and Mathematics Dictionary (A Comprehensive Approach, with "Automatic" Knowledge Acquisition for Easy Learning)**.

Pupils and students must start referring to this important resource to unravel what "seems" to be the mystery around Science and Technology. What pupils and students cannot avoid is the language and culture of Science and Technology, hence, this Dictionary. I am an authority in Physics and Science Education; therefore, I dare say, the terms associated with Physics, for example, can be referred to and learned without any teacher - distance, speed, velocity, acceleration, motion, pressure, viscosity, etc. Thus, a pupil who is encouraged or motivated by a good Science teacher can begin to use this Dictionary as a start. It is important to stress that until a learner's vocabulary grows in Science and Technology, he/she as a person cannot excel in the science subjects - Physics, Chemistry, Agriculture, Biology, ICT, etc. Even though this book is an asset to all students/pupils, those in deprived areas, in particular, will find that it is a key catalyst for bridging the gap between them and their better-resourced counterparts, especially in the developed world.

What is critical for us as citizens is to accept the fact that Science and Technology are closely associated with our lives. These fields are linked to society and studies/developments in these fields are vital for the overall progress of humanity.

Dr. Eugene A. Johnson
Ph.D (Science Education);
M.Phil (Science Education);
M.Phil (Physics-Laser & Fiber Optics);
B.Sc (Hon) Physics;Dip.Ed.
Head of Department
Department of Science Education
Faculty of Science and Technology Education
University of Cape Coast, Ghana
Personal Contact: + 233-243-562189
Office: +233-3321-3489
WhatsApp: +233-243-562189
E-mail: ejohnson@ucc.edu.gh

GHANA ASSOCIATION OF SCIENCE TEACHERS

(GAST)

National Secretariat
Opposite Accra Worker's College
Teacher Education Blk, 1st Floor Rm 10
P.M.B., T.U.C., Accra Ghana, West Africa
E-mail: gast_ghana@yahoo.com
gast@edughana.net

THE GHANA ASSOCIATION OF SCIENCE TEACHERS: ASSESSMENT AND RECOMMENDATION OF SCIENCE, TECHNOLOGY AND MATHEMATICS DICTIONARY BY THE UNIVERSITY CHILD AND YOUTH DEVELOPMENT CENTRE (UCYDE CENTRE)

INTRODUCTION

The misconception that Science as a subject, is difficult to study and pass well has been overemphasized. Unfortunately, this single perception has sunk deep into the subconscious minds of pupils, students and teachers; aggravated by the inability of our nation to cope with the rapid pace of global technological development.

After examinations, one of the things that always dominate reports of Chief Examiners is students' unfamiliarity with most scientific jargons and hence their inability to either spell them correctly or explain and use most basic scientific concepts satisfactorily.

As part of our mandate to make the teaching and learning of science and other STEM (Science, Technology, Engineering and Mathematics)-related subjects easier, the Ghana Association of Science Teachers (GAST) mooted an idea to catalogue all scientific jargons, explain them and give examples of their usage in a single standard document - emerging as a Science Dictionary, or more appropriately, a Science reference book; in the hands of every student. It is upon this backdrop that GAST decided to partner with the UCYDE Centre for the creation of a science dictionary, with a simple but effective new methodology for studying it, to ameliorate the above stated challenges.

IMPORTANCE OF THE DICTIONARY

A critical study of the work reveals that:

- **i.** **The entire document was compiled and written by seasoned professionals with enviable track-record, drawn mainly from GAST, the top Universities and Senior High Schools as well as other top profile institutions/organizations. Most of these consultants have vast experience in teaching, research and curriculum development/assessment, cutting across various levels and countries; with special understanding of the needs of the local terrain.**
- **ii.** **As a result of this project, both students and teachers across the country now have a single standard document to serve as a quick reference for the teaching, learning and understanding of major scientific terminologies, concepts and topics. The book will, therefore, enrich the science vocabularies of all students, which will in turn go a long way to improve upon teaching and learning outcomes of this subject.**
- **iii.** **This is in line with Government's policy of giving all students and pupils equal opportunity in science education.**
- **iv.** **There is a fascinating game attached to this book which can be used to stimulate keen interest in science, help students to master the scientific language at tender ages and make the teaching, learning and passing of science examinations a pleasure with excellent outcomes. This game works through the Osmosis or the Montessori Approach.**
- **v.** **The challenge posed by the study of science to all students is a global canker, affecting even the developed countries. Thus, this unique document which was developed to meet international standards could become a major non-traditional export commodity into other countries; especially the developing ones.**

vi. vi. This unique book will, thus dispel the fear for science and allow as many people as care, to offer science, with resounding results.

RECOMMENDATIONS

It is, therefore, categorically recommended that:

i. Since passing science and mathematics external examinations are mandatory for furthering one's education at the basic and second cycle levels, Government should adopt this dictionary for the use of each and every student; to enhance their understanding in these subjects. If government deems it important to acquire English Dictionary for each and every student, by virtue of English being our official language, it is even more imperative in this age of global technological advancement, to make a comprehensive and specially tailored dictionary such as this, available to each and every student on a continual basis, so that any time they are studying, be it as individuals, a group, at home or in school, they can always refer to this book continually for accurate guidance and as solution to their difficulties in understanding and appreciating Science, Technology and Mathematics.

ii. Science is not English: The Scientific world has its own language, drawn from all over the world, which students find difficult to master early in life, hence their continuous abysmal performance, fear and shunning of it. At the basic level students should focus on continuous playing of the attached science game which works by the Osmosis Approach. This will help them to become very proficient in the scientific language and hence make it very easy for them to learn or be taught science, as they advance further on the academic ladder.

iii. Apart from the effort to get copies of this book for all students in their respective schools, all parents should be persuaded to get copies of this book at home and stimulate their wards to substitute this game for most of their recreational activities; so that science and mathematics become household languages in this era of rapid global technological advancement. The benefits are immense.

iv. As a non-traditional export, this comprehensive and well developed dictionary should be owned as a made-in Ghana science dictionary and promoted as such for the use of all students globally, especially in developing countries, starting from all the countries in the West African Examinations Council zone. This is because, the fear and shunning of science by most students is a global canker that all nations are in continual search for a feasible solution.

We sincerely believe that the adoption of this book by government and parents would go a long way to make the teaching and learning of science and mathematics achieve national growth and development objectives, as well as close the technology gab.

President: Ghana Association Of Scinece Teachers (GAST)

(Rev. Thomas K. Arboh - 0208215281)

National Council for Curriculum & Assessment

In case of reply, the number and date of this letter should be quoted.

Tel. No. 0302770610 / 0302771089
Email: nccagh@yahoo.com
Website: www.nacca.org.gh
My Ref No.NaCCA/G/14/Vol.8/076

Your Ref No. ....

NaCCA
Republic of Ghana

Ministry of Education
Cantonment Branch Post Office
CT PM 77
Cantonments
Accra

25th November, 2018

GODFRIED FUNKOR
P.O. BOX KW 536
ACCRA

BOOK ASSESSMENT
SCIENCE, TECHNOLOGY AND MATHEMATICS DICTIONARY
BY: GODFRIED FUNKOR

The above-named material which was submitted to the Textbook and Educational Equipment Committee (TEEC) through the National Council for Curriculum and Assessment of the Ministry of Education has been assessed.

Decision
The material is recommended for use as a reference for SHS, Technical and Vocational Institutions and for optional purchase.

You are advised to make your own promotional arrangement.

Thank you.

FELICIA BOAKYE-YIADOM (MRS)
ACTING EXECUTIVE SECRETARY, NaCCA
TEXTBOOKS AND EDUCATIONAL
EQUIPMENT COMMITTEE
For: DIRECTOR-GENERAL

Cc: The Hon. Minister, MOE
The Hon. Deputy Minister, MOE
The Chief Director, MOE
The Chairman, GES Council
The Director-General, GES
All Regional and District Directors of Education
The President, CHASS

Conference of Heads of Assisted Secondary Schools (CHASS)

BANKERS:
G.C.B, Ring Road West, Accra

National Secretariat
P.O.Box SD 300,
Stadium - Accra.
Tel: +233-302-256491
+233-28-9535070
GNAT HALL
ADABRAKA

ASSESSMENT AND RECOMMENDATION OF 6-IN-1 SCIENCE, TECHNOLOGY AND MATHEMATICS DICTIONARY PUBLISHED BY THE UNIVERSITY CHILD & YOUTH DEVELOPMENT (UCYDE) CENTRE, ACCRA.

Assessment and Recommendation by Samuel Ofori-Adjei, President of the African Confederation of Principals (ACP), President of the Conference of Heads of Assisted Secondary Schools (CHASS) and Headmaster of the Accra Academy.

Nearly four decades of working experience in formal education, especially at the Secondary level, exposed me to various challenges faced by both students and teachers and the opportunities available to them to excel. These became more pronounced when for a decade and half, I had to supervise teaching and learning of many subjects including Mathematics and the Sciences and later, Information Technology.

The fast pace of development in all facets of life calls for a teaching and learning process which can move in tandem with the demands without compromising on standards. Teaching and learning materials must, thus, be produced to fit into the scheme of things. Assessing information from far and scattered reference materials for teaching and learning, which sometimes are not really available or accurate is gradually becoming a drudgery for both the teacher and the learner. The introduction onto the educational landscape of this quick, daily reference, **6-IN-1 SCIENCE, TECHNOLOGY AND MATHEMATICS DICTIONARY** is timely and it has come to fill a void.

The dictionary, which was compiled with inputs from some of the best brains in the fields covered (Biology, Chemistry, Physics, Agricultural Science, Mathematics and Information Technology) is a veritable one-stop knowledge source for all. The areas covered collectively account for nearly all (or are at the foundation of most of) the knowledge needed to develop the country in this science, technology and mathematics-driven world. The relatively simplified definitions and accompanying illustrations make reading and absorption much easier. As a unique dictionary, the alphabetic sequence of entries was designed to make reference easy and quick. To cap it all, the inclusion of the fun-generating approach to learning tagged, 'The Osmosis Approach' adds another important dimension for improving the assimilation of concepts.

This innovative and comprehensive dictionary must, therefore, be found on the study table of every student, on the shelves of all school libraries and every family must possess a copy for daily reference and playing the osmosis game. This is to make the study of the Sciences and Mathematics a household and institutional affair. Also, based on my experience as President of the African Confederation of Principals (ACP), the content of the book show a good grasp of the fundamental needs of the continent, hence I highly recommend the book to students in all other countries for serious studies in schools and at homes. It is an important and bold step to close the technology gap and thus move developing countries to the developed state.

Samuel Ofori-Adjei

NATIONAL COUNCIL OF PARENT-TEACHER ASSOCIATIONS (NCPTAs)
(GOVERNMENT ASSISTED SCHOOLS P.T.A)

P.O. Box AT 93, Achimota Market Accra Mob: +233 244 366339 / +233 244 841036 / +233 244 092107
Email: info@ncptasghana.org Website: www.ncptasghana.org

ADOPTION AND RECOMMENDATION OF SCIENCE, TECHNOLOGY AND MATHEMATICS DICTIONARY FOR USE BY STUDENTS AND PARENTS

The National Council of Parent-Teacher Associations (NCPTAs) wishes to adopt and recommend the use of the above book by our wards in their respective schools and homes. Every time our wards go out to write examinations, especially external ones, performance in Science and Mathematics have remained low and most of us also look on helplessly because we have little or no idea as to what to do to take them out of their predicaments. Our research showed that the difficulties faced by our students defy national barriers and have become a canker for most students' failure in the aforementioned subjects globally.

We also share the view of the authors that the scientific world has its own language and culture and people who are unable to acclimatize with these quickly become victims in terms of academic performance and application for a better life.

It is our belief that this comprehensively researched dictionary by this team of seasoned professionals can serve as a quick daily reference beside each student as he/she studies these subjects; and can be used to demystify unfamiliar terminologies in their textbooks and notes. We also believe that the Osmosis Game, as an effective knowledge acquisition mechanism, would go a long way to prepare and equip our wards at tender ages with the language, vocabularies and skills needed to effectively study and apply these subjects at any level.

We have reviewed the book extensively and we are satisfied that it is innovative, unique, comprehensive, generates a multiplier effect in learning and covers the thematic areas in Physics, Chemistry, Biology, Mathematics, Information Technology and Agricultural Science. The language chosen to explain the terms and the packaging are designed to suit any category learners.

The explanation of most of the vocabularies are so detailed that students do not need further reference or notes from other sources. It is a one stop, easy-learning shop and parents and school authorities must not only acquire these books for each of their wards, but also get involved in the organisation and playing of the osmosis game with their younger wards; so that they can quickly master the language and skills needed to become effective Science, Technology and Mathematics students, with the prospect of excellent performance in school and professional life.

We share the re-echoing views of teachers and students that accessing information from far and scattered references for teaching and learning, which sometimes are not readily available or accurate, has gradually become drudgery for both the teacher and learner. Thus, the introduction of this 6-in-1, comprehensive material into our educational system is timely and has to come to fill that void. Our wards cannot do without it, if we want to see a dramatic improvement in the teaching and learning of Science, Technology and Mathematics; or at least arrest the downward trend in performance.

The NCPTAs, therefore, fully adopt this comprehensive book; and unreservedly recommends it for your ward's use and expeditious development.

Alexander Yaw Danso
(National President)

TABLE OF CONTENT

ACKNOWLEDGEMENT

We thank God for the idea, strength, people and resources to carry out this work. We are grateful to each one of the 200-member team from the Accra Technical University that collected the initial data that formed the basis of this book. Ms Genevieve Nasser, Mr. Wilberforce Esiah and Dr. Erasmus Owusu, have been instrumental in getting some of the high profile consultants who made their expertise available for this work. We are immensely grateful to the anonymous Professor whose contributions gave this book significant uniqueness.

Our heartfelt gratitude also goes to all the authors and publishers whose work enriched this book. Every effort has been made to acknowledge all key references and contributors, but if some have been inadvertently overlooked, we would be pleased to make the necessary amendments at the first opportunity. We used the masculine pronoun 'he' to represent the feminine as well in some cases, for convenience. A few shortcomings may arise from the fact that the world of science is a vast one and the level of knowledge and precision required, especially in definitions and explanations are very demanding as well as changing with new discoveries and inventions.

Since Technology and Engineering are derived from Science and Mathematics, we sometimes referred to the entire subject matter under discussion as Science and Mathematics or just Science.

The words Gamification and Gamefication refer to the same thing and are both coined from the word game. The synonyms given beside the words defined in this book may not be perfect ones in all cases, especially since the book covers a broad range of areas in Science, Technology and Mathematics. There are also repetitions in attempts to explain the Osmosis/initiation Game for the understanding of all students, especially those at the basic level of the educational system. Similarly, there are inconsistencies in the award of marks for winners in the competition. Organisers of the game are, therefore, at liberty to devise their own awarding system for each stage of the game.

We wish to express our sincere gratitude to all readers in anticipation of useful criticisms and recommendations that would enhance this book.

PREFACE

Importance of Science, Technology, Engineering and Mathematics Education: Science, Technology, Engineering and Mathematics (STEM) are the most important avenues for national growth and development globally and they are the main criteria that divide and give prestigious or under-dog status to countries in the world, with nomenclature such as Super Powers or Developed, Developing, Under Developed, Third World, G7, Industrialized, Rich, Poor, Middle Income and Highly Indebted Poor Countries. In view of this, many countries including Ghana, Nigeria and other African countries are implementing Science, Technology, Engineering and Mathematics Education to give their citizens a head-start in development and growth.

The critical roles Science and Mathematics play in many countries are emphasized by the fact that in such nations, all students are required to obtain, at least, a credit pass in them before advancing from the first to the second cycle institutions and from the second to the tertiary levels. Thus, the study of Science and Mathematics are at the centre of their educational curricula and they are compulsory for all students at the pre-tertiary institutions. Also, since they are so important for national survival, growth and development, science students and teachers are considered as endangered species and given special treatment; with their education being guided by special policies in order to preserve and multiply their numbers at all cost.

Problems Posed by Science, Technology, Engineering and Mathematics Education to Students Globally: Unfortunately, both literature and data support the assertion that Science Education is a developmental bottleneck in every nation and over the years many indigenes in even developed countries shun its study or teaching (Appleton, 2003, Hackling & Rennie, 2001). Thus the success rate in Science Education globally has been very poor, especially in developing countries. For example, according to the Director General of the Council for Scientific and Industrial Research, there are about 500,000 students at the tertiary level in Ghana, yet just about 7% of these students are offering science and its affiliated subjects at the tertiary level (25th December, 2018, https://modernghana.com/). Thus, Jane Opoku–Agyeman (2019), a former Minister of Education and a Vice chancellor in Ghana recommends, "we need many more Mathematicians and Scientists in our schools, workplaces and communities and besides, we can all do with an appreciation and heightened understanding of these subjects."

The progressive decrease in the percentage of students pursuing courses in Science, Technology, Engineering and Mathematics largely correlates with the poor scores they receive in these subjects, which discourage their interest, participation and specialization in them at higher levels. These subjects are reputed to be very difficult to learn and passing examinations in them has been a stumbling block in the educational advancement of many, with casualties too numerous to count. This has degenerated into a situation where STEM subjects are generally seen as exoteric, which only a few, specially gifted people can pursue and succeed in. Even among these few, only a minority are able to move further and reach the very top of the educational ladder, hence limiting greatly the number of effective, problem-solving specialists available in this area globally to assist development and growth.

Reflecting on students' performance in science and mathematics examinations, the West African Examinations Council's Chief Examiners' Report from 2011-2016, also support the view that Senior High School Students in Ghana, for example, have been performing abysmally in Mathematics and Science. The report indicated that 37 per cent of the total number of candidates presented for Science in 2015 and 36 per cent in 2014 scored F9. For Mathematics, 38 per cent of the total number of students presented for the examination in 2016 scored F9, 45 per cent in 2015 and 32 per cent in 2014 (Source: Ghana news Agency, 17th February 2017). Also, in 2015, only 23.63% and 25.4% of students who wrote the West African Senior School Certificate Examinations (WASSCE) in Integrated Science and Mathematics (Core) respectively, obtained grades that tertiary institutions in Ghana would consider as pass grades (A1 – C6) for admission. Performance in previous years in these subjects have generally been abysmal.

Other countries do not have any better results to show, as far as acceptable performance in Science, Technology and Mathematics is concerned. For example, Afolakemi Oredein and Adebisi Awodum (2013) analysed the performance of Nigerian schools in the West African Examinations Council's Senior Secondary Certificate Examinations from 1998 to 2007 in Biology, Chemistry and Physics and concluded that, with the gloomy results in the years under review, "It is imperative for science subject educators and educational managers to look for a better way of improving the situation." Also, analysing the Grade 12 National Senior Certificate results in South Africa, Kriek and Grayson (2009) noted that the national pass rate in Science (particularly Mathematics and Physical Science) has decreased since 2003 and this has been a great concern for stakeholders. Mangwaya Ezron et al (2016), in analysing factors influencing Ordinary Level examination students' academic performance in Integrated Science for the past five years in Zimbabwe (Victoria Falls), noted that just an average of 13% of pupils passed and that as a result of this low pass rate, very few pupils consider admission for the Advanced Level Sciences.

The developed countries also have their fair share of the challenges posed by Science, Technology and Mathematics in their educational systems. For example, in the United Kingdom, it was observed that many indigenes shun Science and Mathematics; while in Australia, research has shown that apart from lack of resources, many beginning primary school teachers did not teach physical sciences because of poor understanding and lack of confidence to teach the subject (Appleton, 2003; Goodrum, Hackling and Rennie, 2001).

Sources of Problem: A major reason for this problem is that the scientific world has its own language, which is drawn from all over the world and quite difficult to master. For example, kwashiorkor, which is a form of malnutrition caused by eating a diet deficient in protein and calories, is a Ghanaian (Ga) terminology and beriberi, which is also a nutritional disorder that is caused by deficiency of vitamin B1 (thiamine) is a Nigerian (Hausa) word. Vertebra comes from the Latin word vetere, which means "to turn." Similarly, Chlorine comes from the Greek word khloros, which means "green". Carat, which is a unit for measuring the purity of gold or the weight of jewels, comes from the Arabic word quirrat, which means "seed". Chernozem, a fertile black earth or soil, rich in humus with a lighter lime-rich layer beneath, is derived from the Russian words chërnyĭ (black) and zemlya (earth). The standard units of measurement in science with the abbreviation S.I. (Units) is from the French word, Système International (d'Unités); Tungsten, which is a hard, lustrous, grey metallic element with a very high melting point is a Swedish word and E-Z Convention, which is the International Union of Pure and Applied Chemistry's method of describing the stereochemistry of double bonds with three or four substituents, derives its root from the first letters of the German words entgegen and zusamme, which mean "opposite" and "together", respectively. Apart from originating from diverse sources, the scientific language is highly technical, with learning, retaining and reproducing most of its terminologies and facts being very difficult. For example, it is not enough to know just the name of a substance; it is also important to know its chemical, structural, molecular or empirical formula to be able to identify it in whatever form it is presented or mentioned. One needs, moreover, to be abreast with and understand the numerous laws and principles which explain natural phenomena.

Research by renowned scientists, educationists and scholars, including a Committee chaired by Professor J. Anamuah Mensah (2014) and a research by Professor J. N. Opoku Agyeman (2019) support our research findings and conclusions that the main causes of the low student interest, abysmal performance and participation/ specialization in Science, Technology and Mathematics are:

i. **The Language of Instruction of these subjects, especially at the basic levels and**

ii. **The Mode of Delivery in the classrooms**

According to the former Minister and Vice Chancellor, "Beyond all the problems concerning Mathematics and Science Education, the most important is the language of instruction." As she puts it, 'we are all yet to see high PISA (Program for International Students Assessment) Scores by any country where their children are taught in a language they do not use regularly or do not use at all." She concludes, "any learner who struggles over the language of instruction, who must think in one language, fully mastered or not, and who must constantly translate concepts over which he/she has little or no grasp and worse still, from a language little understood, is already losing the battle of excellence in education."

On the Mode of Delivery of lessons in the classroom, she noted, "Many are the terrifying stories of learner experience and engagement with these subjects ... The fear of mathematics is real in our schools, however, we use math and science everywhere and nearly at all times... Science is taught in a more or less reified manner to the extent that our learners perceive a subject which is supposed to be all around them as something most remote. The diction, images and equipment employed are usually far removed from the life experience of the learner. Thus, establishing the myth that the subject is not to be comprehended by everyone while the opposite indeed is the truth."

The above problems which tend to militate against the effective teaching, learning and excellent performance in science, technology, engineering and mathematics are aggravated by the fact that:

i. The teaching and learning of supporting subjects such as the language of science are so weak that they are unable to reinforce the study of science; implying lack of adequate vocabulary to express, explain or report observed phenomena or even understand instructions or use tools. The intensity of this problem varies from country to country, with developing countries being the worst affected.

ii. Lack of appropriate resources, especially books, well-fashioned in line with the teaching and examinations syllabi, to help students digest topics covered in them. For example in Ghana, Nigeria and the rest of Africa, teachers and students do not have any credible STEM Dictionary, specially packaged and produced to support the teaching and learning of STEM Subjects or for constant daily reference and general use of students and teachers; giving them competitive disadvantage in their quest for global equality in development through STEM Education.

iii. Inadequacy of the time allotted for instructional period remains a key impediment in the teaching and learning of science, technology and mathematics.

As a result of these and many other daunting challenges, most students are unable to choose science as a course/programme of study or follow its lessons to logical conclusions. They are also unable to retain and apply them, especially during critical periods such as examination times or in solving everyday practical challenges in life.

These problems have defied government policy intervention and spending for a long time, leading to continuously low interest of students in these subjects, high failure rates and low student participation/specialization in them at higher levels. Most countries are, therefore, annually confronted with the problem of finding enough students, especially indigenes, to study science, technology, engineering and mathematics at higher levels. This has negative repercussions for the socio-economic development of any country that finds itself in this situation; keeping these countries perpetually under-developed and heavily indebted. Thus, unless the fundamental problems confronting Science Education are fully addressed, there is no guarantee that STEM Education will achieve the expected results, especially in developing countries. Nothing could be further from this truth.

Goals of Our Project: Thus, in order to solve these major problems which, manifest in various forms, including outright hatred of science and mathematics by many students, this project will ensure that all pre-tertiary students develop unquenchable thirst for the study of STEM, are able to speak and write the languages of these subjects as fluidly as their mother-tongues, have full knowledge of the contents of STEM Education, reduce the abysmal performance, coupled with perennial failures in these subjects and make it easy for policy makers, teachers and students to increase to nationally-desirable levels, the number of students choosing, participating and specializing in them at higher levels to globally competitive levels.

Compilation of the STEM Dictionary to serve as an indispensable, strategic tool in the hand of every pre-tertiary student and teacher: Also, in line with our ambitions to solve these major problems, we have come up with this comprehensive Science, Technology and Mathematics Dictionary; meant to be with every student and teacher, to be students' daily companion, for continuous, quick and detailed reference while studying science privately, in the classroom, laboratory or field; with or without a teacher. That is, since Science and Mathematics are compulsory and examinable at the pre-tertiary level, every student and all teachers of these subjects at the pre-tertiary level need a personal copy of this comprehensive six-in-one STEM Dictionary to support learning, teaching and for constant daily reference by students and teachers. The widespread availability of this Dictionary to all stakeholders, especially every pre-tertiary student and their teachers is a major way of developing their interest in these subjects and multiplying the number of students choosing and specializing in them at higher levels.

Provision of Fun-generating Game to Enhance Easy Mastery of Scientific Language: Instead of developing Junior editions of this book, we have provided various levels of learning the whole book in the form of Games ranging from fundamentality (primary level) through simplicity (Junior High School level) to relatively taxing stages (Senior High School level) because we think the scientific language is a holistic entity and must be acquired as such (rather than in a fragmented forms) to make maximum impact in the life of all beneficiaries of Science Education. This simple, fun-generating approach will enable everyone to find it easy to master the scientific language at an early stage, study science joyfully and with little effort, pass examinations in it with excellent results. This approach is an Automatic Method of Knowledge Acquisition known as "The Osmosis Approach". It takes its root from the biological terminology, Osmosis, and it is a gradual, often unconscious, absorption of knowledge through continual exposure rather than deliberate learning. This is very effective for preparing students at basic levels for the tough challenge of qualifying for and studying science in any of its disciplines at higher levels.

Offers Choice Between Neck-breaking and Very Gentle Learning Gradient: Textbooks, tuition, exercises, examinations and other learning activities, especially in science and mathematics, are structured with supposedly appropriate sets of vocabulary at each level. As one goes higher, new sets of vocabularies are introduced, with the level of sophistication becoming very intense. Apart from the fact that most students have never met most of these vocabularies in their life, they barely have time to digest them in their textbooks, tuition, exercises and other learning activities before being forced to apply them in their examinations and other assessments. This makes the learning gradient in science, technology and mathematics too steep for most students to cope with, leading to unacceptably high failure rates.

The Osmosis Approach to learning science and mathematics offers a very gentle learning gradient to everyone who wants to climb to the peak of his/her education in these fields. Like any spoken language at home, it consciously or unconsciously exposes learners to virtually all the vocabularies needed to master the scientific language at a tender age. Apart from the early exposure, it gives them an easy way of mastering these vocabularies and their meanings by playing the Osmosis Game and being exposed to all these vocabularies over and over again for a prolonged period. This leads to mastering each vocabulary and its meaning long before students/pupils reach the actual level that they may need it in the course of their education. When they eventually meet them in their textbooks, tuition, exercises, examinations and other fields of learning, the foregone conclusion is excellent performance at the end of the day.

These steps will enable countries using this dictionary to consciously or unconsciously produce enough effective problem-solving scientists and critical thinkers for rapid development, growth and economic empowerment to close the gap between them and the developed world.

Scope and Sources of Materials Used: The book covers Biology, Chemistry, Physics, Mathematics, Information and Communications Technology and Agricultural Science comprehensively. It is based on the study of trends in numerous competitive past Science, Mathematics and Information Technology examination questions, marking schemes and syllabi of reputable international examination bodies, over a long period, as well as very good text books, chief examiners' reports, pamphlets and other dictionaries written by renowned authors. We have also combed the World Wide Web extensively to equip users with rare knowledge.

Standard (Examination-oriented) Definitions: Definitions are based on examiners' perspective to yield maximum marks for students. For example, in defining Osmosis we noted that: (i) It involves a **movement of molecules** of a solvent, usually water; (ii) from a solution of lower concentration. (iii) to a solution of higher concentration. (iv) through a semi-permeable membrane. (v) until the two solutions balance in concentration. In most cases, vocabularies and topics which serve as favourites of examiners are comprehensively explained, with numerous relevant examples and illustrations tailored to meet the requirements of competitive examination bodies. The diagrams are generally drawn and labelled in ways that examiners would expect. This is a reference book par-excellence, for quick and easy access to a broad range of very important, systematically captured facts, which reflect examiners' views as much as possible. The book is such that its contents are often adequate to answer lots of questions and the aim is to make it directly relevant to all students working towards passing examinations with distinction. Since examiners may use different concepts of the same terminology (which students may not be familiar with) synonyms and explanation of difficult concepts within definitions are used to give a multiplier effect in learning.

Competence of Authors: The contributors and editors are highly qualified academics, experienced research scientists, authors, lecturers, teachers, consultants to governments, local and international organisations, senior examiners of reputable international examination bodies and heads of very prominent institutions.

Category of Users: It is very suitable for, and recommended to all students in Senior High Schools, Technical Schools and Colleges of Education, however, its comprehensiveness, simplicity, modernity and the Automatic Knowledge Acquisition Mechanism makes it easy for those in Primary and Junior High Schools to use very beneficially in building a strong base to become distinguished scientists. This book is meant for long term education of young people of all ages because, with time, learning is accelerated by the fact that certain features of the book as well as the context of learning become catalysts for recollecting facts once encountered. As a student or pupil, your teacher may not always be with you, but this book will always be with you to assist you or bail you out in difficult learning situations relating to Science, Technology and Mathematics. Tertiary institutions as well as ordinary people who desire to widen their horizon in science and related subjects will find it very useful. Teachers and examiners, especially those who set science examination questions, prepare marking schemes, mark and write reports on competitive science examinations will find this book indispensable for putting students on their toes.

Extra Attention: World Rankings in science and mathematics released by the International Association for the Evaluation of Educational Achievement observed that students from poor backgrounds are three times more likely to perform poorly in Science, Technology and Mathematics than their counterparts from better-endowed backgrounds. This book tries to put all students on an equal footing to excel in science by including features such as modernity, comprehensiveness and the Osmosis Game. Thus, as a student/pupil, your first priority is to grab your own copy of this dictionary immediately, irrespective of your background. For the same reason, and without excuses, all heads of educational institutions, parents and guardians are entreated to give a copy each, to all their students and wards, create an enabling environment for playing the Osmosis Game, encourage them in other ways of using this unique dictionary and they will obtain results which will pleasantly surprise all concerned. Also, in communities that are deficient in Science and Mathematics teachers as well as educational materials at the pre-tertiary level, this book gives their students a very strong foundation to compete favourably at higher levels with their more fortunate counterparts, if they devote time to playing the Osmosis Game on a daily basis, making this book their final authority in their search for excellent definitions, explanations, diagrams, examples, etc. and how to access the limitless opportunities available in science, technology, engineering and mathematics education.

FUN TIME WITH SCIENCE

AN EFFECTIVE KNOWLEDGE ACQUISITION MECHANISM FOR EVERY STUDENT & FOR EASY LEARNING

POLICY ISSUES

The countries currently implementing Science, Technology, Engineering and Mathematics (STEM) Education in Africa include Ghana, Nigeria, Cameroon, South Africa, Ethiopia, Senegal, Rwanda, Cape Verde, Chad, Malawi, Tanzania, Togo, Uganda, Kenya, Zambia, Zimbabwe, Angola, Madagascar, Burundi, Botswana and Mali (https://www.forbes.com).

In all these countries the curricula review corresponds to national priorities of shifting the structure and content of the educational system from merely passing examinations to building character, nurturing values, raising literate, confident, and engaged citizens who can think critically, as well as follow curricula which address the inherent challenges in the existing curricula as well as ensure that the contents of the new curricula can be internationally benchmarked everywhere. The new approach to learning these subjects also involve inclusion, classroom level autonomy, integrated and coherent teaching and learning, flexibility, responsibility and accountability.

In Ghana, for example, government policy initiatives include development of new syllabi at all levels, development of new textbooks, training of teachers, building new structures, equipping existing ones and generally building the capacity of all personnel and institutions in the educational system. The Ministry of Education's policies, therefore, hinges on six core competencies as follows:

- Critical thinking and problem solving,
- Creativity and Innovation,
- Communication and collaboration,
- Cultural Identity and Global Citizenship
- Digital Literacy and
- Leadership and Personal Development.

Even though mathematics and science are used everywhere, everyday, by everyone and for various purposes (but with varying sophistication), many are the harrowing stories of students' experience and engagement with these subjects, especially at the pre-tertiary levels. According to a former Vice Chancellor of the University of Cape Coast and former Minister of Education, Prof. Jane Naana Opoku Agyeman (29th April, 2019), the fear of mathematics and science is real in all schools and in all countries, thus, these subjects continue to be important areas for effective policy interventions and innovations. In her opinion, having studied trends both locally and internationally, it became apparent that Schools' Curriculum and Study Programmes, ESPECIALLY IN MATHEMATICS AND SCIENCE REQUIRE AN URGENT OVERHAUL, NOT SO MUCH IN CONTENT AS IN THE LANGUAGE OF INSTRUCTION AND THE MODE OF DELIVERY IN THE CLASSROOMS.

Thus, the main focus of this project is to compile an accurate and comprehensive Science, Technology, Engineering and Mathematics Dictionary for use by all students, teachers and other stakeholders and then introduce Gamification in all classrooms globally as an educational tool in addressing the problems of inappropriate language of instruction and mode of delivery in science and mathematics education at the basic and second cycle schools.

This Game and Dictionary reduce all the different subjects making up STEM into one subject, making teaching and learning of STEM easy for all stakeholders, providing the linkages among all the various STEM subjects and removing the perception among stakeholders, especially teachers and students, that each component of STEM is a totally different subject, with little or no link to each other

FUN TIME WITH SCIENCE GAME

Yu-kai Chou (2012) defines gamification as the craft of deriving all the fun and addicting elements found in games and applying them to real world or productive activities, while Wang (2011) sees it as a series of designed principles, processes and systems used to influence, engage and motivate individuals, groups and communities to drive behaviours and influence desired outcomes. Other scholars see games as interactive play that teach participants goals, rules, adaptation, problem solving and interaction. They satisfy our fundamental need to learn by providing enjoyment, passionate involvement, motivation, ego gratification, adrenaline flow, creativity, social interaction and emotion in the game itself while the learning takes place. – en.m.wikipedia.org. As educators, governments and parents realize the psychological needs and benefits of gaming on learning; thus, this educational tool has become mainstream in learning. The Game type chosen for this project utilises the Osmosis and Initiation Approaches to education.

Osmosis is a scientific way of learning derived from a biological concept: In this sense, it is defined as the process by which molecules of a solvent (often, water) tend to pass through a semi-permeable membrane from a less concentrated solution into a more concentrated one, until the two solutions become equal in terms of concentration. Applying this to the Psychology of learning, osmosis is defined as the process of gradual, often unconscious assimilation of ideas, knowledge, etc., through continual exposure rather than deliberate learning. Osmosis is an efficient, enjoyable and social way to learn. It is an all in one learning platform which really provides a lot of memory hooks for you. The base idea is that using spaced repetition everyday creates the foundation for the learner, in this case, in science, technology and mathematics. When you are about to forget something, osmosis helps you to remember and understand. Confucius said: I do and I understand: I see and I only remember: I hear and I forget. Thus, Osmosis is the best tool for staying organized and integrating classwork with high yield resources, all on one platform.

A related concept of osmosis is the immersion or initiation approach to learning. Immersion is the act of dipping something in a liquid, with the liquid completely covering it. Thus, Immersion learning refers to any educational approach that teaches by placing a student directly in an environment, where the activity to be learnt is naturally taking place, and the learner then becomes part of the process and learns the activity through participation in it. There have been many studies showing positive results from immersion learning, especially if a student also receives explicit instruction on the target information at some level. Learning through initiation is also a natural, organic way to learn a language, in this case, the scientific language. For example, children learn their mother tongue through initiation and one only has to look at children who are bi-lingual to see that this is the best way to learn. They can learn to speak both languages at once just by listening to their parents speak in the different languages and participating or imitating them. Shockingly, they assimilate all or most of the vocabulary and terminologies in those languages unconsciously and within a few years you would hear them rattling those languages, even more efficiently than an adult who has spent many years learning those languages. Thus, one of the most common uses of immersion learning is in teaching foreign languages.

Using the Osmosis and immersion Approaches, this project teaches students how to speak the scientific language at very tender ages, so that as they progress on the academic ladder, they would have no difficulty opting for the science programme, mastering science, becoming scientists and applying science in their professional and everyday life. Learners would be encouraged to speak, think and live science by playing the game. The sheer mastery of the language acquired from playing the game from Basic and Senior High School levels enhances the students' interest and confidence in Science, Technology and Mathematics as they pursue their academic and career goals. Science is fun our young ones must be encouraged to discover this.

The Osmosis/Initiation Game demystify science and relieves the usual tension associated with the subject on most peoples' minds. The process enables individuals to acquire knowledge, skills and attitudes, which leads to the development of their faculties in full. When these dimensions of education are achieved, then learning becomes broad-based, rapid, meaningful, fulfilling and very beneficial to the learners, their countries and the world.

Most of what the children may be learning today through initiation/ osmosis may appear too high for their level, but just as children learn their mother-tongues, today's peripheral knowledge becomes tomorrows core knowledge and by that time, learners do not have to struggle, cramming anything into their cranium: the knowledge is already there, just like their mother tongues; in their subconscious minds. Anytime they need it, all they do is to go and fetch it, simply by recalling and using it the way they want.

Lots of human and other animal behaviours, including apprenticeship, are learnt through Immersion and Osmosis, thus these are highly beneficial tools to be deployed in studying the myriad of problems in Science, Technology and Mathematics Education. The UCYDE Centre believes that students can achieve a lot of learning and understanding of science and mathematics through immersion and osmosis. At the UCYDE Centre, the student is sat down and impacted with high doses of science, technology and mathematics languages, simply by playing competitive games with the 6-in-1 Science, Technology and Mathematics Dictionaries.

This game can be used for competitive entertainment at all levels in schools, at any social setting or at homes. The aim is not to learn; it is just to play a game; have fun; but alongside as you are playing the game, you are learning; and in this case you are acquiring high level scientific knowledge for your examinations; for your everyday work, and so on, without knowing. In the case of examinations, multiple choice questions become easy and questions which ask students to define, explain, compare, distinguish between etc., also become very easy for students as a results of their mastery of the scientific language.

The game is particularly meant for junior users, to learn how to speak and write the scientific and mathematics languages as accurately as possible at tender ages. The aim is not to learn but children playing and directly absorbing high level scientific knowledge and ideas without knowing. A prominent educationist once said, if you want children to learn something, immerse them as much as possible in it as a way of learning seamlessly: there is a text to memory conversion issue in this theory.

No child should be denied the right to use this dictionary on the basis of being too young, since all that everyone needs to participate in this high level, effective, learning process is to be able to count from 1 to 62. This is because the longest word in the dictionary has 62 letters and every word must have a fair chance of being encountered or discovered by students.

Playing the game from Primary 1 to Junior High School 3 exposes one to (and familiarizes one with) 85,410 science, ICT and Mathematics terminologies and their meanings

Playing the game from Primary 1 to SHS 3 exposes one to (and familiarizes one with) 113,880 Science, ICT and Mathematics terminologies and their meanings; thus, automatically making one to assimilate high-level science, technology and mathematics knowledge happily without much effort and without knowing when and how, at a very tender age.

With Play at The Centre of Learning, the Game begins with an assertion of the ease and joy with which mathematics, science and technology can be understood. Thus, the results of the learning process are fantastic.

Before students write their final examinations in their final year (BECE), they would have been well-groomed in the vocabularies, concepts and meanings in at least three of their key subjects at the basic level: i. **SCIENCE**. Ii. **MATHEMATIC**S. iii. **ICT**.

Before they make their choices of subjects, courses and professions at the second cycle level, they would have been well-groomed in **PHYSICS, CHEMISTRY, BIOLOGY, ELECTIVE MATHEMATICS, INFORMATION AND COMMUNICATION TECHNOLOGY, AGRICULTURAL SCIENCE, CORE MATHEMATICS AND INTEGRATED SCIENCE (8 SUBJECTS)**. This lays a very firm foundation for students to be developed into all round powerful scientists, acting as incentives for many more to choose science and science-related programmes and subjects. Even those who veer into non-science programmes or professions will be well equipped with powerful science backgrounds to analyze and solve problems in their respective fields by deploying scientific tools and knowledge acquired from this book – the 6-in-1 Science, Technology and Mathematics Dictionary.

This project will change the learning environment of science, technology and mathematics and put a smile on the faces of all students studying these subjects.

HOW TO PLAY THE FUN TIME WITH SCIENCE GAME

Instead of junior or basic editions of this book, we have provided various levels of learning the whole book in the form of Games ranging from fundamentality (primary level) through simplicity (Junior High School level) to relatively taxing stages (Senior High School level) because we think the scientific language is holistic and must be acquired as such to make maximum impact in the life of all beneficiaries of Science Education.

1. The Osmosis/Initiation Game is played in seven stages and all that anybody needs to know in order to start playing the game is to be able to count from 1-62. To start playing the game, you may use or develop a scoring chart similar to the one shown on the next two pages, i.e., xxv-xxvi - or the one at Appendix 3 which you can detach and photocopy as many times as possible for daily use. Since the game can be played by any convenient number of persons the last column is denoted as player … n.
2. Next, you prepare 62 cards of the same size and numer them, as the ones shown at **Appendix 1 attached to the inside back cover of this book - which you can detach, cut and use for playing the game.** The 62 cards represent the number of letters of the longest entry in this dictionary **(ELECTRONIC DATA INTERCHANGE FOR ADMINISTRATION, COMMERCE AND TRANSPORT).** This is to give equal chances of selection to all the entries covered in the book. You then place the numbered cards in any convenient container and shake it well to mix up the cards. Alternatively, you can put the cards together and shuffle them as in playing cards.
3. Next, you prepare 26 cards, labelled A, B, C, ..., Z of any convenient size but same as the ones shown in **Appendix 2 which you can detach and use for playing the game.** These 26 cards represent the 26 letters of the alphabet and are meant to give every entry in this dictionary an equal chance of being selected. After separating/cutting the various letters, you place them also in any convenient container and shake them well to mix up, for the purpose of random selection. You can also arrange them in one pack for the purpose of shuffling, and random selection. Next, register each person or group taking part in the game. To ensure competitiveness, efficiency and fairness, a coordinator for the game is selected.
4. The coordinators are mostly fellow pupils/students. However, the assessment of written revision of definitions is mainly by teachers/senior students even though the participants can select some of the competitors, but all using the dictionary as a guide.

LEVELS OF THE FUN TIME WITH SCIENCE GAME AND THE LENGTH/COMPLEXITY OF WORDS:

The longest entry in this dictionary is 62 letters. However, at the Primary 1-3 stage we would start with a 1, 2, 3, 4, … up to a maximum of 10 letter words. If the pupils prove extraordinary, we then increase the number of letters in a word gradually towards the 62. At the Primary 4-6 level we would start with 15 letters, at the Junior High School we would start with 20 letters and at the Senior High School and above, we would start with 25 letters and work towards the maximum of 62 letters. However, coordinators and score masters/mistresses are advised to exercise their discretion at each level based on the ability of the class they are working with each time. This is not to say that longer words are necessarily more complex than shorter ones but rather that they may generally be easier to learn or remember.

Stage 1: Competing in ability to locate all sections of the STEM Dictionary easily: Letters of the alphabet are written on a cardboard and each letter is detached, put into a container and shaken to mix. A letter of the alphabet is picked at random without replacement, announced and students are asked to compete in opening the dictionary to any section of the dictionary whose words start with the picked letter. Any student or member of a group participating in the game, who is able to do this first wins a point for himself or his group. This continues until the last letter is picked. The winner could be announced at the end of this session or at the end of the entire game. This part is mainly for lower primary students and could be skipped by those in upper primary and above.

Stage 2: Competing in Accumulation and Familiarization with STEM Vocabularies and the Scientific language. In this case numbers are written on a cardboard from 1-62 - because the longest word in the STEM dictionary is 62 letters. These numbers are also detached and put into a different container and shaken to mix up. A number (eg. 4) and a letter (eg. B are randomly picked up and announced as 4B. This implies that competitors should look for any 4 letter word in the B section of the Dictionary. Whoever finds such a word first (eg. Base) wins the point for himself or for his group. The picked number is replaced however the picked letter is not replaced. The process is repeated until the last letter of the alphabet is picked. The points are summed up and the winners announced at the end of this session or at the end of the final stage of the game.

Note: The points earned per word depends on the length of the word. For example, if the length of a word is 5 letters, the one who found it earns five points, if it is 10 letters, he earns 10 points and so on. Thus, based on the level of the competitors and their competence, the game coordinators could determine that the maximum length of the words should not exceed a certain length or the opinion of the participants should be sought. For example, for those in Lower Primary, the numbers put into the container should be up to a maximum of 10, meaning they should not be working with words above 10 letters; For Upper primary students, the numbers in the container should be up to a maximum of 15, meaning working with words up to 15 letters; for JHS a maximum of 20 letter words, for SHS a maximum of 25 or 30 letter words and above this level they can work with words up to 62 letters. More daring or highly competitive groups or individuals at each stage could introduce longer words into their game or compete at higher levels of the competition.

STAGE 3: Competing in ability to Retain STEM Dictionary Words Learnt at Stage 2: The container with the letters of the alphabet is shaken for the letters to mix up thoroughly. A letter is picked up at random without replacement and announced or shown to the competitors. Competitors are asked to write down (or mention) the word it represented from the list of words picked at Stage 2. Another letter is picked at random and the process continues until the last letter is picked. Alternatively, participants are given blank sheets and asked to write all or as many of the words found in Stage 2, as they can remember from the words picked at Stage 2 of the Game. The answer sheet of each participant or group of participants is collected and marked. The winners could be announced at the end of this session or at the end of the entire Game.

Stage 4: Competing in ability to Find the meaning of the words found in Stage 2: The words found in stage 2 are written on separate pieces of papers, put in a container and shaken thoroughly to mix up. The words are then picked at random without replacement and announced or shown to competitors to look for the meaning. The individual or group that located the meaning in the dictionary first scores the point. This continues until the last word in the container is picked up and the meaning found. The scores of each group or individual can be totaled and announced or done at the end of the entire Game.

Stage 5-7: Competing in ability to Retain the Meaning of the Words found in Stage 2: The meaning of each of the words found in Stage 2 are made available to participants and they are given time to go through thoroughly. The time allocated for this is based on the level of participants, their ability and also on the discretion of the coordinator or referee in the Game. After the time given by the coordinator is up the meaning of the words are taken away from participants and they are tested on their ability to recall the meaning and all that are said about the word, including all definitions of the same word, diagrams if any, and every detail given about the word. In view of this the marks may be allocated according to the details given. These could be done in one of three ways or all done as separate stages:

Stage 5: Competing in ability to Define words orally: The container is shaken and a word is picked at random without replacement by a competitor and he or his group is asked to define the word in detail, with marks being awarded according to the details given by him or supplied by the other members of the group. If a group is unable to define its word to the satisfaction of the coordinator, the word is passed on to be defined by the other competitors for a bonus mark, before having their turn. The container is passed on to the other members to have their chance and the scores totaled. The winner is declared at the end of the session or at the end of the entire game.

Stage 6: Competing in ability to Retain Definition of words through Filling in Gaps left in Definitions: Key words or phrases are picked out of the definitions of each word and participants are asked to fill in the missing words or phrases. Competitors are timed. The answers are collected and marked at the end of the given time. The results are announced at the end of the session or competitors wait until the end of all stages of the Game.

Stage 7: Competing in ability to Retain Definition of words through Writing All Definitions Totally from Memory: Participants are given blank sheets of paper and given time to write all they know about each word based on the definition given by the STEM Dictionary. At the end of the given time the answers are collected, marked and marks awarded based on details such as definition, explanation, examples, diagrams, formulae, etc.

Stage 8: Announcement of Final Results and Interview of Participants: The various marks scored at each stage of the Coopetition are totaled and the secondary and overall winners are announced and rewarded or applauded. The participants are then interviewed to Express their Opinion in terms of whether they enjoyed the activities and what they have learnt in the process of participating in the entire game.

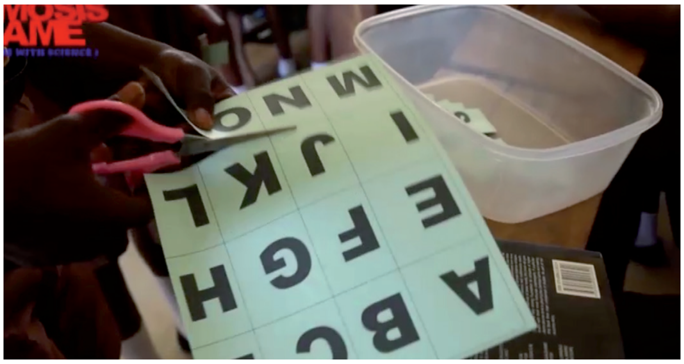
A: PRIMARY CATEGORY: PRIMARY 1-3:

1. Stage 1: How to locate different sections of the dictionary in terms of the letters forming the alphabet
2. Stage 2: Discovering the Science terminologies: that is, how to locate words in the dictionary.
3. Stage 3 and 4:Oral and written revision of the words found
4. Stage 5: Finding the meaning of the words found
5. Stage 6 & 7: Oral and written revision of the meaning of words

GETTING READY TO START THE OSMOSIS GAME: FUN TIME WITH SCIENCE!

1. Ensure that participants first practice how to count at least from 1-62 and revise their A, B, C, D, …, Z, thoroughly.

2. Let them cut the card with the alphabets to separate each letter from the others and put them into a container.



3. Let them cut the card with the numbers to separate each number from the others and put them into another container

4. Divide the participants into any convenient number of groups and give a copy of the dictionary to each participant. Preferably, each competitor in each group should have a table

SCORING CHART FOR PLAYING THE OSMOSIS GAME OF "FUN TIME WITH SCIENCE"

THE SCORING CHART FOR PLAYING THE OSMOSIS GAME OF "FUN TIME WITH SCIENCE"

No of Games Played	Dictionary Column for word search	Dictionary Skills Score	WORDS FOUND	NAME OF PLAYER 1:							NAME OF PLAYER n:							
				Word Finding Score	Oral Revision of Word Score	Written Revision of Word Score	Word Defining Score	Oral Revision of Def'n Score	Written Revision of Def'n Score	Total Score	Dictionary Skills Score	Word Finding Score	Oral Revision of Word Score	Written Revision of Word Score	Word Defining Score	Oral Revision of Def'n Score	Written Revision of Def'n Score	Total Score
14TH	A																	
15TH	B																	
16TH	C																	
17TH	D																	
18TH	E																	
19TH	F																	
20TH	G																	
21ST	H																	
22ND	I																	
23RD	J																	
24TH	K																	
25TH	L																	
26TH	M																	
TOTAL SCORES																		

SCORING CHART FOR PLAYING THE OSMOSIS GAME OF "FUN TIME WITH SCIENCE" (CONTINUED)

No of Games Played	Dictionary Column for word search	Dictionary Skills Score	WORDS FOUND	NAME OF PLAYER 1:							NAME OF PLAYER n:							
				Word Finding Score	Oral Revision of Word Score	Written Revision of Word Score	Word Defining Score	Oral Revision of Def'n Score	Written Revision of Def'n Score	Total Score	Dictionary Skills Score	Word Finding Score	Oral Revision of Word Score	Written Revision of Word Score	Word Defining Score	Oral Revision of Def'n Score	Written Revision of Def'n Score	Total Score
14TH	N																	
15TH	O																	
16TH	P																	
17TH	Q																	
18TH	R																	
19TH	S																	
20TH	T																	
21ST	U																	
22ND	V																	
23RD	W																	
24TH	X																	
25TH	Y																	
26TH	Z																	
TOTAL SCORES																		

in front of him/her, so that he/she can put his/her dictionary on the table, instead of holding it in his/her hand

5. Appoint Coordinators and Score masters/mistresses to act as referees or judges to administer the game.

STAGES OF THE OSMOSIS GAME:

STAGE 1: COMPETING IN LOCATING VARIOUS SECTIONS OF THE DICTIONARY

1. The coordinator should Shake the container with the 26 letters of the alphabet to mix well and ask the participants to get ready to start the competition
2. The coordinator then picks any letter at random from the container with the alphabets, announces the letter and ask the competitors to open their dictionaries to any section of the dictionary starting with the picked letter.
3. Anybody in any of the groups who is able to locate that section of the dictionary first by opening the dictionary to the section indicated wins 5 points for the group.

4. The picked letter is not replaced. Another letter is picked and the process continues without replacement of the letter. When the last letter is picked by the coordinator and the participants located it, this brings this stage to an end. The winners can be announced or the Score Master/Mistress waits until all the stages of the game are completed before announcing the scores.

STAGE 2: COMPETING IN WORD LOCATION IN THE DICTIONARY

1. **Pick the first 10 numbers from its container** and put it into a separate container. The size of the numbers picked should increase as the competitors gain confidence in playing the game.
2. Put the container with the letters of the alphabet and the container with the 10 numbers in front of you
3. Select a number, then a letter at random and announce them simultaneously. For example, if you select the number 4 and the letter C, you announce it as 4C.
4. You then give the command for the participants to look for any 4-lettered word from the C section of the dictionary
5. Any participant who finds such a word first wins points for his/her group equivalent to the number of letters of the word found.

6. The picked letter of the alphabet is not replaced in the container but the picked number is replaced and the process continues until the last letter of the alphabet is picked out. Again, the winner can be announced or the Score Master/ Mistress waits until the game is completed.

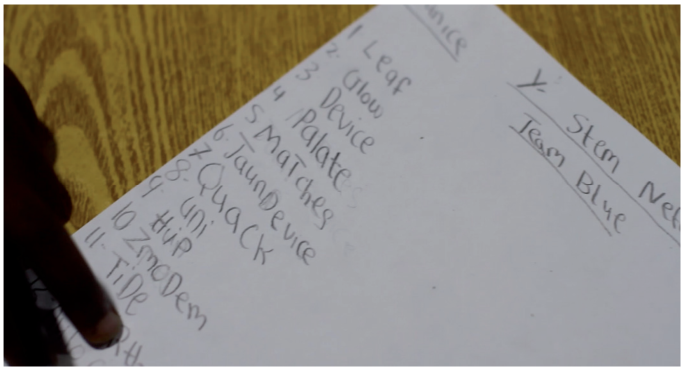
STAGE 3: COMPETING IN ORAL REVISION OF WORDS FOUND

1. The coordinator writes the found words on the board and gives participants 15 minutes break to examine the words.
2. After cleaning the board, the coordinator takes the dictionary away from all the participants.
3. The coordinator then picks the container with the letters of the alphabet and asks the students to get ready.
4. The coordinator shakes the container well to make sure the letters are well mixed up. He then picks a letter at random and announces it. For example, if the letter C is picked he announces it as C
5. The coordinator then asks the participants to mention or spell the word that the picked letter represents. The participant who is able to do so first wins 5 points for his/her group.

6. The process continues until the last letter is picked out of the container. Again, the winner can be announced or the Score Master/Mistress waits until the game is completed.

STAGE 4: COMPETING IN WRITTEN REVISION OF WORDS FOUND

1. After the oral revision, the coordinator then gives pen/pencil to the participants to write all or as many of the 26 words found in the course of the game as they can remember from memory.



2. Each correct word attracts 5 marks. These are marked and all the scores for each group are totalled.
3. The results may be announced immediately after the written revision test or at the end of the contest.

STAGE 5: COMPETING IN FINDING THE MEANING OF WORDS

1. Shake the container with the words found in them well.
2. Pick each word at random and ask the competitors to look for the meaning. Any participant that finds the meaning of the picked word first wins 5 points for the group.

3. The coordinator can read the meaning of the word to the participants or ask them to attempt reading it themselves. Any group member who successfully reads the meaning earns a bonus of 2 marks for the group.
4. The process is repeated until the meaning of the last word is found. Again, the winner can be announced or the Score Master/Mistress waits until the game is completed.

FINAL SCORING AND DECLARATION OF WINNERS

1. Note: Participants in Primary 1-3 do not compete in the oral and written revision of the meaning of the words unless they are extraordinarily sharp or have become very used to playing the game.
2. All the scores obtained from the Game are added.
3. The winning groups at the various Stages of the Competition are first Announced.
4. The Grand Total of each group and the Overall winner is announced, applauded and rewarded.

NB: The Pictures for Primary 1-3 are Pupils Selected from Various Schools in Ashongman Estate, Accra

PRIMARY 4-6

1. **Stage 1:** How to locate different sections of the dictionary in terms of the letters forming the alphabet
2. **Stage 2:** How to locate terminologies or words.
3. **Stage 3:** Oral and written revision of the words found
4. **Stage 4:** Finding the meaning of the words found
5. **Stage 5:** Oral and written revision of the meaning of words found (This stage may be introduced when the players become competent)

PREPARATIONS FOR THE GAME:

1. Let the pupils cut the card with

a. The alphabets to separate each letter from the others and put them into a container.

b. The numbers to separate each number from the others and put them into another container

2. Divide the participants into any convenient number of groups and give a copy of the dictionary to each participant. Preferably, each competitor in each group should have a table in front of him/her, so that he/she can put his/her dictionary on the table, instead of holding it in his/her hands
3. Appoint Coordinators and Score masters/mistresses to act as referees or judges to administer the game.

STAGE 1: DICTIONARY SKILLS COMPETITION

(This Stage May Be Ignored For Upper Primary Upwards If They Are Already Skilled In Dictionary Use)

1. The coordinator should Shake the container with the 26 letters of the alphabet to mix well and ask the participants to get ready to start the competition
2. The coordinator then picks any letter at random from the container with the alphabets, announces the letter and asks the competitors to open their dictionaries to any section of the dictionary starting with the picked letter.
3. Anybody in the group who is able to locate that section of the dictionary first by opening the dictionary to the section indicated wins 5 points for the group.
4. The picked letter is not replaced. Another letter is picked and the process continues without replacement of the letter. When the last letter is picked by the coordinator and the participants located it, this brings stage 1 to an end. The winners can be announced or the Score Master/Mistress waits until all the stages of the game are completed before announcing the scores.

STAGE 2: WORD FINDING COMPETITION

1. **Pick the first 15 numbers from its container** and put it into a separate container. The size of the numbers picked should increase as the competitors gain confidence in playing the game.
2. Put the container with the letters of the alphabet and the container with the 15 numbers in front of you
3. The **coordinator(s)** select a number, then a letter at random and announce them simultaneously. For example, if you select the number 4 and the letter C, you announce it as 4C.

4. You then give the command for the participants to look for any 4-lettered entry/word from the C section of the dictionary

5. Any participant who finds such a word first wins 10 points for his/her group
6. The picked letter of the alphabet is not replaced in the container but the picked number is replaced and the process continues until the last letter of the alphabet is picked out. Again, the winner can be announced or the Score Master/Mistress waits until the game is completed.

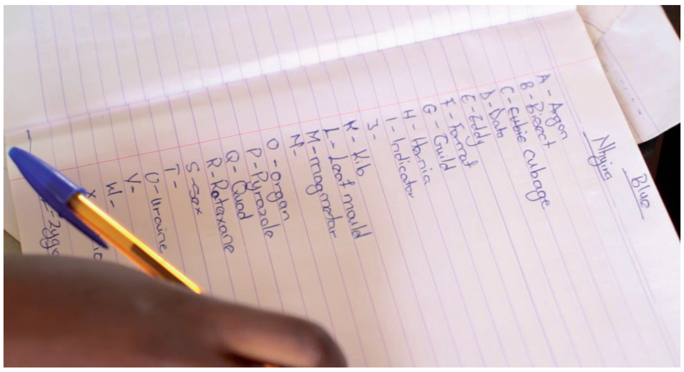
STAGE 3: ORAL REVISION OF WORDS COMPETITION

1. After the words are found the coordinator takes the dictionaries away from all the competitors and puts the words found also into a different container.
2. He then picks the container with the letters of the alphabets and shakes it.
3. He then picks a letter at random and asks the participants to mention the word that the letter picked represents. The group or individual who mentions the correct word from the list of words found at stage 2 first wins the point.

4. If they are unable to mention the word that the letter represents or get it wrong, it is passed on to another group or individual for a bonus.
5. The letter picked is not replaced and the process continues until the last letter is picked from the container.
6. The coordinator sums up the scores and announces the winner or announces it at the end of the entire game.

STAGE 4: WRITTEN REVISION OF WORDS

1. The coordinator writes the found words on the board and gives participants 10-15 minutes break to examine the words.
2. After cleaning the board, the coordinator takes the dictionary away from all the participants. He then gives blank sheets of paper and pen/pencil to each participant.
3. The coordinator then picks the container with the letters of the alphabet and asks the students to get ready. The coordinator gives the signal for the competitors to write all or as many of the 26 words found, as they can remember.



4. When the time given for work to be completed is up, the papers are collected and marked. The scores of the members of each group are totalled and the winners declared immediately or at the end of all the stages.

STAGE 5: COMPETING IN FINDING THE MEANING OF WORDS FOUND

1. Shake the container with the words found in them well.
2. Pick each word at random and ask the competitors to look for the meaning. Any participant that finds the meaning of the picked word first wins 10 points for the group.
3. The coordinator can read the meaning of the word to the participants or ask them to attempt reading it themselves. Any group member who successfully reads the meaning earns a bonus of 5 marks for the group.

4. The process is repeated until the meaning of the last word is found and read. Again, the winner can be announced or the Score Master/Mistress waits until the game is completed.

STAGE 6 & 7: ORAL AND WRITTEN REVISION OF MEANING OF WORDS

1. Where the competitors are capable, the coordinator can take the books way from them and test them on their ability to reproduce the meaning of the words found orally and in written form, from memory.

FINAL SCORING AND DECLARATION OF WINNERS

1. All the scores obtained from the Game are added.
2. The winning groups at the various Stages of the Competition are first Announced.
3. The Grand Total of Each group and the Overall winner is announced, applauded and rewarded.

NB: The pictures for primary 4-6 are pupils from the Golden Gate International School, Accra

JUNIOR HIGH SCHOOL

1. Stage one of the game comprises of finding/discovering science terminologies or words from the dictionary.
2. Stage 2 & 3 comprises of oral and written revision of the words found
3. Stage 4: comprises of finding the meaning of the words found
4. Stage 5: comprises of oral revision of the meaning of the words found
5. Stage 6: comprises of written revision of the meaning of the words found

HOW TO PLAY THE GAME

1. Let them cut the card with

a. The alphabets to separate each letter from the others and put them into a container.

b. The numbers to separate each number from the others and put them into another container

2. Divide the participants into any convenient number of groups and give a copy of the dictionary to each participant. Preferably, each competitor in each group should have a table in front of him/her, so that he/she can put his/her dictionary on the table, instead of holding it in his/her hand
3. Appoint Coordinators and Score masters/mistresses to act as referees or judges to administer the game.

STAGE 1: WORD FINDING COMPETITION

1. **Pick the first 20 numbers from its container** and put it into a separate container. The size of the numbers picked should increase as the competitors gain confidence in playing the game.
2. Put the container with the letters of the alphabet and the container with the 20 numbers in front of you
3. Select a number, then a letter at random and announce them simultaneously. For example, if you select the number 4 and the letter Q, you announce it as 4Q.

4. You then give the command for the participants to look for any 4-lettered entry/word from the Q section of the dictionary
5. Any participant who finds such a word first wins 10 points or mark could be number of letters forming the found word.

6. The picked letter of the alphabet is not replaced in the container but the picked number is replaced and the process continues until the last letter of the alphabet is picked out. Again, the winner can be announced or the Score Master/Mistress waits until the game is completed.

STAGE 2: ORAL REVISION OF WORDS COMPETITION

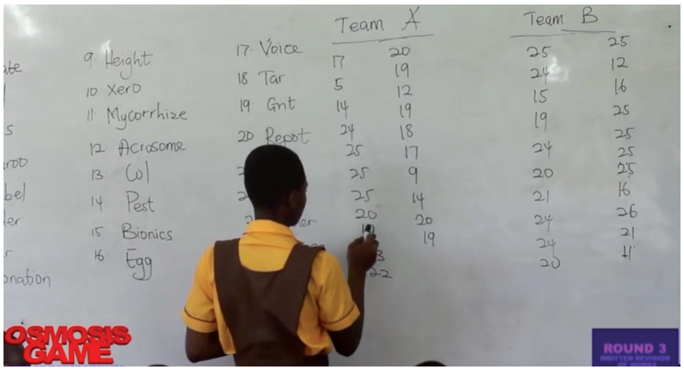
1. The coordinator writes the found words on the board and gives participants 10-15 minutes break to examine the words.
2. After cleaning the board, the coordinator takes the dictionary away from all the participants.
3. The coordinator then picks the container with the letters of the alphabet and asks the students to get ready.
4. The coordinator shakes the container well to make sure the letters are well mixed up. He then picks a letter at random and announces it. For example, if the letter C is picked he announces it as C
5. The coordinator then asks the participants to mention or spell the word that the picked letter represents. The participant who is able to do so wins 20 points.

6. If a group is unable to identify a word, it is passed on as a bonus to the other group.
7. The process continues until the last letter is picked out of the container. The scores of members of each group are totalled. Again, the winner can be announced or the Score Master/Mistress waits until the game is completed.

STAGE 3: WRITTEN REVISION COMPETITION OF WORDS FOUND

1. Papers and pens are given to all the competitors after the dictionary has been taken from them or closed.
2. The coordinator gives the command for each participant to write all or as many of the 26 words found, as possible

3. At the end, the scripts are collected and marked and the scores totalled. The winner can be declared or after the entire game.



STAGE 4: WORD DEFINITION COMPETITION

1. Shake the container with the words found in them well.
2. Pick each word at random and ask the competitors to look for the meaning.
3. Any participant that finds the meaning of the picked word first and reads wins 10 points for the group.

4. The coordinator can read the meaning of the word to the participants or ask them to attempt reading it themselves. Any group member who successfully reads the meaning earns a bonus of 5 marks for the group.
5. The process is repeated until the meaning of the last word is found. Again, the winner can be announced or the Score Master/Mistress waits until the game is completed.

STAGE 5 & 6: ORAL AND WRITTEN REVISION OF MEANING OF WORDS COMPETITION

1. About 20 minutes are given to the participants to revise the meaning of the words
2. Each group is divided into two, with the objective of one taking part in the Oral Revision Game and the Other in the Written Revision Game

STAGE 5: THE ORAL REVISION COMPETITION OF MEANING OF WORDS

1. While those doing the Written Revision are out of the room and competing those left to take part in the Oral Revision get themselves ready.
2. The container with the words are shaken before the competitors remaining with the audience.
3. The coordinator holds the container among each group of competitors and asks any member of the group to pick a word at random

4. The word picked is read and any member of the group is given the chance to define the word from memory.

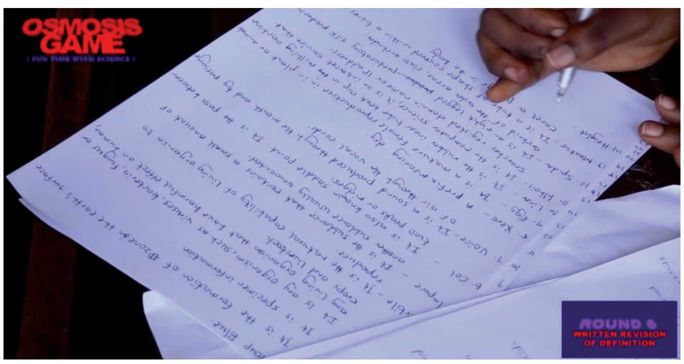
5. All members are allowed to contribute to the definition from memory, in case the one doing the definition is wrong or left something out. When they are exhausted, the Score Master/Mistress gives a mark up to a maximum of 30 or part thereof, depending on the effort put in.
6. The same question is passed on to the other groups, if the first group was wrong or left out some important elements in the definition. This bonus yields a maximum of 10 marks or part thereof.
7. There is no replacement of the word picked from the container.
8. Instead, the container is shaken and passed on to the other groups to have their turn.
9. This continues until the last word is picked out of the container
10. The marks obtained by each group from the Oral Revision are added and the results announced immediately or after all the stages.

STAGE 6:THE WRITTEN REVISION COMPETITION OF MEANING OF WORDS

This stage could be divivded into two parts: A, keywords are picked from definitions of each of the word found and competitors are given time and ask to fill in those keywords in the definition. All participant can take part in this.

Those taking part in the second part of the written revision Game are taken into a separate room, outside the purview of the audience to give them tranquility to compete. Each group is further divided into convenient sub groups of about 3 participants so that they can work for thier mother group.

1. They (the Written Revision Contestants) are given sheets of paper and pen (or computer, if available) as well as all the words selected.
2. They are given time and asked to define each of those words.



3. When the time is up, the Coordinator in charge collects the papers and marks, using the dictionary as a guide. The total score for each question is 25 marks based on i. word definition. Ii. Further explanation. Iii. Diagrams. Iv. Examples. V. uses or application of concept. Vi. Any additional point which can be proved to be right by the coodinators will attract a bonus of 5 additional marks.
4. All adding up to 30 marks per question well answered.

FINAL SCORING AND DECLARATION OF WINNERS

5. All the scores obtained from the Game are added.
6. The winning groups at the various Stages of the Competition are first Announced.
7. The Grand Total of Each group and the Overall winner is announced, applauded and rewarded.

NB: The pictures for the Junior High School are pupils from the Kwabenya Atomic M/A 5 JHS, Accra

SENIOR HIGH, TECHNICAL AND VOCATIONAL SCHOOLS, COLLEGES AND OTHERS

1. Stage 1 of the game comprises of discovering/finding the science terminologies or words.
2. Stage 2 & 3 comprise of oral and written revision of the words found
3. Stage 4 comprises of finding the meaning of the words found
4. Stage 5: comprises of oral revision of the meaning of the words found
5. Stage 6: comprises of written revision of the meaning of the words found

PLAYING THE OSMOSIS (OR FUN TIME WITH SCIENCE) GAME

Let them cut the card at the Appendix of this book with

a. The alphabets to separate each letter from the others and put them into a container.

b. The numbers to separate each number from the others and put them into another container.

3. Divide the participants into any convenient number of groups and give a copy of the dictionary to each participant. Preferably, each competitor in each group should have a table in front of him/her, so that he/she can put his/her dictionary on the table, instead of holding it in his/her hand.
4. Appoint Coordinators and Score masters/mistresses to act as referees or judges to administer the game.

STAGE 1: WORD FINDING

1. **Pick the first 25 numbers (OR ALL THE 62 NUMBERS) from their container** and put them into a separate container. The size of the numbers picked should increase as the competitors gain confidence in playing the game.
2. Put the container with the letters of the alphabet and the container with the numbers in front of you. Select a number, then a letter at random and announce them simultaneously. For example, if you select the number 4 and the letter C, you announce it as 4C.
3. You then give the command for the participants to look for any 4-lettered entry/word from the C section of the dictionary.

4. Any participant who finds such a word first wins points equivalent to the number of letters forming that word for his/her group
5. The picked letter of the alphabet is not replaced in the container but the picked number is replaced and the process continues until the last letter of the alphabet is picked out. Again, the winner can be announced or the Score Master/ Mistress waits until the game is completed.

STAGE 2: ORAL REVISION STAGE

1. The coordinator writes the found words on the board and gives participants about 10 minutes to revise the words found in stage 1.
2. After cleaning the board, the coordinator takes the dictionary away from all the participants.
3. The coordinator then picks the container with the letters of the alphabet and asks the students to get ready.
4. The coordinator shakes the container well to make sure the letters are well mixed up. He then picks a letter at random and announces it. For example, if the letter C is picked he announces it as C.
5. The coordinator then asks the participants to mention or spell the word that the picked letter represents. The participant who is able to mention or spell the word first wins 20 points. If the group is unable to identify the word, the question is passed on to another group for a bonus of 10marks
6. The process continues until the last letter is picked out of the container and the last word is identified. Again, the winner can be annnounced or the Score Master/Mistress waits until the game is completed.

STAGE 3: WRITTEN REVISION OF WORDS FOUND

1. Papers and pens are given to all the competitors after the dictionary has been taken from them or closed.
2. The coordinator gives the command to participants to write all or as many of the 26 words found, as possible
3. At the end, the scripts are collected and marked and the scores totalled. The winner can be declared or after the entire game

STAGE 4: WORD DEFINITION COMPETITION

1. Shake the container with the words found in them well. Pick each word at random and ask the competitors to look for the meaning.
2. Any participant that finds and starts reading the meaning of the picked word first wins 10 points for the group

3. The process is repeated until the meaning of the last word is found. Again, the winner can be announced or the Score Master/Mistress waits until the game is completed.

STAGE 5 & 6: ORAL AND WRITTEN REVISION OF MEANING OF WORDS

Each group is divided into two, with the objective of one taking part in the Oral Revision Game and the Other in the Written Revision Game

STAGE 5: THE ORAL REVISION COMPETITION OF MEANING OF WORDS FOUND

1. While those doing the Written Revision are out of the room and competing those left to take part in the Oral Revision get themselves ready.
2. The container with the words are shaken before the competitors remaining with the audience.
3. The coordinator holds the container slightly above the head of each group and asks any member of the group to pick a word at random
4. The word picked is read and any member of the group that picked the word is given the chance to define the word. All members are allowed to contribute to the definition, in case the one doing the definition is wrong or left something out.

When they are exhausted, the Score Master/Mistress gives a mark up to a maximum of 30 or part thereof, depending on the effort put in.

5. The same question is passed on to the other groups if the first group was wrong or left out some important details in the definition. This bonus yields a maximum of 10 marks or part thereof.

6. There is no replacement of the word picked from the container.
7. Instead, the container is shaken and passed on to the other groups to have their turn.
8. This continues until the last word is picked out of the container
9. The marks obtained by each group from the Oral Revision competition are added and result declared immediately or after all the stages of the game.

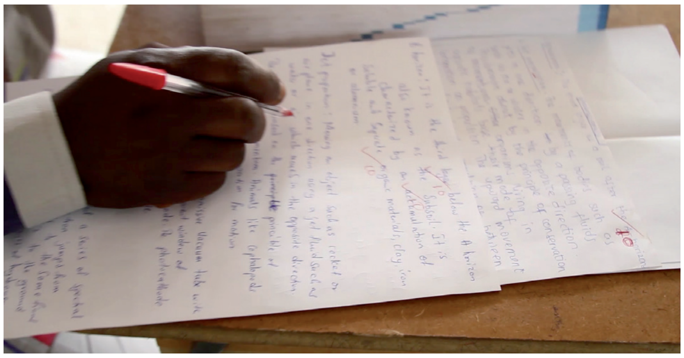
STAGE 6: THE WRITTEN REVISION COMPETITION OF MEANING OF WORDS FOUND

The written revision is in 2 parts: with the first part, key words are omitted from the definition of each of the words found and all participants are asked to supply or fill in those missing words in each of the definitions. These are collected and marked, then each group's marks are totaled and the results declared. With the second part of the written revision of the meaning of the words participants are expected to define each of the words found from memory.

Those taking part in the second part of the written revision Game are taken into a separate room, outside the view of the audience, to give them tranquility to compete.

1. They (the Written Revision Contestants) are given sheets of paper and pen as well as all the words selected.
2. They are given time and asked to define each of those words from memory.

3. Each team of senior high school or other senior students playing the game are divided into sub groups of about 3 participant each to put their heads together for ideas as they compete at the written revision stage of the osmosis game.
4. When the time is up, the Coordinator in charge collects the papers and marks, using the dictionary as a guide. Total score for each question is 30 marks based on i. word definition. Ii. Further explanation. Iii. Diagrams. Iv. Examples. V. uses or application of concept. Vi. Any additional point not in the dictionary attracts 5 additional marks. All adding up to 30 marks per question well answered.



The winners at this stage can be declared seperately or declared alongside all the other results from the various stages.

FINAL SCORING AND DECLARATION OF WINNERS

1. All the scores obtained from the various stages of the Game are added.
2. The winning groups at the various Stages of the Competition are first Announced.
3. The Grand Total of Each group and the Overall winner is announced, applauded and rewarded.

NB: The pictures for the Senior High School are students from West Africa Senior High School, Accra

NOTE: The Oral Revision Aspects of the Game can be Omitted in Serious Competitions so that in such Cases Only the Following Stages can Feature:

i. Word Finding Competition
ii. Written Revision of Words Found Competition
iii. Finding Meaning of Words Competition
iv. Written Revision of Meaning of Words Competition by Filling in the Gaps for Missing Words/Phrases
v. Written Revision of Words Competition by Writing out the Full Text of Definitions

On the other hand, Organizers can also choose to ignore the Written Revision Competitions and Focus Rather on the Oral Revision Competitions.

BENEFITS OF GAMES IN EDUCATION AND LEARNING ACTIVITIES

The benefits of using games in the classroom include:

6. **Making learning very effective:** According to Diane Ackerman, ‘’Play is our brain’s favourite way of learning.’’ Evidence is mounting that the benefits of play go well beyond entertainment. In fact, they help set up engaging scenarios, establish educational goals and help evaluate learners in many ways. According to Yann Teyssier (Nov.20 2016), Games for learning are trendy, and can be seen as the training for the future. Psychologists advise that when planning lessons, teachers should incorporate at least one game a day into one of their key learning areas as either a teaching and learning tool, assessment strategy or classroom motivator. This is because, when playing games, students become more engaged in their learning; thought content is reinforced and class productivity is increased.

7. **Motivating Learning:** When met with difficulties, people may feel depressed, devastated, discouraged or cynical; such feelings are not present in the gaming environment. This is where gamification steps in (Daniel Boateng Appiah, July, 2015). Muntean (2011) reiterates that gamification constitutes a powerful boost to make students determined to study or read more. By playing games, students become more motivated to learn, pay attention and participate in tasks. Games help students to become a part of a team, as well as take responsibility of their own learning. The concept is that a designer takes the motivational properties of games and layers them on top of other learning activities, integrating the human desire to communicate and share accomplishments with goal-setting to direct the attention of learners and motivate (Landers and Callan, 2011, page 421). Thus gamification helps to motivate students towards studying: because of the positive feed back, they are encouraged, they show interest and they are stimulated to learn.

8. **Classroom management tool:** Games are also a great classroom management tool, helping to motivate a class. The notion of Games is important for learners. In fact, this is often the reason why they will be motivated and interested in learning. Everybody likes to play and have fun, even more, when you learn skills at the same time. Moreover, the learners would want to win in the training and have the best results they can. When rewards are added to the training, this enhances the motivation more than a simple game. Even if it is a small gift, it will keep the desire to win for learners and they will have the impression you care about their training.

9. **Strategy Simulator:** Most games require problem-solving strategies and planning. By applying a range of strategies in a game, students are able to use their working memories to solve problems, thus increasing their mental cognition. Stimulating the brain with strategies in game can be a great brain workout.

10. **Powerful tool for New knowledge acquisition:** Games are a great tool to use in the classroom to acquire and consolidate new knowledge. Thus, before and after teaching new content to the class, a teacher must provide students with a game that will stimulate learning, consolidate their understanding and make connections with what they already know. Improvising or asking students to create their own content specific games can also be a great way to assess students at the end of a unit work.

11. **Development of Positive Learning Environment:** Playing games in the classroom is always great fun. When playing a game, endorphins are produced that stimulate the brain and gives students a feeling of euphoria. This feeling of euphoria creates a great sense of happiness and excitement for students in the classroom, thus, making them develop a positive learning environment. Also, using games in a lesson as part of teaching and learning, helps to create positivity around the lesson, motivating students with their participation and creating a positive attitude towards each other and the learning process.

12. **Getting Continuous Alertness and Focused Attention from Students:** Playing games require students to pay great attention to detail. As games can move quickly when playing, a student needs to be alert and attentive. This attentiveness when playing a game can help students to stay focused on their tasks in the classroom throughout the day.

13. **Enhancing Great Competition and Class Cooperation:** Games are a great way to promote healthy competition in learning among peers and they also increase class cooperation. Students need to work together as a team when playing either as a class against each other in small teams or as individuals against each other. Through games students learn how to take turns, build respect, listen to others and play fairly.

14. **Serve as a Strong Memory Builder:** Playing a range of content specific games increase memory greatly. As they play a game, students need to remember important details about a topic and also use their working memory to think and act quickly. As students construct a game, they are required to use their memory of specific content to create questions and answers suitable for the game, then use their memory of the topic to play the game. Games can also create a positive memory and experience of learning for students in the classroom.

15. **Giving Less Stress in Learning:** Having to answer questions on worksheet or produce a page of text can be quite daunting and stressful for many students. It can also create a negative perception of students learning environment and give a true indication of their own learning. However, in playing games students learn a lot for a long time without noticing any mental or bodily fatigue.

16. **Opportunity to learn while engaging in fierce, harmless competition:** By stepping away from the traditional lecture and teaching methods of decades past, educators can now allow their students to benefit from high interest, interactive games. In fact, Games are often significantly more effective in promoting student involvement in the lesson. Participants in games have intrinsic motivation to win. This drive keeps them tuned into the lesson and learning throughout the activity. These games can take a host of forms and can be applied to nearly any subject.

17. **Close Interaction with Learning Materials:** Games also allow students to interact with the material in a hands-on fashion, instead of simply being presented with the information and asked to retain it.

18. **Promoting team work:** Most games require teamwork. By allowing students to engage in a game play, teachers are providing them the opportunity to practice working cooperatively. Success in many classroom games require problem solving. To win, students must figure out and answer out or navigate a puzzle. When teachers have students participate in these games, they are providing them with the opportunity to practice their problem-solving skills. The more they practice these complex problem solving skills the better they will become critical thinkers. The decision-making processes required to play certain games makes the brain to work hard. These cognitive processes can range from simple decisions to the formulation of complex strategies. Children and adults of all ages can benefit from the mental stimulation that game-based learning provides.

19. **Enhance Great Retention**. Educational games can result in higher retention rates, compared to book-learning. In fact, the knowledge and skills acquired through game-based learning are retained longer than information from other learning methods. This is because game-based learning allows kids to develop cognitive, social and physical skills simultaneously. This learning enhances essential life skills such as cooperation and team work. Also, game-based learning allows learners to explore different parts of games as a form of learning. Games are typically designed at different ability levels and with the goal of helping the players to retain the information that they learn and apply it to other problem solving situations. Many of these games are relevant to real life situations and helps children to make informed decisions when doing so many other things.

20. **Games feel more like a form of entertainment than a method of learning.** Because games include rules, definitive objectives, measurable goals, and competition, they deliver an interactive experience that promote a sense of achievement for all of the participants. Learners are motivated by hands-on and active learning opportunities. Students are able to work on accomplishing a goal by choosing specific actions. They experience the consequences of the actions, which is one of the ways that a game-based learning experience is similar to real life. The engagement between learners keeps them coming back to learn even more. The on-going practices of decision making, planning and learning in a game environment are easy to translate to everyday situations that children will face as they become older. Educational games allow kids to practice and develop physical skills such as hand-eye coordination. In such an integrated learning environment, educational game players can also work on spatial and fine motor skills.

21. **In game learning, Digital literacy is present and it is an important skill for a life time use.** Students need to acquire basic skills such as problem solving, developing creativity, analytic thinking, collaboration, etc. with others. Other important skills kids must have in digital age include strong communication skills, ethics and accountability, which they acquire easily while playing the game. Each time students play the same game, they perform cognitive actions such as recalling the rules, keeping track of hazards, and remembering how the sequence of play works. Kids utilize their strategic thinking skills including using logic to make sound decisions and to plan ahead by making predictions about what might happen next. Students also develop strong problem solving stills. They will need to think quickly on the spot without being able to hesitate, which is a skill that will stay with and serve them throughout their life. Learners also learn how to think creatively and plan out their moves a few steps ahead.

22. **Learners benefit from the immediate feedback that takes place during game playing.** Instead of having to wait days or even weeks for assignments or test grades, students get instantaneous results about whether or not they made a good decision. They also get to find out the long term effects of their decision making. One decision at the beginning of a game could have lasting effects throughout play. The rapid feedback helps kids to realize quickly when they made a good decision or bad one. Educators are also able to get rapid feed back by watching how the children engage and react. While playing a game, children also have the freedom to make mistakes without any major consequences of physical or mental harm. They can experiment in a safe environment while playing games. Any mistakes that are made can be discussed in a group setting afterward. This allows students to reflect on what they did and perhaps change or maintain their strategy for the next time.

This is what UNICEF also has to say in a book entitled,

learning through play, developed by the Education Section of UNICEF's Headquarters office:

1. **Play is a major strategy by which young people gain essential knowledge and skills.** For this reason, creating opportunities and environments that promote play, exploration and hands on learning are core values for effective primary education. Thus educators should be rethinking how to teach young people to tap their enormous learning potential through play.

2. **Children or adults playing are often smiling and laughing**. The overall feeling is one of enjoyment, motivation, thrill and pleasure.

3. **Play is actively engaging:** Young people playing become deeply involved, often combining physical, mental and verbal engagements.

4. **Play is Iterative:** Young people play to practice skills, try out possibilities, revise hypotheses and discover new challenges, leading to deeper learning. Thus play and learning are dynamic methods of learning fast difficult and impossible topics.

5. **Play is socially interactive:** Play allows young people to communicate ideas, to understand others through social interaction, paving the way to build deeper understanding and more powerful relationships.

6. **In the light of these, UNICEF recommends that: Both public and private investments should take into account essential financing for the appropriate learning materials, equipment and professional support that lay a strong emphasis on play as much as possible**.

OTHER WRITERS' VIEWS ON IMPORTANCE OF GAMES AND PLAY IN LEARNING AND THE CLASSROOM IN GENERAL:

1. **Play is important in learning because play is the true work of a child.** Learning through play is crucial for positive, healthy development, regardless of a child's situation.

2. **Learning through play is about continuity; bringing together children's spheres of life** – home, school, the wider world and doing so over time – Susan MacKay, Director of Research and Teaching at Portland Children Museum'.

3. Research has shown that children in direct instruction programs can experience negative effects such as stress, decreased motivation for learning and behavior problems. **However, in recent studies, children's learning outcomes are shown to be more effective in a play-based program,** compared to children's learning outcomes in direct instruction approaches.

4. **Involvement in a play stimulates a child's drive for exploration and discovery.** This motivates the child to gain mastery over their environment, promoting focus and concentration. It also enables the child to engage in the flexible and higher level thinking processes deemed essential for the 21st century learner. This include inquiry processes of problem solving, analyzing, evaluating and applying knowledge and creativity.

5. **Play also supports positive attitudes to learning.** This includes imagination, curiosity, enthusiasm and persistence. The types of learning processes and skills fostered in play cannot be replicated through rote learning where there is an emphasis on remembering facts.

6. Research indicates increased complexity of language and learning processes used by children. **Play-based programs are, however, linked to important literacy skills.** This includes understanding the structure of words and the meaning of words. Play-based programs can provide a strong basis for later success at school. They support the development of socially competent learners, who are able to face challenges and create solutions

7. As an adult **Albert Einstein** remembered a pivotal event in his life in which merely playing with a compass on his sick bed inspired his interest in scientific discovery. He wrote that, ''this experience made a deep and lasting impression on me. **Something deeply hidden had to be behind things."**

8. **Thus, play is far more powerful for children than many teachers, parents and policy makers realize.** It is the key to learning. Researchers and educators all over the world have found out that play enriches learning and develops key skills such as inquiry, expression, experimentation and teamwork. This dictionary, with the Osmosis/Immersion Game attached to help you find science, mathematics and technology the easiest and most fascinating subjects to study is a one-stop accurate walking encyclopedia, for all students and pupils.

BENEFITS OF FUN TIME WITH SCIENCE USING THE OSMOSIS AND INITIATION APPROACHES FOR EASY LEARNING

1. An activity-based student teaching and learning programme, such as the Fun Time with Science (or the Osmosis Game), makes every classroom very ACTIVE. Here the teacher's role is mostly that of a facilitator rather than an authority in knowledge. He engages learners in tasks and makes abstract ideas into concrete ones. Even without the teacher, some of the students could play this facilitator or coordinator role. The most important feature is learning by doing rather than learning by listening, so this method of instruction fulfills the natural urge of growing children to learn more and in the process, they end up absorbing high doses of scientific facts and ideas, without knowing when and how they did it. It is at its best in deprived schools, even though its impact in endowed schools is no less important.
2. Learning through **PLAY** as the Osmosis Game does:

a. Offers a pressure-free environment for students to learn: No matter the size, background or ability of a class, students can engage in the various stages of the Osmosis Game at their own pace, with immediate feed-back.

d. Promotes better understanding of lessons among students as they learn the lesson by practising the tasks themselves. It is Student-centred and they play the lead role by formulating strategies to compete and win.

c. Inspires the students to apply their creative ideas, knowledge and minds in solving problems as well as promoting competitive spirit among them.

d. Helps learners psychologically as they can express their emotions through active participation in very useful activities. This helps in developing their personalities, social traits and inter-personal management skills.

In fact, many renowned educationists are of the opinion that this activity based method of learning is very suitable for subjects involving the experimental sciences.

3. The Osmosis Game is based on long term planning, with minute details of the process. The planning is centrally done by experts, with rigorous but simple instructions. The objectives of the osmosis game are achieved by making the planning of the lessons flawless. As a result, teaching flaws and teacher ignorance of the subject matter does not affect students. In fact, one can judge the progress of many students instantly through their performance in the Osmosis Game, as it progresses through the various stages.

4. Even though Learners have varied levels of understanding and performance, the Osmosis approach sooner or later encourages and brings the less-endowed students to the level of the meritorious ones, especially if they are made to work in the same team. The Spirit of competition pushes participants to the limit of their endurance as well as promotes development of personality and self-confidence.

5. Fear of punishment from teachers and parents for poor performance in examinations, which creates pressure, exhaustion, stress, fear and hatred of science and mathematics as well as loss of interest in the subject and faith in the teachers are avoided. Also, failure in class tests, examinations and other forms of assessment which can lead to loss of self-confidence, habouring of low self estimation, which also leads to various health problems and sometimes inducing suicide tendencies in students are avoided in the Osmosis Approach.

6. Continuous group activities such as this creates strong bonds of companionship, minimizes deviant behaviour and promotes good corporate behaviour as well as social harmony.

7. Mass Education and Entertainment: The Fun Time with Science or the Osmosis Game electrifies and throws any crowd into a frenzy; hence it could be used to effectively entertain or educate large numbers of people. It can also generate keen interest in the study of science among large numbers of people, especially youngsters. All that the Coordinator needs to do is to arrange for the participants to sit according to the desired groupings for effective competition and then select the level of the Game to be played, based on the level and type of education of the participants.

The key issue here is not the pupil pronouncing, understanding fully or memorising the words and their meanings, but being exposed to and playing with the words and their meanings often may result in their being stored in the sub-conscious mind, such that the next time the pupil sees or hears such words in any context, they can be recalled fully or partially for any purpose. By the Osmosis Approach, therefore, the pupil gains high-level scientific knowledge steadily without making any conscious effort towards learning (just as osmosis occurs naturally). All things being equal, the Osmosis Approach gives so much competitive advantage to those who start playing the Game early, especially from very basic levels.

The game gives all categories of players an unquenchable thirst for Science, Technology and Mathematics as they discover the mysteries of these subjects and the fact that they can be studied with fun and ease through mastering the languages (of Science, Technology and Mathematics). The game also makes it possible for students/pupils to be more knowledgeable than their teachers in certain concepts or topics from time to time.

As they try to use these new words in their everyday lives, be it in parlance, classwork, examinations or otherwise, they stand the chance of conquering their fear of Science, Technology and Mathematics and on their way to becoming powerful Scientists.

Even though this game could be computerised, the gains resulting from humans playing it with one another in a competitive spirit adds that crucial impetus and fun which comes from social and psychological effects. Parents, siblings and teachers of younger users of the book, especially those in Primary Schools, and Junior High Schools are encouraged to actively get involved in playing the game with pupils under them and they would sooner or later be fascinated by their unconscious ability to produce competent scientists easily.

No one restricts a growing child as to how many and what type of words of their spoken language the child learns at any time; except when it is badly used; and since the users of this book are not going to be making any effort to learn, but rather learning by engaging in an interesting competitive game, we urge all stakeholders to encourage all categories of pupils/students to learn quickly and speak fluent Science, Technology and Mathematics languages with the adult users, so that their development into great scientists will occur at very tender ages than the world has ever been used to.

Fun Time with Science is relevant always and takes you to any level of your education and professional life.

We are all born with an immature brain, which is why using the right learning tools in early childhood development is so important. In fact, children remember better if the teaching is associated with an object that they could relate to. Thus, when they play the Osmosis Game, they imbibe the contents relating to the Science, Technology and Mathematics Dictionary, the Cards, Scoring Sheets and other objects.

No matter how it looks like initially or your preconceptions about Fun Time with Science or Osmosis Game, introduce it to all the children at all levels. They always start with reciting A, B, C, D, …, Z; follow it with revising their counting from 1, 2, …, 100; then start the Game with a 1-letter, 2-letter, 3-letter …, x-letter words until they reach multiple letter words and then, everything in the Dictionary, with their meaning and application gradually rehearsed/revised and known.

When you recycle the same knowledge again and again into your system in such an easy and flexible manner as the Osmosis Game does, it becomes fully digested and absorbed into your entire make-up in such a way that it becomes part of you; a catalyst which precipitates new knowledge; which, working through the multiplier effect, fills you with so much valuable information in all fields of science, technology and mathematics; such that you can never forget them. Our heart burns with passion to churn out numerous problem-solving, highly professional scientists, who will lift the world onto another pedestal with their inventions and discoveries. Our approach is mass mobilization of every youngster into the Fun Time with Science or the Osmosis Game.

Register yourself, school, organization or home now to participate. Give your little ones a chance in life by introducing the Osmosis Game into your school and home. It works. Do not be left out.

IMPACT OF THE STEM DICTIONARY AND OSMOSIS GAME ON STUDENTS' PROGRESS AND SUCCESS

Play is important in learning because, play is the true work of a child. What then is the impact of the Osmosis Game and Dictionary on students' progress and success in life? Playing the Osmosis Game from Primary 1 to Junior High School Form 3, exposes students to as many as 85,410 STEM Vocabularies, their meanings and applications, while playing it from the same level to Senior High School Form 3 exposes them to about 113,880.

Since examinations remain the most potent way of assessing students' academic progress globally, we would apply the same principle, without loosing sight of the fact that this learning activity permeates all aspects of students' life and progress; especially the need to be able to speak and write the scientific and mathematics languages fluently, in order to function seamlessly in a STEM-Driven world. Most questions involving Theory and Multiple Choice Responses are answered directly by playing the Osmosis Game or using this Dictionary as a constant, personal reference book while questions involving Calculations and Practicals are mostly answered indirectly.

Below are samples of some of the numerous questions answered directly by this STEM Dictionary and the pages to find them. The subjects selected are Integrated Science, Biology, Chemistry and Physics. The questions are from the West African Examinations Council's Basic Education Certificate Examinations (BECE) and its Senior School Certificate Examinations (WASSCE); however, questions from any pre-tertiary Syllabus or Examinations Body could be used. The Dictionary and Game are indispensable for excellent grooming of students at the pre-tertiary level globally for any of the branches of Science, Technology, Engineering and Mathematics.

INTEGRATED SCIENCE QUESTIONS FROM THE WEST AFRICAN BASIC EDUCATION CERTIFICATE EXAMINATIONS (B.E.C.E.) AND STEM DICTIONARY REFERENCE PAGES

1. What is a FOOD CHAIN? [BECE 2019, Integrated Science, Q2 a(i)] Reference for Answer in Science, Technology and Mathematics (STEM) Dictionary: page 528.
2. What is an ELEMENT? [BECE 2019 Integrated Science Q6 d(ii)], [BECE 2011 Integrated Science Q6 c(i) and [BECE 2001 Integrated Science Q3 b(i)] Ref: STEM Dic. page 445.
3. Why is the plant in any food chain referred to as a PRODUCER? [BECE 2019 Integrated Science Q2 a(ii)] Ref: STEM Dic. page 1038.
4. What is the term given to the other organisms that DEPEND DIRECTLY OR INDIRECTLY ON THE PRODUCER FOR FOOD? [BECE 2019 Integrated Science Q2 a(iii)] Ref: STEM Dic. page 302 (CONSUMERS,)
5. What is MATTER? [BECE 2019 Integrated Science Q3 b(i)] Ref: STEM Dic. page 799.
6. State two of the STATES OF MATTER. [BECE 2019 Integrated Science Q3 b(ii)] Ref: STEM Dic. page 799.
7. Describe the nature of LOAMY SOIL. [BECE 2019 Integrated Science Q3 c(i)] Ref: STEM Dic. page 761.
8. Name any two PLANTS NUTRIENTS [BECE 2019, Integrated Science Q3 c(ii)] Ref: STEM Dic. page 999
9. What is a FORCE? [BECE 2019 Integrated Science Q4 d(i)] Ref: STEM Dic. page 530.
10. What is POLLUTION? [BECE 2019 Integrated Science Q5 c(i), c(ii)] Ref: STEM Dic. page 1012.
11. What is PHOTOSYNTHESIS? [BECE 2019 Integrated Science Q6 c(i)] Ref: STEM Dic. page 986.
12. The diagram below is an illustration of a fish. Study the diagram carefully and answer the questions that follow: i. Identify the fish. ii. Name each of the parts labeled. iii. Name the habitat of the fish. [BECE 2018 Integrated Science Q1 a(i)] Ref: STEM Dic. page 516 (Fish) page 1293 (Tilapia).
13. What is an ION? [BECE 2018 Integrated Science Q2 a(i)] Ref: STEM Dic. page 687.
14. Differentiate between PESTS and PARASITES as used in agriculture. [BECE 2018 Integrated Science Q2 b] Ref: STEM Dic. page 971 (Pest) and page 947 (Parasite)
15. What is WORK? [BECE 2018 Integrated Science Q2 c(i)] Ref: STEM Dic. page 1384.
16. Name two diseases associated with the CIRCULATORY SYSTEM OF HUMANS. [BECE 2018 Integrated Science Qn 2(d)] Ref. STEM Dic. page 742 (Leukaemia or Leukemia), page 599 (Haemorrhoids or Hemorrhoids).

17. What is MALNUTRITION? [BECE 2018 Integrated Science Q3 a(i)] Ref: STEM Dic. page 787.

18. State one symptom each of the following deficiency diseases a. SCURVY b. RICKETS Ref. STEM Dic. page 1145 and page 1112

19. Explain the term HAZARD. [BECE 2018 Integrated Science Q4a] Ref: STEM Dic. page 608.

20. Differentiate between OSMOSIS and DIFFUSION. [BECE 2018 Integrated Science Q4 (b)] Ref: STEM Dic. page 925 (Osmosis) and page 380 (Diffusion).

21. What is WEATHER? [BECE 2018 Integrated Science Q4 c(i)] Ref: STEM Dic. page 1372.

22. What is a FERTILE SOIL? [BECE 2018 Integrated Science Q4 d(i) and 2014, Q2 c(i)]. Ref: STEM Dic. page 1190 (soil nutrition) .

23. What is a MAGNETIC FIELD? [BECE 2018 Integrated Science Q5 a(i)] Ref: STEM Dic. page 780.

24. State two methods of MAGNETISATION. [BECE 2018 Integrated Science Q5 a(ii)] Ref: STEM Dic. page 782.

25. What are DERIVED QUANTITIES? (BECE 2018 Integrated Science Q6 a(i)] Ref: STEM Dic.page 365

26. Explain each of the following terms: i. SOFT WATER ii. HARD WATER [BECE 2018 Integrated Science Q6 c(i),(ii)] Ref: STEM Dic. page 605(Hard Water), page 1189 (soft water).

27. Draw a labeled diagram of the
    i. TOOTH (Human) [BECE 2017 Integrated Science Q1 a(ii)] Ref: STEM Dic. page 1298.
    ii. REFRACTION OF LIGHT [BECE 2017 Integrated Science Q1 b(ii)] Ref: STEM Dic. page 1091
    iii. FILTRATION [BECE 2019 Integrated Science Q1 c Ref: STEM Dic. page 513

28. What is DISPERSAL OF SEEDS? [BECE 2017 Integrated Science Q2 c(i)] Ref: STEM Dic. page 1150.

29. Explain the term FORWARD BIAS of p-n junction diode. [BECE 2017 Integrated Science Q2 (d)] Ref: STEM Dic. page 533.

30. What is an ACID? [BECE 2017 Integrated Science Q3 a(i)] Ref: STEM Dic. page 18 .

31. Give two differences between an ACID AND a BASE, in terms of taste and feel. [BECE 2017 Integrated Science Q3 a(ii)] Ref: STEM Dic. page 18 (Acid). and page 134 (Base).

32. Define PRESSURE. [BECE 2017 Integrated Science Q3 b(i)] Ref: STEM Dic. page 1030.

33. Explain the following terms as associated with living organisms. i. UNICELLULAR ii. MULTICELLULAR [BECE 2017 Integrated Science Q3 (i) (ii)]Ref: STEM Dic. page 1330 (unicellular) and page 853 (multicellular).

34. What is CHLOROPLAST? [BECE 2017 Integrated Science Q4 b(ii)] Ref: STEM Dic. page 246.

35. Differentiate between AEROBIC RESPIRATION and ANAEROBIC RESPIRATION. [BECE 2017 Integrated Science Q4 b(ii)] Ref: STEM Dic. page 36 (aerobic respiration), page 67 (anaerobic).

36. Differentiate between EGESTION and DIGESTION in nutrition. (BECE 2017 Integrated Science Q5) Ref: STEM Dic. page 427 (egestion) and page 381 (digestion).

37. Define the term, SOIL PROFILE. [BECE 2017 Integrated Science Q5 c(i)] Ref: STEM Dic. page 630.

38. What is an ALLOY? [BECE 2017 Integrated Science Q6 a(i] Ref: STEM Dic. page 53.

39. State two causes of CORROSION of metal. [BECE 2017 Integrated Science Q6 a(ii)] Ref: STEM Dic. page 313(corrosion).

40. What is PLANET? [BECE 2017 Integrated Science Q6 b(i)] Ref: STEM Dic. page 999.

41. Define the term, CROP ROTATION. [BECE 2017 Integrated Science Q6 d(i)] Ref: STEM Dic. page 323.

42. What is a SATELLITE? [BECE 2016 Integrated Science Q4 a (i)] Ref: STEM Dic. page 1137.

43. What is a FORCE? [BECE 2016 Integrated Science Q5 a(i)] Ref: STEM Dic. page 530.

44. What is a CHEMICAL CHANGE? [BECE 2016 Integrated Science Q5 a(ii)] Ref: STEM Dic.page 238.

45. State three differences between CHEMICAL and PHYSICAL CHANGE [BECE 2016 Integrated Science Q5 b(ii)] Ref: STEM Dic. page 238 and page 989.

46. What is TECHNOLOGY? [BECE 2015 Integrated Science Q2 a(i)] and [BECE 2012 Integrated Science Q2 d(i)] Ref: STEM Dic. page 1272.

47. Mention the junction of a TRANSISTOR. [BECE 2015 Integrated Science Q3 a(i)] Ref: STEM Dic. page 1306.

48. What is DENSITY of a body? [BECE 2015 Integrated Science Q4 d(i)] Ref: STEM Dic. page 361.

49. What is a VEGETABLE CROP? [BECE 2015 Integrated Science Q5 (a)] Ref: STEM Dic.page 1350 .

50. What is AIR POLLUTION? [BECE 2015 Integrated Science Q5 b(i)] Ref: STEM Dic.page 43.

51. What is ELECTRICAL CONDUCTOR [BECE 2016 Integrated Science Q6 b(i)] Ref: STEM Dic. page 434 (Electrical Conductor) and page 293 (Conductor).

52. What is GERMINATION OF SEED? [BECE 2014 Integrated Science Q2 c(i)] Ref: STEM Dic. page 570 (Germination).

53. State 3 differences between METAL and NON-METAL (BECE 2014 Integrated Science Q3(d) and BECE 2010, Q3 b(i) Ref: STEM Dic. page 815 (Metal), page 888 (Non-Metal).

54. What is REFRACTION of light? (BECE 2014 Integrated Science Q5 [i] Ref: STEM Dic. page 1091 (Refraction).

55. What is a FRUIT? [BECE 2013 Integrated Science Q5 c(i)] Ref: STEM Dic. page 543.

56. State one use each of the following instruments used in the study of the weather. i. RAIN GAUGE ii. HYGROMETER iii. ANEMOMETER (BECE 2013 Integrated Science) Ref: STEM Dic. page 1075 (Rain Gauge), page 642 (Hygrometer) and page 71 (Anemometer).

57. Name two types of transistors. [BECE 2013 Integrated Science, Qn 6. (c) (i)]Ref. STEM Dic. page 1306.

58. State one function of each of the following components of a typical cell: i. NUCLEUS. ii. CHLOROPLAST. iii. MITOCHONDRION (BECE 2012 Integrated Science Q2 c)] Ref: STEM Dic. page 896 (Nucleus), page 246 (Chloroplast) and page 832 (Mitochondrion).

59. Give two differences between (i) ELECTRICAL INSULATORS and (ii) ELECTRICAL CONDUCTORS [BECE 2012 Integrated Science Q3 (b) (i)] Ref: STEM Dic. page 293 (Conductor) and page 434 (Electrical Conductor), page 673 (Insulator).

60. Explain each of the following terms as used to describe change of state of matter: (i) CONDENSATION (ii) FREEZING [BECE 2012 Integrated Science Q3 c(i)] Ref: STEM Dic. page 292 (Condensation) and page 540 (Freezing).

61. What is TRANSISTOR? [BECE 2012 Integrated Science Q4 a] Ref: STEM Dic. page 1306

62. What is RESPIRATORY ORGAN? [BECE 2012 Integrated Science Q4 c(i)] Ref: STEM Dic. page 1104.

63. What are STARS? [BECE 2012 Integrated Science Q5 a(i)] Ref: STEM Dic. page 1223.

64. State two differences between (i) PLANTS and (ii) ANIMALS. [BECE 2012 Integrated Science Q5 b(i)] Ref: STEM Dic. page 999 (Plants) and page 75 (Animals).

65. State two similarities between (i) PLANTS and (ii) ANIMALS. [BECE 2012 Integrated Science Q5 b(ii)] Ref: page 999 (Plants) page 75 (Animals).

66. What are ANNUAL PLANTS? [BECE 2012 Integrated Science Q6 a(i)] Ref: STEM Dic. page 77.

67. What are PERENNIAL PLANTS? [BECE 2012 Integrated Science Q6 a(ii)] Ref: STEM Dic. page 963

68. What are RUMINANTS? [BECE 2011 Integrated Science Q2 a] Ref: STEM Dic. page 1124

69. What is FORCE? [BECE 2011 Integrated Science Q2 b(i)] Ref: STEM Dic. page 530.

70. What is a MIXTURE? [BECE 2011 Integrated Science Q3 b(i)] Ref: STEM Dic. page 834.

71. What is REFLECTION OF LIGHT? [BECE 2011 Integrated Science Q3 b(i)] Ref: STEM Dic. page 1090

72. What is a FERTILIZER? [BECE 2011 Integrated Science Q3 c(i)] Ref: STEM Dic. page 508

73. What is i. RUSTING ii. FOOD NUTRIENTS iii. SOIL EROSION? [BECE 2011 Integrated Science Q5 a(i)] Ref: STEM Dic. page 1189 (Soil Erosion). page 899 (Nutrients) and page 293 (Rust).

74. Explain each of the following terms as used in ecology. (i) ADAPTATION (ii) ENDANGERED SPECIES [BECE 2011 Integrated Science Q5 b(i)] Ref: STEM Dic. page 28 (Adaptation) and page 453 (Endangered Species).

75. What is MAGNETIC FIELD? [BECE 2011 Integrated Science Q5 d(i)] Ref: STEM Dic. page 875.

76. Define SOLVENT [BECE 2011 Integrated Science Q6 a(i)] Ref: STEM Dic. page 1196.

77. Define SOLUTE [BECE 2011 Integrated Science Q6 a[ii] Ref: STEM Dic. page 1195.

78. Explain the following terms as used in animal production. i. RATION ii. DEHORNING [BECE 2011 Integrated Science Q6 b,(i),(ii)] Ref: STEM Dic. page 1079 (Ration) and page 357 (Dehorning).

79. What is NEUTRALIZATION REACTION? [BECE 2010 Integrated Science Q2 a(i)] Ref: STEM Dic. page 875 (neutralization)].

80. Explain WEAVING as used in animal production (BECE 2010 Integrated Science Q2 (b) Ref: STEM Dic.page 1372 .

81. What is a MILKY WAY? [BECE 2010 Integrated Science Q2 c(i)] Ref: STEM Dic. page 828 (milky way).

82. What is a HABITAT? [BECE 2010 Integrated Science Q2 d(i),(ii)] Ref: STEM Dic. page 597.

83. Define PRESSURE. [BECE 2010 Integrated Science Q3 a(i)] Ref: STEM Dic. page 1030.

84. What is an ALLOY? [BECE 2010 Integrated Science Q3 b(ii)] Ref: STEM Dic. page 53

85. Define POWER. [BECE 2010 Integrated Science Q4 c(i)]Ref: STEM Dic. page 1026.

86. What is RESPIRATION [BECE 2010 Integrated Science Q5 a(i)] Ref: STEM Dic. page 1103.

87. What is FUSE? [BECE 2010 Integrated Science Q5 d(i)] Ref: STEM Dic. page 548.

88. What is the difference between i. UNICELLULAR and ii. MULTICELLULAR ORGANISM? [BECE 2010 Integrated Science Q6 a(i)] Ref: STEM Dic. page 1330 (unicellular) and page 853 (multicellular).

89. What is the difference between i. CLIMATE and WEATHER? [BECE 2010 Integrated Science Q6 b(ii)] Ref: STEM Dic. page 260 (climate) and page 1372 (weather).

90. Explain each of the following processes: i. CORROSION ii. SUBLIMATION [BECE 2010 Integrated Science Q6 d(i) d(ii)] Ref: STEM Dic. page 313 (corrosion) and page 1242 (sublimation).

91. State the laws of REFRACTION. [BECE 2009 Integrated Science Q2 c(i)] Ref: STEM Dic. page 733 (laws of refraction).

92. What is an ATOM? [BECE 2009 Integrated Science Q2 c(i)] Ref: STEM Dic. page 108.

93. What is FRICTION? [BECE 2009 Integrated Science Q3 c(i)] Ref: STEM Dic. page 542.

94. Explain HEREDITY. [BECE 2009 Integrated Science Q4 a(i)] Ref: STEM Dic. page 616.

95. What is a SIMPLE MACHINE? [BECE 2009 Integrated Science Q4 c(i)] Ref: STEM Dic. page 1173 (simple machine).

96. State the difference between an (i) OPAQUE object and a (ii) TRANSLUCENT object. [BECE 2008 Integrated Science Q2 a(i)] Ref: STEM Dic. page 287 (Opaque) and page 1308 (Translucent).

97. What is RECYCLING? [BECE 2008 Integrated Science Q2 c(i)] Ref: STEM Dic. page 1087.

98. What is an ECHO? [BECE 2008 Integrated Science Q4 c(i),(ii)] Ref: STEM Dic. page 421.

99. What is BIOTECHNOLOGY? [BECE 2008 Integrated Science Q4 c(i)] Ref: STEM Dic. page 158.

100. Define PRESSURE. [BECE 2008 Integrated Science Q5 a(i)] Ref: STEM Dic. page 1030.

101. What is a COMPOUND? [BECE 2008 Integrated Science Q5 c(i)] Ref: STEM Dic. page 287.

102. What is a LIVING CELL? [BECE 2007 Integrated Science Q2 (a)] Ref: STEM Dic. page 223 (cell).

103. Give one function of each of the following components of a living cell. i. CELL MEMBRANE ii. CHLOROPLAST iii. NUCLEUS [BECE 2007 Integrated Science Q2 (b)] Ref: STEM Dic. page 224 (cell membrane), page 246 (chloroplast) and page 896 (nucleus).

104. What is the difference between (i) EGESTION and (ii) EXCRETION [BECE 2007 Integrated Science Q4 (b)] Ref: STEM Dic. page 427 (egestion) and page 485 (excretion).

105. What is SURFACE TENSION? [BECE 2007 Integrated Science Q4 d (i)] Ref: STEM Dic. page 1254 (surface tension).

106. What is a Satellite? [BECE 2006 Integrated Science Q2 c(i)] Ref: STEM Dic. page 1137.

107. Define each of the following terms and give one example each in each case. (i) COMPOUND. (ii) ELEMENT [BECE 2006 Integrated Science Q5 (b)] Ref: STEM Dic. page 287 (compound) and page 445 (element).

108. Define (i) SELF-POLLINATION. (ii). CROSS-POLLINATION [BECE 2006 Integrated Science Q3 a(i),(ii)] Ref: STEM Dic. page 324 (cross-pollination), page 1011 (pollination), page 1154 (self-pollination).

109. What is a VECTOR of disease? [BECE 2006 Integrated Science Q4 a(i)] Ref: STEM Dic. page 1349.

110. Give two differences between an (i) ELECTRICAL INSULATOR and a (ii) CONDUCTOR. [BECE 2006 Integrated Science Q5 c(i)] Ref: STEM Dic. page 293 (Conductor) and page 434 (Electrical Conductor), page 673 (Insulator).

111. Define RESPIRATION. [BECE 2005 Integrated Science Q2 a (i)] Ref: STEM Dic. page 1103.

112. Give two differences between i. PHOTOSYNTHESIS and RESPIRATION [BECE 2005 Integrated Science Q2 a (i); 2005 Q2 a (ii)] Ref: STEM Dic. page 1103(respiration) and page 986 (photosynthesis).

113. Give two differences between a (i) VEIN and an (ii) ARTERY [BECE 2005 Integrated Science Q3 a (i)] Ref: STEM Dic. page 1351 (vein) and page 99 (artery).

114. State the LAWS OF REFLECTION [BECE 2005 Integrated Science Q c(i)] Ref: STEM Dic. page 733 (Laws of Reflection).

115. Explain VEGETATIVE REPRODUCTION [BECE 2005 Integrated Science Q4 a(i)] Ref: STEM Dic. page 1350 (Vegetative Propagation).

116. State two differences between (i) SEXUAL REPRODUCTION and (ii) VEGETATIVE REPRODUCTION. [BECE 2005 Integrated Science Q4 a(i)] Ref: STEM Dic. page 1162 (sexual reproduction) and page 1350 (Vegetative Reproduction/ Propagation).

117. Give two differences between (i) a PARASITE and ii. a VECTOR. [BECE 2005 Integrated Science Q5 a (i)] Ref: STEM Dic. page 1349 (vector) and page 947 (parasite).

118. What is DISPERSION OF LIGHT? [BECE 2004 Integrated Science Q2 a(i)] Ref: STEM Dic. page 395 (dispersion).

119. What is SOIL EROSION? [BECE 2004 Integrated Science Q3 a(i), (ii)] Ref: STEM Dic. page 1189

120. Distinguish between (i) HEAT and (ii) TEMPERATURE [BECE 2004 Integrated Science Q3 c(i)] Ref: STEM Dic. page 609 (Heat) and page 1274 (Temperature).

121. What is a HORMONE? [BECE 2004 Integrated Science Q4 a(i)] Ref: STEM Dic. page 630.

122. Distinguish between a RHIZOME and a STEM TUBER. [BECE 2004 Integrated Science Q5 a(i)] Ref: STEM Dic. page 1227 (stem tuber) and page 1109 (rhizome).

123. Define DENSITY OF SUBSTANCE. [BECE 2004 Integrated Science Q5 c(i)] Ref: STEM Dic. page 361 (density).

124. What is an ELECTRICAL INSULATOR? [BECE 2003 Integrated Science Q3 c(i); 2002 Q4] Ref: STEM Dic. page 673 (insulator).

125. What is DEFICIENCY DISEASE? [BECE 2003 Integrated Science Q4 a(i)] Ref: STEM Dic. page 354.

126. Explain the difference between (i) PHYSICAL CHANGE and (ii) CHEMICAL CHANGE.[BECE 2002 A Integrated Science Q1 d(i)] Ref: STEM Dic. page 989 (physical change) and page 238 (chemical change).

127. What is a MACHINE? [BECE 2002 A Integrated Science Q1 d(i)] Ref: STEM Dic. page 775.

128. Define each of the following terms: (i) COMPOUND (ii) ELEMENT [BECE 2002 A Integrated Science Q2 b(i), (ii)] Ref: STEM Dic. page 287 (compound) and page 445 (element).

129. Define WORK. [BECE 2002 A Integrated Science Q3 c(i),(ii)] Ref: STEM Dic. page 1384.

130. Distinguish between HEAT and TEMPERATURE. [BECE 2002 A Integrated Science Q3 d(i),(ii)] Ref: STEM Dic. page 609 (Heat) and page 1274 (Temperature).

131. Define VALENCY. [BECE 2002 B Integrated Science Q1 b(i)] Ref: STEM Dic. page 1344.

132. What is a SOLUTION? [BECE 2002 B Integrated Science Q2 (a)] Ref: STEM Dic. page 1195.

133. Name the parts of a FLOWER. [BECE 2002 B Integrated Science Q3 (a) Ref: STEM Dic. page 287.

134. What is an EMBRYO? [BECE 2002 Integrated Science Q4 a(i)] Ref: STEM Dic. page 449.

135. What is SURFACE TENSION? [BECE 2001 Integrated Science Q1 a(i)]Ref: STEM Dic. page 1254.

136. What is CONSTIPATION? [BECE 2001 Integrated Science Q1 b(i]),(ii)] Ref: STEM Dic. page 300.

137. What is an ANION? [BECE 2001 Integrated Science Q2 b(i)] Ref: STEM Dic. page 76.

138. What is DIGESTION? [BECE 2001 Integrated Science Q4 a (i)] Ref: STEM Dic. page 381.

139. What is meant by ECLIPSE OF THE SUN? [BECE 2001 Integrated Science Q4 b(ii)] Ref: STEM Dic. page 421.

140. With the aid of a labeled diagram, distinguish between TOTAL ECLIPSE and PARTIAL ECLIPSE. [BECE 2001 Integrated Science Q4 b(ii)] Ref: STEM Dic.page 421.

141. What is an ELECTROSTATIC FORCE? [BECE 2001 Integrated Science Q2 (a) Ref: STEM Dic. page 444.

142. What is FERTILIZATION? (BECE 2001 Integrated Science Q3) Ref: STEM Dic. page 508.

143. Define an/a (i) ALLOY (ii) DIFFUSION (iii) COLLOID [BECE 2001 Integrated Science Q4 (a)] Ref: STEM Dic. page 53 (alloy), page 380 (diffusion) and page 275 (colloid).

144. What is POLLUTION? [BECE 2001 Integrated Science Q4 c(i)] Ref: STEM Dic. page 1012.

INTEGRATED SCIENCE QUESTIONS FROM THE WEST AFRICAN SENIOR SCHOOL CERTIFICATE EXAMINATIONS (W.A.S.S.C.E.) AND STEM DICTIONARY REFERENCE PAGES

1. Define each of the following terms. (i) COMPOUND (WASSCE 2019 - JULY INT. SCIENCE Q1a[i] Ref: STEM Dic. Page 267). (ii) MIXTURE (WASSCE 2019 - JULY INT.SCIENCE Q1a[i] Ref: STEM Dic.page 287 (compound) page 834 (mixture))
2. State two differences between an ELEMENT and a COMPOUND. (WASSCE 2019 - JULY INT.SCIENCE Q1a [ii] Ref: STEM Dic. page 445 (element), page 287 (compound).
3. Explain each of the following terms: (i) GREENHOUSE EFFECT (WASSCE 2019 - JULY INT.SCIENCE Q1b[i] Ref: STEM Dic. page 589)
4. (ii) CLIMATE CHANGE (WASSCE 2019 - JULY INT.SCIENCE Q1b [ii] Ref: STEM Dic. page 260).
5. What are DECOMPOSERS? (WASSCE 2019 - JULY INT. SCIENCE Q2 c[i] Ref: STEM Dic. page 352).
6. What is POLLUTION? (WASSCE 2019 - JULY INT.SCIENCE Q3 c[i] Ref: STEM Dic. page 1012).
7. List three major ATMOSPHERIC POLLUTANTS. (WASSCE 2019 - JULY INT.SCIENCE Q3 c[ii] Ref: STEM Dic. page 1012 (pollution and pollutants).
8. What are TIDAL WAVES? (WASSCE 2019 - JULY INT.SCIENCE Q4 b[i] Ref: STEM Dic. page 1293).
9. Define MECHANICAL ADVANTAGE of simple machine (WASSCE 2019 - JULY INT.SCIENCE Q6 b[i] Ref: STEM Dic. page 804).
10. Give two examples of SECOND CLASS LEVERS (WASSCE 2019 - JULY INT.SCIENCE Q6 b[ii] Ref: STEM Dic. page 300).
11. a. State three differences between HYPOGEAL GERMINATION and EPIGEAL GERMINATION (WASSCE 2019 - JULY INT.SCIENCE Q6d [i]. b. Distinguish between EPIGEAL GERMINATION and HYPOGEAL GERMINATION, giving one example in each case (WASSCE 2001 – JULY INT.SCIENCE Q3 d[i]Ref: STEM Dic. Pages 436 and 608) Ref: STEM Dic. page 646 (Hypogeal Germination), page 300 (Epigeal Germination).
12. State the significance of each of the following stages in water treatment. (i) SCREENING (WASSCE 2019 - JULY INT. SCIENCE Q5 d[ii]Ref: STEM Dic. page 1144) (ii) FILTRATION (WASSCE 2019 - JULY INT.SCIENCE Q5d[i] Ref: STEM Dic. page 513) (iii) CHLORINATION (WASSCE 2019 - JULY INT. SCIENCE Q5 d[ii] Ref: STEM Dic. page 245).
13. State two functions of the PLACENTA in mammals (WASSCE 2019 - JULY INT.SCIENCE Q5 a[i] Ref: STEM Dic. page 997).
14. What is meant by the term, MODERN TECHNOLOGY? (WASSCE 2018 - JULY INT.SCIENCE Q1a[i Ref: STEM Dic. page 1272 (Technology).
15. Explain the term CONVECTION as used in heat energy. (WASSCE 2018 - JULY INT.SCIENCE Q1b[i] Ref: STEM Dic. page 306).
16. Differentiate between SOIL EROSION and SOIL DEPLETION. (WASSCE 2018 - JULY INT.SCIENCE Q1c[i] Ref: STEM Dic. page 1189(soil erosion) and page 1189 (soil depletion)).
17. Explain the term SUBLIMATION. (WASSCE 2018 - JULY INT. SCIENCE Q2 b[i] Ref: STEM Dic. page 1242).
18. 17 Name three substances that can SUBLIME. (WASSCE 2018 – JULY INT.SCIENCE Q2 b[ii] Ref: STEM Dic. page 1242).
19. Distinguish between GREEN MANURE and FARM MANURE (WASSCE 2018 - JULY INT.SCIENCE Q2 d[i] Ref: STEM Dic. page 588 (Green Manure), page 791(Manure – Farm Manure).
20. What is a REFLEX ACTION (WASSCE 2018 - JULY INT. SCIENCE Q3 b[i] Ref: STEM Dic. page 1090).
21. List two groups of HYDROCARBONS. (WASSCE 2018 – JULY INT.SCIENCE Q3 c[i]. Ref: STEM Dic. page 637).
22. Distinguish between EXCRETION and EGESTION. (WASSCE 2018 - JULY INT.SCIENCE Q3 b[ii] Ref: STEM Dic. page 427 (egestion) and page 485 (excretion)
23. Define the term PRESSURE. (WASSCE 2018 - JULY INT. SCIENCE Q4a[i] Ref: STEM Dic. page 1030).
24. Name four ENDOCRINE GLANDS found in humans. (WASSCE 2018 - JULY INT.SCIENCE Q6 b Ref: STEM Dic. page 453 and page 454 (Endocrine gland and endocrine system)
25. What is SHADOW? (WASSCE 2018 - JULY INT.SCIENCE Q6c[i] Ref: STEM Dic. page 1163).
26. a. Explain the term ARTIFICIAL INSEMINATION as used in animal production (WASSCE 2018 - JULY INT.SCIENCE Q6 d[i] b. What is ARTIFICIAL INSEMINATION? (WASSCE 2001 - JULY INT.SCIENCE Q4 d[i]. Ref: STEM Dic. page 99 (Artificial Insemination) page 671 (Insemination).
27. State three differences between COMPOUND and MIXTURES (WASSCE 2018 - NOV INT.SCIENCE Q1b[i] Ref: STEM Dic. page 287 (compound) page 834 (mixture)
28. Define RESISTANCE of a Conductor. (WASSCE 2018 - NOV INT.SCIENCE Q1c[i] Ref: STEM Dic. page 1101).
29. Explain the term OESTRUS (WASSCE 2018 - NOV INT. SCIENCE Q1 d[i] Ref: STEM Dic. page 906 (oestrous cycle).
30. List three characteristics of IONIC COMPOUNDS (WASSCE 2018 - NOV INT.SCIENCE Q2 a[i] Ref: STEM Dic.page 689).
31. What is INDUSTRIAL WASTE (WASSCE 2018 - NOV INT. SCIENCE Q2 b[i] Ref: STEM Dic. page 664).
32. Give two uses of TRANSISTORS: (WASSCE 2018 - NOV INT. SCIENCE Q2 a[ii]Ref: STEM Dic. page 1306).
33. State the LAW OF FLOTATION. (WASSCE 2018 - NOV INT. SCIENCE Q4 a[i] Ref: STEM Dic. page 732).
34. What is GLOMERULAR FILTRATE? (WASSCE 2018 - NOV INT. SCIENCE Q5 a[i] Ref: STEM Dic. page 576).
35. What is a MAGNETIC FIELD? (WASSCE 2018 - NOV INT. SCIENCE Q5c[i] Ref: STEM Dic. page 875).
36. Explain the term FLUSHING as used in animal production. (WASSCE 2018 - NOV INT.SCIENCE Q5 d[i] Ref: STEM Dic. page 525).
37. What is RUSTING? (WASSCE 2018 - NOV INT.SCIENCE Q6a[i] Ref: STEM Dic.page 1125).
38. Explain the term SICKLE CELL ANAEMIA [WASSCE 2018 - NOV INT.SCIENCE Q6 Ref: STEM Dic. page 1167 (Sickle cell & sickle cell disease)].
39. What is meant by the term SEED DISPERSAL (WASSCE 2017 - JUNE INT.SCIENCE Q1a[i] Ref: STEM Dic. page 1150).

40. What is POTENTIAL ENERGY? (WASSCE 2017 - JUNE INT. SCIENCE Q1b [i] Ref: STEM Dic. page 300).

41. Explain the term RATION as used in animal production. (WASSCE 2017 - JUNE INT.SCIENCE Q2 a[i] Ref: STEM Dic. page 1079).

42. Explain each of the following terms: (i) COMMUNITY (WASSCE 2017 - JUNE INT.SCIENCE Q2 d[i] Ref: STEM Dic. page 281): (ii) POPULATION (WASSCE 2017 - JUNE INT.SCIENCE Q2 d[ii] Ref: STEM Dic. page 1018).

43. What is an ENZYME? (WASSCE 2017 - JUNE INT.SCIENCE Q3 b[i] Ref: STEM Dic. page 463).

44. a. Define the term HYDROCARBON (WASSCE 2017 - JUNE INT.SCIENCE Q3 d[i]. b. What is a HYDROCARBON? (WASSCE 2001 - JULY INT.SCIENCE Ref: STEM Dic. page 637).

45. What is meant by the POLE OF A MAGNET? (WASSCE 2017 - JUNE INT.SCIENCE Q4 a[i] Ref: STEM Dic. page 781 (magnetic pole).

46. Distinguish between:

47. (i) An ATOM and an ION (WASSCE 2017 - JUNE INT.SCIENCE Q4 d[i] Ref: STEM Dic. page 108 and page 687). (ii) NEUTRALIZATION AND ESTERIFICATION. (WASSCE 2017 - JUNE INT.SCIENCE Q4 b[ii] Ref: STEM Dic. page 875 (Neutralisation), page 475 (Ester).

48. Define each of the following terms as associated with reproduction in humans
(i) COPULATION (WASSCE 2017 - JUNE INT.SCIENCE Q5 a[i] Ref: STEM Dic. page 310). (ii)EJACULATION (WASSCE 2017 - JUNE INT.SCIENCE Q5 a[ii] Ref: STEM Dic. page 430). (iii) OVULATION (WASSCE 2017 - JUNE INT.SCIENCE Q5 a[iii] .Ref: STEM Dic. page 929). (iv) MENSTRUATION (WASSCE 2017 - JUNE INT.SCIENCE Q5 a[i]
Ref: STEM Dic. page 810).

49. What is SHORT-SIGHTEDNESS (WASSCE 2017 - JUNE INT. SCIENCE Q5 c[i]. Ref: STEM Dic. page 1166).

50. List three emissions of RADIOACTIVITY (WASSCE 2017 - JUNE INT.SCIENCE Q b[i] Ref: STEM Dic. page 1072).

51. What is the WATER CYCLE (WASSCE 2017 - NOV INT.SCIENCE Q1b[i] Ref: STEM Dic. page 1367.)

52. Explain THINNING as used in maize production. (WASSCE 2017 - NOV INT.SCIENCE Q2 a[ii] Ref: STEM Dic. page 1288).

53. Distinguish between PERMANENT MAGNETS and TEMPORARY MAGNETS (WASSCE 2017 - NOV INT.SCIENCE Q2 d[i] Ref: STEM Dic. page 779 (Magnet). page 1275 (Temporary Magnetism), page 968 (Permanent Magnet).

54. State one function of each of the following mammalian teeth (i) INCISOR (WASSCE 2017 - NOV INT.SCIENCE Q5 b[i] Ref: STEM Dic. page 657).

55. (ii)CANINE (WASSCE 2017 - NOV INT.SCIENCE Q5 b[i] Ref: STEM Dic. page 201).

56. (iii) MOLAR (WASSCE 2017 - NOV INT.SCIENCE Q5 b[i] Ref: STEM Dic. page 837).

57. Define MOMENTUM of a body (WASSCE 2017 - NOV INT. SCIENCE Q6 a[i] Ref: STEM Dic. page 841).

58. State one use of (i) MICROWAVE (WASSCE 2016 – JUNE INT. SCIENCE Q1b [ii] Ref: STEM Dic. page 827). (ii) X-RAYS (WASSCE 2016 - JUNE INT.SCIENCE Q1b [ii] Ref: STEM Dic. page 1387).

59. What is IONIC BOND? (WASSCE 2016 - JUNE INT. SCIENCE Q1c[i] Ref: STEM Dic. page 689)

60. What is CENTRE OF GRAVITY of a body? (WASSCE 2016 - JUNE INT.SCIENCE Q2 b[i] Ref: STEM Dic. page 227)

61. What is MOMENTUM (WASSCE 2016 - JUNE INT.SCIENCE Q3 a[i] Ref: STEM Dic. page 841).

62. Differentiate between VOLUNTARY ACTION and INVOLUNTARY ACTION (WASSCE 2016 - JUNE INT.SCIENCE Q4 b[i] Ref: STEM Dic. page 1363 (Voluntary Action) , page 686 (Involuntary Action).

63. State one function of the following parts of the human eye. (i) PUPIL (WASSCE 2016 - JUNE INT.SCIENCE Q4 b[ii] Ref: STEM Dic. page 1052. ii.LENS (WASSCE 2016 - JUNE INT. SCIENCE Q4 b [ii] Ref: STEM Dic. Pages 697. (ii) RENTINA (WASSCE 2016 - JUNE INT.SCIENCE Q4 b[ii Ref: STEM Dic. page 1106

64. State the general molecular formula for each of the following families of organic compounds (i) ALKANES (WASSCE 2016 - JUNE INT.SCIENCE Q4 c[i] Ref: STEM Dic. page 50).(ii) ALKENES (WASSCE 2016 - JUNE INT.SCIENCE Q4 c[ii] Ref: STEM Dic. page 50). (iii) ALKYNES (WASSCE 2016 - JUNE INT.SCIENCE Q4 c[iii] Ref: STEM Dic. page 51). (iv) ALKANOLS (WASSCE 2016 - JUNE INT.SCIENCE Q4 c[iv] Ref: STEM Dic. page 50).

65. Explain the term SOIL CONSERVATION as used in soil management. (WASSCE 2016 - JUNE INT.SCIENCE Q5 a[i] Ref: STEM Dic. page 1189).

66. What is a MUSICAL NOTE (WASSCE 2016 - JUNE INT. SCIENCE Q5 b[i] Ref: STEM Dic. page 892 (refer to NOTE).

67. What is WEATHERING ROCKS? (WASSCE 2016 - JUNE INT. SCIENCE Q6 b[i] Ref: STEM Dic. page 1372).

68. What is a FERTILE SOIL? (WASSCE 2016 - NOV INT.SCIENCE Q1a[i] Ref: STEM Dic. page 1190 (soil nutrition)

69. State one function each of the following structures in the human body. (i) ALVEOLI (WASSCE 2016 - NOV INT.SCIENCE Q1 c[ii] Ref: STEM Dic. page 58). (ii) HEPATIC PORTAL VEIN (WASSCE 2016 - NOV INT.SCIENCE Q1 c[ii] Ref: STEM Dic. page 615). (iii) GALL BLADDER (WASSCE 2016 - NOV INT.SCIENCE Q1 c[ii] Ref: STEM Dic. page 553).

70. What is ESTERIFICATION? (WASSCE 2016 - NOV INT. SCIENCE Q2 a[i] Ref: STEM Dic. page 475).

71. Define DISPLACEMENT. (WASSCE 2016 - NOV INT.SCIENCE Q4 d[i] Ref: STEM Dic. page 395).

72. What are RADIOISOTOPES? (WASSCE 2016 - NOV INT. SCIENCE Q5 c[i] Ref: STEM Dic. page 1073).

73. What is an INORGANIC FERTILIZER? (WASSCE 2015 - JUNE INT.SCIENCE Q1 a[i] Ref: STEM Dic.page 670.

74. List three ORGANELLES found in an animal cell (WASSCE 2015 - JUNE INT.SCIENCE Q2 b[i] Ref: STEM Dic.page 920)

75. What are RADIOISOTOPE (WASSCE 2015 - JUNE INT. SCIENCE Q2 d[i] Ref: STEM Dic. page 1073).

76. Explain how each of the following respiratory structures in humans is adapted to its function. (i) NASAL CHAMBER (WASSCE 2015 - JUNE INT.SCIENCE Q3 b[i] Ref: STEM Dic. page 863). (ii) DIAPHRAGM (WASSCE 2015 - JUNE INT. SCIENCE Q3 b[ii] Ref: STEM Dic. page 374). (iii) ALVEOLI (WASSCE 2015 - JUNE INT.SCIENCE Q3 b[iii] Ref: STEM Dic. page 58).

77. Define MECHANICAL ADVANTAGE as applied to simple machines (WASSCE 2015 - JUNE INT.SCIENCE Q3 d[i] Ref: STEM Dic. page 804).

78. What is an UNSATURED HYDROCARBON (WASSCE 2015 - JUNE INT.SCIENCE Q4 b[i] Ref: STEM Dic. page 1336).

79. Give two examples of UNSATURED HYDROCARBONS (WASSCE 2015 - JUNE INT.SCIENCE Q4 b[i] Ref: STEM Dic. page 1336).

80. Give two gases which contribute to GREENHOUSE EFFECT (WASSCE 2015 - JUNE INT.SCIENCE Q5 b[i] Ref: STEM Dic. page 589).

81. What is HEAT? (WASSCE 2015 - JUNE INT.SCIENCE Q6 a[i] Ref: STEM Dic. page 609).

82. What is a LIVING CELL? (WASSCE 2015 - NOV INT.SCIENCE Q1 a[i] Ref: STEM Dic. page 223 (Cell).

83. What is meant by OXIDATION NUMBER of an atom? (WASSCE 2015 - NOV INT.SCIENCE Q1 b[i] Ref: STEM Dic. page 930).

84. What is SOIL EROSION (WASSCE 2015 - NOV INT.SCIENCE Q1 c[i] Ref: STEM Dic. page 1189).

85. Define the term ALLOY (WASSCE 2015 - NOV INT.SCIENCE Q2 d[i] Ref: STEM Dic. page 53).

86. Explain the term HOMEOSTASIS (WASSCE 2015 - NOV INT. SCIENCE Q3 c[i] Ref: STEM Dic. page 627).

87. Distinguish between STRONG ACID and WEAK ACID (WASSCE 2015 - NOV INT.SCIENCE Q4 a[ii] Ref: STEM Dic. page 1238 andpage 1371)

88. Explain the term TILLAGE as used in crop production (WASSCE 2015 - NOV INT.SCIENCE Q4 b[i] Ref: STEM Dic. page 1294).

89. What is a PATHOGEN (WASSCE 2015 - NOV INT.SCIENCE Q4 d[i] Ref: STEM Dic. page 955).

90. What is a FUNCTIONAL GROUP (WASSCE 2015 - NOV INT. SCIENCE Q5 c[i] 323; (WASSCE 2001 - NOV INT.SCIENCE Q2 a[i] Ref: STEM Dic. page 546).

91. Define CASTRATION as used in animal production (WASSCE 2015 - NOV INT.SCIENCE Q6 d[i] Ref: STEM Dic. page 216).

92. State two functions each of the following structures during pregnancy in humans. (i) PLACENTA (WASSCE 2014 - JUNE INT.SCIENCE Q1 a[i] Ref: STEM Dic. page 997). (ii) AMNIOTIC FLUID (WASSCE 2014 - JUNE INT.SCIENCE Q1 a[ii] Ref: STEM Dic. page 63).

93. Explain each of the following terms used in poultry production (i) CANDLING (WASSCE 2014 - JUNE INT.SCIENCE Q1 b[i] Ref: STEM Dic. page 201). (ii) CULLING [WASSCE 2014 - JUNE INT.SCIENCE Q1 b[ii], (WASSCE 2007 - JUNE INT. SCIENCE Q3 b[i] Ref: STEM Dic. page 329 (Cull)]. (iii) a. BROODING (WASSCE 2014 - JUNE INT.SCIENCE Q1 b[iii], (WASSCE 2007 - JUNE INT.SCIENCE Q3 b[ii], page 181.

94. B. What is BROODING in poultry production (WASSCE 2002 - NOV INT.SCIENCE Q1 d[i] Ref: STEM Dic. page 181 (Brood).

95. Define TEMPERATURE. (WASSCE 2014 - JUNE INT.SCIENCE Q1 d[i] Ref: STEM Dic. page 1274).

96. State OHM'S LAW (WASSCE 2014 - JUNE INT.SCIENCE Q2 d[i] Ref: STEM Dic. page 907).

97. What is REFRACTION OF LIGHT? (WASSCE 2014 - JUNE INT. SCIENCE Q3 d[i] Ref: STEM Dic. page 1091 (Refraction)).

98. Define VELOCITY. (WASSCE 2014 - JUNE INT.SCIENCE Q5 a[i] Ref: STEM Dic. page 1351).

99. Explain the term, BIOTECHNOLOGY. (WASSCE 2014 - JUNE INT.SCIENCE Q6 c[i] Ref: STEM Dic. page 158).

100. Explain two applications of BIOTECHNOLOGY in health and medicine . (WASSCE 2014 - JUNE INT.SCIENCE Q6 c[ii] Ref: STEM Dic. page 158).

101. Explain the term, SHORT SIGHTEDNESS. (WASSCE 2014 - NOV INT.SCIENCE Q1 c[i] Ref: STEM Dic. page 1166).

102. What is IMPLANTATION in humans? (WASSCE 2014 - NOV INT.SCIENCE Q2 a[i] Ref: STEM Dic. page 655).

103. Explain the term, VECTOR as used in animal production. (WASSCE 2014 - JUNE INT.SCIENCE Q2 b[i] Ref: STEM Dic. page 1349).

104. What are PLASTICS? (WASSCE 2014 - NOV INT.SCIENCE Q2 c[i] Ref: STEM Dic. page 1001).

105. Define RELATIVE DENSITY. (WASSCE 2014 - NOV INT. SCIENCE Q3 a[i]. Ref: STEM Dic. page 1095).

106. Explain the term NUTRITION. (WASSCE 2014 - NOV INT. SCIENCE Q4 b[i] Ref: STEM Dic. page 899).

107. What is a BASE? (WASSCE 2014 - NOV INT.SCIENCE. Q4 c[i] Ref: STEM Dic. page 134).

108. Explain the term DIFFUSION (WASSCE 2014 - NOV INT. SCIENCE Q5 a[i] Ref: STEM Dic. page 380).

109. What are CHOROMOSOMES (WASSCE 2014 - NOV INT. SCIENCE Q6 b[i] Ref: STEM Dic. page 249).

110. Define FORCE. (WASSCE 2014 - NOV INT.SCIENCE Q6 d[i] Ref: STEM Dic. page 530).

111. Explain the term TISSUE as applied to living organisms (WASSCE 2013 - JUNE INT.SCIENCE Q2 a[i] Ref: STEM Dic. page 1295).

112. State the LAW OF CONSERVATION of energy. (WASSCE 2013 - JUNE INT.SCIENCE Q2 b[i] Ref: STEM Dic. page 732).

113. What is a CHEMICAL COMPOUND? (WASSCE 2013 – JUNE INT. SCIENCE Q2 c[i] Ref: STEM Dic. page 238 (Chemical Compound). page 287 (Compound).

114. Explain the term ARTIFICIAL INSEMINATION as used in animal production (WASSCE 2013 - JUNE INT.SCIENCE Q3 c[i] STEM Dic. page 99 (Artificial Insemination) page 671 (Insemination).

115. Explain the following ecological terms (i) COMMUNITY (WASSCE 2013 - JUNE INT.SCIENCE Q4 b[i] Ref: STEM Dic. page 281) (ii) POPULATION (WASSCE 2013 - JUNE INT.SCIENCE Q4 b[ii] Ref: STEM Dic. page 1018) (iii) HABITAT (WASSCE 2013 - JUNE INT.SCIENCE Q4 b[iii] Ref: STEM Dic. page 597).

116. What is DISPERSION OF LIGHT (WASSCE 2013 - JUNE INT. SCIENCE Q4 c[i] Ref: STEM Dic. page 395 (Dispersion).

117. What is CROP PEST (WASSCE 2013 - JUNE INT.SCIENCE Q5 d[i] Ref: STEM Dic. page 971(Pest).

118. Define FORCE (WASSCE 2013 - JUNE INT.SCIENCE Q6 c[i] Ref: STEM Dic. page 530).

119. Define ENERGY. (WASSCE 2013 - NOV INT. SCIENCE Q1 a[i], Ref: STEM Dic page 457).

120. Explain the term SPECIFIC HEAT OF FUSSION (WASSCE 2013 - NOV INT. SC. Q2 d[i] Ref: STEM Dic. page 1204 (Specific Latent Heat of Fusion).

121. What is PARASITE? (WASSCE 2013 - NOV INT.SCIENCE Q3 c[i] Ref: STEM Dic. page 947).

122. What are FOSSILS. (WASSCE 2013 - NOV INT.SCIENCE Q4 b[i] Ref: STEM Dic. page 534).

123. State NEWTON'S SECOND LAW of MOTION (WASSCE 2013 - NOV INT.SCIENCE Q5 c[i] Ref: STEM Dic. page 877).

124. What is a FUNCTIONAL GROUP (WASSCE 2013 - NOV INT. SCIENCE Q5 c[i] Ref: STEM Dic. page 546).

125. What is a SIMPLE MACHINE (WASSCE 2013 - JUNE INT. SCIENCE Q6 a[i]; (WASSCE 2008 - JULY INT.SCIENCE Q2 b[i] Ref: STEM Dic. page 1173 (simple machine) and page 775 (machine).

126. Explain each of the following terms as associated with simple machine (i) LOAD (WASSCE 2013 - JUNE INT.SCIENCE Q6 a[ii] Ref: STEM Dic. page 760).

127. (ii) EFFICIENCY (WASSCE 2013 - JUNE INT.SCIENCE Q6 a[ii] Ref: STEM Dic. page 427).

128. Explain each of the following terms as used in animal production (i) CALVING (WASSCE 2013 - JUNE INT.SCIENCE Q6 c[i] Ref: STEM Dic. page 199 (Calving Cycle).

129. (ii) GESTATION PERIOD (WASSCE 2013 - JUNE INT.SCIENCE Q6 c[ii] Ref: STEM Dic. page 570).

130. iii) FLUSHING (WASSCE 2013 - JUNE INT.SCIENCE Q6 c[iii] Ref: STEM Dic. page 525).

131. What are PLASTICS? (WASSCE 2012 - JUNE INT.SCIENCE Q1 a[i] Ref: STEM Dic. page 1001).

132. Explain the term DE-HORNING as used in animal production. (WASSCE 2012 - JUNE INT.SCIENCE Q1 c[i] Ref: STEM Dic. page 357).

133. State two differences between BOILING and EVAPORATION (WASSCE 2012 - JUNE INT.SCIENCE Q2 a[i]Ref: STEM Dic. page 167 & page 482).

134. What is a FUNCTIONAL GROUP? (WASSCE 2012 - JUNE INT. SCIENCE Q2 c[i] Ref: STEM Dic. page 546).

135. What is RELATIVE DENSITY? (WASSCE 2012 - JUNE INT. SCIENCE Q3 d[i] Ref: STEM Dic. page 1095).

136. What is a TRANSFORMER? (WASSCE 2012 - JUNE INT. SCIENCE Q4 c[i] Ref: STEM Dic. page 1305).

137. What is meant by MOLAR MASS? (WASSCE 2012 - JUNE INT. SCIENCE Q5 c[i]Ref: STEM Dic. page 838).

138. What is a THERMOSTAT? (WASSCE 2012 - JUNE INT.SCIENCE Q6 a[i] (WASSCE 2002 - JULY INT.SCIENCE Q5 b[i]. Ref: STEM Dic. page 1287)

139. What are LAYERS in poultry production (WASSCE 2012 - NOV INT. SCIENCE Q1 d[i] Ref: STEM Dic. page 733).

140. What is DISCONTINUOUS VARIATION (WASSCE 2012 - NOV INT.SCIENCE Q2 a[i] Ref: STEM Dic. page 393(Discontinuous Variation) and page 1347 (Variation).

141. Define ELECTRIC CURRENT (WASSCE 2012 - NOV INT. SCIENCE Q2 b[i] Ref: STEM Dic. page 331 (Current) and page 431 (Electric Current).

142. What is COVALENT BOND? (WASSCE 2012 - NOV INT. SCIENCE Q2 d[i] Ref: STEM Dic. page 318).

143. List two examples of chemicals compounds which contain COVALENT BOND. (WASSCE 2012 - NOV INT.SCIENCE Q2 d[ii] Ref: STEM Dic. page 318).

144. Name two cations which cause HARDNESS IN WATER (WASSCE 2012 - NOV INT.SCIENCE Q3 d[i] Ref: STEM Dic. page 1367 (Water Hardness) and page 605 (Hard water).

145. Define the following terms (i) ISOTOPES (WASSCE 2012 - NOV INT.SCIENCE Q4 c[i] Ref: STEM Dic. page 293).

146. (ii) MASS NUMBER (WASSCE 2012 - NOV INT.SCIENCE Q4 c[ii] Ref: STEM Dic. page 796).

147. What is CROSS-POLLINATION as used in pant reproduction? (WASSCE 2012 - NOV INT.SCIENCE Q4 c[ii] Ref: STEM Dic. page 324(Cross Pollination) and page 1011 (Pollination).

148. Distinguish between a NORMAL SALT and an ACID SALT (WASSCE 2012 - NOV INT.SCIENCE Q5 a[i] Ref: STEM Dic. page 891 (Normal Salt) and page 19 (Acid Salt).

149. What is CEREAL as used in crop production. (WASSCE 2012 - NOV INT.SCIENCE Q5 b[i] Ref: STEM Dic. page 229).

150. What is REFRACTION OF LIGHT? (WASSCE 2012 - NOV INT. SCIENCE Q5 c[i] Ref: STEM Dic. page 1091 (Refraction).

151. What is VEGETATIVE REPRODUCTION? (WASSCE 2012 – NOV INT.SCIENCE Q5 d[i] Ref: STEM Dic. page 1350 (Vegetative Propagation/Vegetation).

152. What is VACCINATION ?(WASSCE 2012 - NOV INT.SCIENCE Q6 a[i] Ref: STEM Dic. page 1343 (Vaccine).

153. a. Define KINETIC ENERGY, (WASSCE 2012 - NOV INT. SCIENCE Q1 d[i] b. What is KINETIC ENERGY? (WASSCE 2001 – JULY INT.SCIENCE Q4 a[i] Ref: STEM Dic. page 712).

154. What is VENTILATION? (WASSCE 2012 - NOV INT.SCIENCE Q6 d[i] Ref: STEM Dic. page 1352).

155. What is a FERTILIZER? (WASSCE 2011 - JUNE INT.SCIENCE Q1 b[i] Ref: STEM Dic. page 293).

156. a. What is EXCRETION? (WASSCE 2011 - JUNE INT.SCIENCE Q1 c[i] b. What is meant by EXCRETION? (WASSCE 2001 - NOV INT. SCIENCE Q4 a[i]Ref: Ref: STEM Dic. page 485).

157. What is meant by MASS of a body? (WASSCE 2011 - JUNE INT.SCIENCE Q1 d[i] Ref: STEM Dic. page 795 (Mass).

158. What is FOOD PRESERVATION? (WASSCE 2011 - JUNE INT.SCIENCE Q2 a[i] Ref: STEM Dic. page 529 (Food Preservation).

159. What is a PATHOGEN? (WASSCE 2011 - JUNE INT.SCIENCE Q2 b[i] Ref: STEM Dic. page 955).

160. What are X-RAYS? (WASSCE 2011 - JUNE INT.SCIENCE Q2 c[i] Ref: STEM Dic. page 1387).

161. What is DIRECT CURRENT? (WASSCE 2011 - JUNE INT. SCIENCE Q3 b[i] Ref: STEM Dic. page 389).

162. Explain FREQUENCY as applied to sound waves. (WASSCE 2011 - JUNE INT.SCIENCE Q4 a[i] Ref: STEM Dic. page 541).

163. Explain the term POPULATION DENSITY. (WASSCE 2011 - JUNE INT.SCIENCE Q4 c[i] Ref: STEM Dic. page 1019).

164. Distinguish between NORMAL SALT and ACID SALT (WASSCE 2011 - JUNE INT.SCIENCE Q5 a[i]. Ref: STEM Dic. page 891 (Normal Salt) and page 19(Acid Salt)

165. What is CROP ROTATION? (WASSCE 2011 - JUNE INT. SCIENCE Q6 a[i]. Ref: STEM Dic. page 323).

166. What is a TRANSISTOR? (WASSCE 2011 - JUNE INT.SCIENCE Q6 c[i] Ref: STEM Dic. page 1306).

167. Explain the term, OXIDATION NUMBER of an atom (WASSCE 2011 - NOV INT.SCIENCE Q1 c[i] Ref: STEM Dic. page 930).

168. Explain the term, POTENTIAL ENERGY. (WASSCE 2011 - NOV INT.SCIENCE Q1 d[i] Ref: STEM Dic. page 300).

169. What is an ALLOY? (WASSCE 2011 - NOV INT.SCIENCE Q2 a[i] Ref: STEM Dic. page 53).

170. What is OVULATION? (WASSCE 2011 - NOV INT.SCIENCE Q3 b[i] Ref: STEM Dic. page 929).

171. Explain the term SOLUTION? (WASSCE 2011 - NOV INT. SCIENCE Q2 a[i]. Ref: STEM Dic. page 1195).

172. State one function each of the following cells in blood. (i) RED BLOOD CELLS (WASSCE 2011 - NOV INT.SCIENCE Q4 a[i] Ref: STEM Dic. page 1087). (ii) WHITE BLOOD CELLS (WASSCE 2011 - NOV INT.SCIENCE Q4 a[i] Ref: STEM Dic. page 1377).

173. Explain the term CULLING as applied to poultry management practice. (WASSCE 2011 - NOV INT.SCIENCE Q4 b[i] Ref: STEM Dic. page 329)

174. Explain the term RADIOACTIVITY. (WASSCE 2011 – NOV INT.SCIENCE Q4 c[i] Ref: STEM Dic. page 1072).

175. What is an ELECTROMAGNETIC WAVE? (WASSCE 2011 - NOV INT.SCIENCE Q4 d[i] Ref: STEM Dic. page 438).

176. Explain DATABASE as applied to computers. (WASSCE 2011 - NOV INT.SCIENCE Q5 e Ref: STEM Dic. page 346).

177. What is SALT? (WASSCE 2011 - NOV INT.SCIENCE Q6 a[i] Ref: STEM Dic. page 1133).

178. What is INERTIA? (WASSCE 2011 - NOV INT.SCIENCE Q6 d[i] Ref: STEM Dic. page 664).

179. What is meant by the term, RATE OF CHEMICAL REACTION? (WASSCE 2010 - NOV INT.SCIENCE Q1 a[i] Ref: STEM Dic. page 1079 (rate of reaction).

180. Define WORK. (WASSCE 2010 - NOV INT.SCIENCE Q1 b[i] Ref: STEM Dic. page 1384).

181. Describe the mechanism of INHALATION in humans. (WASSCE 2010 - NOV INT.SCIENCE Q1 c[i] Ref: STEM Dic. page 668).

182. What is IMMUNITY? (WASSCE 2010 - NOV INT.SCIENCE Q2 c[i] Ref: STEM Dic. page 654).

183. Explain the term, PARTURITION as applied to animal product. (WASSCE 2010 - NOV INT.SCIENCE Q3 a[i] Ref: STEM Dic. page 952)

184. What is the CENTRE OF GRAVITY of a body? (WASSCE 2010 - NOV INT.SCIENCE Q3 d[i] Ref: STEM Dic. page 227)

185. What is a COMPUTER HARD DISK? (WASSCE 2010 - NOV INT.SCIENCE Q5 a[i] Ref: STEM Dic. page 605 (HARD DISKDRIVE).

186. What is WEATHERING OF ROCKS? (WASSCE 2010 - NOV INT. SCIENCE Q1 a[i] Ref: STEM Dic. page 1372).

187. Explain briefly how each of the following process occurs: (i) MELTING OF SOLID (WASSCE 2010 - NOV INT.SCIENCE Q6 d[i] Ref: STEM Dic. page 808)

188. (ii) CONDENSATION OF GAS (WASSCE 2010 - NOV INT. SCIENCE Q6 d[ii] Ref: STEM Dic. page 292 (CONDENSATION).

189. What is an ACID? (WASSCE 2009 - JUNE INT.SCIENCE Q1 e[i] Ref: STEM Dic. page 18).

190. Define the FOCAL LENGTH of a convex lens. (WASSCE 2009 - JUNE INT.SCIENCE Q2 a[i] Ref: STEM Dic. page 526)

191. What is a HEDGE (WASSCE 2009 - JUNE INT.SCIENCE Q3 c[i] Ref: STEM Dic. page 612).

192. Explain the LIMING as applied to soil (WASSCE 2009 - JUNE INT.SCIENCE Q4 b[i] Ref: STEM Dic. page 749).

193. What is TRANSPIRATION? (WASSCE 2009 - JUNE INT. SCIENCE Q5 a[i] Ref: STEM Dic. page 1309).

194. What is a STANDARD SOLUTION? (WASSCE 2009 - JUNE INT. SCIENCE Q4 d[i] Ref: STEM Dic. page 1222).

195. What is BREATHING? (WASSCE 2009 - JUNE INT.SCIENCE Q6 c[i] Ref: STEM Dic. page 178).

196. Explain the term, FEMALE CIRCUMCISION. (WASSCE 2009 - JUNE INT.SCIENCE Q2 d[i] Ref: STEM Dic. page 254 (Circumcision), page 505 (female circumcision).

197. What is an ECHO?(WASSCE 2009 - JUNE INT.SCIENCE Q5 c[i] Ref: STEM Dic. page 421).

198. State one function each of the following reproduction hormones (i) OESTROGEN (WASSCE 2009 - JUNE INT.SCIENCE Q3 b[i] Ref: STEM Dic. page 906).

199. (ii) TESTOSTERONE (WASSCE 2009 - JUNE INT.SCIENCE Q3 b[ii] Ref: STEM Dic. page 1280).

200. (iii) PROGESTERONE (WASSCE 2009 - JUNE INT.SCIENCE Q3 b[iii] Ref: STEM Dic. page 1038).

201. (iv) PROLACTIN (WASSCE 2009 - JUNE INT.SCIENCE Q3 b[i] Ref: STEM Dic. page 1040.

202. What are FATS? (WASSCE 2009 - JUNE INT.SCIENCE Q3 d[i] Ref: STEM Dic. page 502).

203. Distinguish between a SCALAR QUANTITY and a VECTOR QUANTITY (WASSCE 2009 - NOV INT.SCIENCE Q1 a[i] Ref: STEM Dic. page 1139 and page 1350).

204. What is FOOD PRESERVATION? (WASSCE 2009 - NOV INT. SCIENCE Q1 b[i] Ref: STEM Dic. page 529).

205. Explain the following defects of a simple electric cell: (i) POLARIZATION (WASSCE 2009 - NOV INT.SCIENCE Q2 b[i] Ref: STEM Dic. page 1010). (ii) LOCAL ACTION (WASSCE 2009 - NOV INT.SCIENCE Q2 b[i] Ref: STEM Dic. page 761).

206. What are PLASTICS? (WASSCE 2009 - NOV INT.SCIENCE Q2 c[i] Ref: STEM Dic. page 1001).

207. State one use each of the following fractions of petroleum. (i)BITUMEN (WASSCE 2009 - NOV INT. SCIENCE Q3 a[i] Ref: STEM Dic. page 161). (ii) DIESEL (WASSCE 2009 - NOV INT.SCIENCE Q3 a[ii] Ref: STEM Dic. page 377 (diesel fuel). (iii) PETROLEUM GAS (WASSCE 2009 - NOV INT.SCIENCE Q3 a[iii] Ref: STEM Dic. page 972).

208. What is meant by DISPERAL OF SEEDS? (WASSCE 2009 - NOV INT.SCIENCE Q3 b Ref: STEM Dic. page 395 (dispersal), page 1150 (seed dispersal).

209. Explain the following terms (i) HAY (WASSCE 2009 - NOV INT.SCIENCE Q3 c[i] Ref: STEM Dic. page 608). (ii) SILAGE (WASSCE 2009 - NOV INT.SCIENCE Q3 c[i] Ref: STEM Dic. page 1170).

210. Define MASS NUMBER of an atom. (WASSCE 2009 - NOV INT. SCIENCE Q4 b[i] Ref: STEM Dic. page 796).

211. Define RELATIVE DENSITY (WASSCE 2009 - NOV INT. SCIENCE Q5 a[i] Ref: STEM Dic. page 1095).

212. Define FOCAL LENGTH as applied to a diverging lens (WASSCE 2009 - NOV INT.SCIENCE Q6 b[i] Ref: STEM Dic. page 526).

213. What is meant by BOILING POINT of a liquid (WASSCE 2008 - JULY INT.SCIENCE Q1 a[i] Ref: STEM Dic. Page page 167).

214. What are PERIPHERAL DEVICES of a computer? (WASSCE 2008 - JULY INT.SCIENCE Q d[i] Ref: STEM Dic. page 967).

215. What is CONVECTION as applied to fluids? (WASSCE 2008 - JULY INT.SCIENCE Q1 e[i] Ref: STEM Dic. page 306).

216. Define HEAT. (WASSCE 2008 - JULY INT.SCIENCE Q3 b[i] Ref: STEM Dic. page 609)

217. Give the meaning of each of the following abbreviations in computer language: (i) CD (WASSCE 2008 - JULY INT.SCIENCE Q3 c[i] Ref: STEM Dic. page 222) (ii) RAM (WASSCE 2008 - JULY INT.SCIENCE Q3 c[ii] Ref: STEM Dic. page 1075)

218. What is an ESTER? (WASSCE 2008 - JULY INT.SCIENCE Q5 e[i] Ref: STEM Dic. page 475)

219. What is MOMENT OF A FORCE about a point? (WASSCE 2008 - JULY INT.SCIENCE Q6 a[i] Ref: STEM Dic. page 841).

220. What are PROCESSING DEVICES of a computer? (WASSCE 2008 - JULY INT.SCIENCE Q6 d Ref: STEM Dic. page 1037)

221. What are VITAMINS? (WASSCE 2008 - NOV INT.SCIENCE Q1 a[i] Ref: STEM Dic. page 1360).

222. What is a FERTILE SOIL? (WASSCE 2008 - NOV INT.SCIENCE Q1 c[i] Ref: STEM Dic. page 1190 (soil nutrition).

223. What is a COLLOID? (WASSCE 2008 - NOV INT.SCIENCE Q1 e[i] Ref: STEM Dic. page 275).

224. What are LEGUMINOUS CROPS? (WASSCE 2008 - NOV INT. SCIENCE Q3 b [i]Ref: STEM Dic. page 739).

225. What is a SEMI CONDUCTOR? (WASSCE 2008 - NOV INT. SCIENCE Q3 c[i]Ref: STEM Dic. page 1156).

226. State one function each of the following structures of morale joint. (i) SYNOVIAL FLUID (WASSCE 2008 - NOV INT. SCIENCE Q4 c[i] Ref: STEM Dic. page 1261). (ii) SYNOVIAL MEMBRANE (WASSCE 2008 - NOV INT.SCIENCE Q4 c[ii] Ref: STEM Dic. page 1261). (iii) LIGAMENT (WASSCE 2008 - NOV INT SCIENCE Q4 c[i] Ref: STEM Dic. page 746). (iv) CARTILAGE (WASSCE 2008 - NOV INT.SCIENCE Q4 c[iv] Ref: STEM Dic. page 213).

227. Explain the term MULCHING (WASSCE 2007 - JUNE INT. SCIENCE Q1 a[i]Ref: STEM Dic. page 853 (Mulch).

228. Give two advantages of MULCHING (WASSCE 2007 - JUNE INT.SCIENCE Q1 a[ii]Ref: STEM Dic. page 853 (Mulch).

229. Explain each of the following terms (i) MASS NUMBER (WASSCE 2007 - JUNE INT.SCIENCE Q1 c[i] Ref: STEM Dic. page 796). (ii) RELATIVE ATOMIC MASS (WASSCE 2007 - JUNE INT.SCIENCE Q1 c[ii] Ref: STEM Dic. page 1094).

230. What is BIOTECHNOLOGY? (WASSCE 2007 - JUNE INT. SCIENCE Q1 d[i] Ref: STEM Dic. page 158).

231. What is THERMOSTAT? (WASSCE 2007 - JUNE INT.SCIENCE Q2 a[i] Ref: STEM Dic. page 1287).

232. What is a POLYMER? (WASSCE 2007 - JUNE INT. SCIENCE Q3 a[i] Ref: STEM Dic. page 1014).

233. What is LYMPH? (WASSCE 2007 - JUNE INT.SCIENCE Q3 d[i] Ref: STEM Dic. page 771).

234. State the PRINCIPLE OF FLOTATION. (WASSCE 2007 - JUNE INT.SCIENCE Q3 e[i] Ref: STEM Dic. page 732 (Law of Flotation).

235. What is MALNUTRITION (WASSCE 2007 - JUNE INT.SCIENCE Q5 d[i] Ref: STEM Dic. page 787).

236. Explain the term, GENETIC ENGINEERING (WASSCE 2007 - NOV INT.SCIENCE Q1 b[i] Ref: STEM Dic. page 565).

237. Define the following terms as applied to sound waves: (i) FREQUENCY (WASSCE 2007 - NOV INT. SCIENCE Q3 a[i]Ref: STEM Dic. page 541). (ii) AMPLITUDE (WASSCE 2007 - NOV INT.SCIENCE Q3 a[i] Ref: STEM Dic. page 66).

238. Explain the following terms (i) RELATIVE ATOMIC MASS (WASSCE 2007 - NOV INT.SCIENCE Q4 d[i] Ref: STEM Dic. page 1094). (iii)ISOTOPES (WASSCE 2007 - NOV INT.SCIENCE Q4 d[ii] Ref: STEM Dic. page 293)

239. What is a COMPUTER NETWORK? [(WASSCE 2007 - NOV INT.SCIENCE Q6 e[i] Ref: STEM Dic. page 872 (network)].

240. What is RHESUS FACTOR? (WASSCE 2006 - JULY INT.SCIENCE Q1 a[i] Ref: STEM Dic. page 1109).

241. What is a SOLVENT? (WASSCE 2006 - JULY INT.SCIENCE Q1 b[i] Ref: STEM Dic. page 1196).

242. What is a COMPUTER PERIPHERAL? (WASSCE 2006 - JULY INT.SCIENCE Q1 e[i] Ref: STEM Dic. page 967).

243. Define POWER. (WASSCE 2006 - JULY INT.SCIENCE Q2 b[i] Ref: STEM Dic. page 1026).

244. What is a FOOD WEB? (WASSCE 2006 - JULY INT.SCIENCE Q2 c[ii] Ref: STEM Dic. page 529).

245. What is OESTROUS in a farm animal? (WASSCE 2006 - JULY INT.SCIENCE Q2 c[ii] Ref: STEM Dic. page 906 (Oestrous cycle).

246. What is meant by NEUTRAL POINT in a magnetic field? (WASSCE 2006 - JULY INT.SCIENCE Q3 a[i] Ref: STEM Dic. page 875).

247. What is COLOSTRUM? (WASSCE 2006 - JULY INT.SCIENCE Q3 b[i] Ref: STEM Dic. page 276).

248. What is meant by GRAFTING? (WASSCE 2006 - JULY INT. SCIENCE Q5 a[i] Ref: STEM Dic. page 582).

249. What is a software as applied to the computer? (WASSCE 2006 - JULY INT.SCIENCE Q5 e[i] Ref: STEM Dic. page 1189).

250. What is INTEGRATED PEST MANAGEMENT? (WASSCE 2006 - NOV INT.SCIENCE Q1 c[i] Ref: STEM Dic. page 674).

251. Define DISPERSION OF LIGHT. (WASSCE 2006 - NOV INT. SCIENCE Q1 d[i] Ref: STEM Dic. page 395 (dispersion).

252. What is FIBROID? (WASSCE 2006 - NOV INT.SCIENCE Q2 d[i] Ref: STEM Dic. page 509).

253. Define the following terms as applied to simple machines (i)VELOCITY RATIO (WASSCE 2006 - NOV INT.SCIENCE Q3 b[i] Ref: STEM Dic. page 1351) (ii) EFFICIENCY (WASSCE 2006 - NOV INT.SCIENCE Q3 b[ii] Ref: STEM Dic. page 427).

254. What is a COMPUTER VIRUS? (WASSCE 2006 - NOV INT. SCIENCE Q4 e[i]Ref: STEM Dic. page 1384 (worm).

255. Define the term VARIATION. (WASSCE 2006 - NOV INT. SCIENCE Q5 a[i] Ref: STEM Dic. page 1347).

256. What is meant by the term MOLE (WASSCE 2006 - NOV INT. SCIENCE Q5 b[i] Ref: STEM Dic. page 838).

257. What are MACRONUTRIENTS? (WASSCE 2006 - NOV INT. SCIENCE Q5 c[i] Ref: STEM Dic. page 777).

258. What are FLOPPY DISK? (WASSCE 2006 - NOV INT.SCIENCE Q5 c[i] Ref: STEM Dic. page 522).

259. State the three STATES OF MATTER. (WASSCE 2005 - JULY INT.SCIENCE Q1 a[i] Ref: STEM Dic. page 1224).

260. Distinguish between CONTINUOUS VARIATION and DISCONTINUOUS VARIATION (WASSCE 2005 - JULY INT. SCIENCE Q2 a[i] Ref: STEM Dic. page 304 and page 393).

261. Explain the importance of each of the following devices in household wiring. (i)FUSE (WASSCE 2005-JULY INT.SCIENCE Q3 a[i] Ref: STEM Dic. page 548). (ii) EARTHING (WASSCE 2005 - JULY INT.SCIENCE Q3 a[ii] Ref: STEM Dic. page 418). (iii) SWITCH (WASSCE 2005 - JULY INT.SCIENCE Q3 a[iii] Ref: STEM Dic. page 1257).

262. What is TISSUE RESPIRATION? (WASSCE 2005 - JULY INT. SCIENCE Q3 c[i] Ref: STEM Dic.page 1296.

263. State two characteristics of X-RAYS. (WASSCE 2005 - JULY INT.SCIENCE Q4 b[ii] Ref: STEM Dic. page 1387).

264. What is a FLOWER? (WASSCE 2005 - JULY INT.SCIENCE Q5 a[i] Ref: STEM Dic. page 523).

265. What is a JOINT? (WASSCE 2005 - NOV INT.SCIENCE Q2 b[i] Ref: STEM Dic.page 701).

266. What is a COMPUTER DISKETTE? (WASSCE 2005 - NOV INT. SCIENCE Q2 a[i] Ref: STEM Dic. page 522).

267. What is RELATIVE DENSITY? (WASSCE 2005 - NOV INT. SCIENCE Q2 a[i]] Ref: STEM Dic. page 1095).

268. What is HARD WATER? (WASSCE 2005 - NOV INT.SCIENCE Q2 d[i] Ref: STEM Dic.page 605.

269. What is meant by COMPUTER HARDWARE? (WASSCE 2005 - NOV INT.SCIENCE Q3 e[i] Ref: STEM Dic.page 606)

270. Define SPECIFIC HEAT CAPACITY. (WASSCE 2005 - NOV INT. SCIENCE Q4 a[i] Ref: STEM Dic.page 1203).

271. Define the term OXIDATION (WASSCE 2005 - NOV INT. SCIENCE Q4 c[i] Ref: STEM Dic. page 930)

272. Explain the term VISCOSITY (WASSCE 2005 - NOV INT. SCIENCE Q5 b Ref: STEM Dic.page 1359).

273. Distinguish between a SATURATED HYDROCARBON and UNSATURATED HYDROCARBON (WASSCE 2004 - JULY INT. SCIENCE Q1 b[i] Ref: STEM Dic. page 1137 and page 1336).

274. What is a MAGNETIC FIELD? (WASSCE 2004 - JULY INT. SCIENCE Q1 d[i] Ref: STEM Dic. page 875).

275. What is an ALLOY? (WASSCE 2004 - JULY INT.SCIENCE Q2 a[i] Ref: STEM Dic. page 53).

276. What is meant by VENTILATION? (WASSCE 2004 - JULY INT. SCIENCE Q2 b[i] Ref: STEM Dic. page 1352).

277. Explain the following terms. (i) PHENOTYPE (WASSCE 2004 - JULY INT.SCIENCE Q2 c[i]. (WASSCE 2012 - JUNE INT. SCIENCE Q2 c[i] Ref: STEM Dic. page 977).

278. (ii) GENOTYPE (WASSCE 2004 - JULY INT.SCIENCE Q2 c[ii] Ref: STEM Dic.page 566).

279. Define ELECTRIC POWER. (WASSCE 2004 - JULY INT. SCIENCE Q1 d[i]Ref: STEM Dic. page 433).

280. What is meant by a DAIRY CATTLE. (WASSCE 2004 - JULY INT.SCIENCE Q3 a[i] Ref: STEM Dic. page 342 (Dairy).

281. Define the term ISOTOPE. (WASSCE 2004 - NOV INT. SCIENCE Q1 a[i] Ref: STEM Dic. page 293).

282. Distinguish between HAY and SILAGE. [WASSCE 2004 - NOV INT.SCIENCE Q2 c[i] Ref: STEM Dic. page 608(hay), page 1170 (silage)]

283. What is FORCE as applied in science? (WASSCE 2004 - NOV INT.SCIENCE Q2 d[i] Ref: STEM Dic. page 530).

284. What is RUST? (WASSCE 2004 - NOV INT.SCIENCE Q2 e[i] Ref: STEM Dic. page 293).

285. What is COLOSTRUM? (WASSCE 2004 - NOV INT.SCIENCE Q3 c[i] Ref: STEM Dic. page 276).

286. What is VELOCITY? (WASSCE 2004 - NOV INT.SCIENCE Q3 d[i] Ref: STEM Dic. page 1351).

287. What is a HORMONE? (WASSCE 2004 - NOV INT.SCIENCE Q4 b[i] Ref: STEM Dic. page 630).

288. What is meant by MOLAR MASS? (WASSCE 2004 - NOV INT. SCIENCE Q4 c[i] Ref: STEM Dic. page 838).

289. What is SEA BREEZE? (WASSCE 2004 - NOV INT.SCIENCE Q5 b[i] Ref: STEM Dic. page 1145).

290. What is FAMILY PLANNING? (WASSCE 2004 - NOV INT. SCIENCE Q5 c[i]Ref: STEM Dic. page 501).

291. What is DETERGENT? (WASSCE 2004 - NOV INT.SCIENCE Q5 d[i],Ref: STEM Dic. page 369).

292. What is meant by BOOTING A COMPUTER? (WASSCE 2004 - NOV INT.SCIENCE Q5 e[i] Ref: STEM Dic. page 170 (boot).

293. What is OSMOSIS? (WASSCE 2003 - JULY INT.SCIENCE Q1 a[i] Ref: STEM Dic. page 925).

294. What is an ESTER? (WASSCE 2003 - JULY INT.SCIENCE Q1 b[i] Ref: STEM Dic. page 475).

295. What is HEDGE? (WASSCE 2003 - JULY INT.SCIENCE Q2 a[i] Ref: STEM Dic. page 612).

296. a. State the FIRST AND SECOND LAWS OF REFLECTION OF LIGHT [WASSCE 2003 - JULY INT.SCIENCE Q2 b[i]. b. State the LAWS OF REFLECTION OF LIGHT (WASSCE 2002 - JULY INT. SCIENCE Q2 a[i] Ref: STEM Dic. page 733).

297. What is an ALKALI? (WASSCE 2003 - JULY INT.SCIENCE Q1 a[i] Ref: STEM Dic. page 49).

298. What is meant by the term PHOTOSYNTHESIS? (WASSCE 2003 - JULY INT.SCIENCE Q2 b[i]Ref: STEM Dic. page 986).

299. What is INHALATION? (WASSCE 2003 - JULY INT.SCIENCE Q3 c[i] Ref: STEM Dic. page 668).

300. What is a MIRAGE? (WASSCE 2003 - JULY INT.SCIENCE Q4 c[i],Ref: STEM Dic. page 831).

301. What is a FLOWER? (WASSCE 2003 - JULY INT.SCIENCE Q4 c[i] Ref: STEM Dic. page 287).

302. What SAPONIFICATION? (WASSCE 2003 - JULY INT.SCIENCE Q5 a[i] Ref: STEM Dic. page 1136).

303. 287. What is FRICTION? (WASSCE 2003 - JULY INT.SCIENCE Q5 c[i] Ref: STEM Dic. page 542).

304. 288. Give one function of each of the following parts of a computer. (i) THE KEYBOARD (WASSCE 2003 - JULY INT.SCIENCE Q2 e[i] Ref: STEM Dic. page 709). (ii) THE MODEM (WASSCE 2003 - JULY INT.SCIENCE Q2 e[ii] Ref: STEM Dic. page 835).

305. Draw and label a large diagram of a named TOOTH (WASSCE 2003 - JULY INT.SCIENCE Q4 a[i] Ref: STEM Dic. page 1298).

306. State three functions of the mouse of a computer (WASSCE 2003 - JULY INT.SCIENCE Q5 e Ref: STEM Dic. page 66).

307. What is CORROSION? (WASSCE 2003 - NOV INT.SCIENCE Q1 a[i] Ref: STEM Dic. page 313).

308. What is BIOTIC FACTOR? (WASSCE 2003 - NOV INT.SCIENCE Q1 a[i] Ref: STEM Dic. page 158).

309. What is the CENTRE OF GRAVITY OF a body? (WASSCE 2003 - NOV INT.SCIENCE Q1 b[i] Ref: STEM Dic. page 227).

310. What is a RECTIFIER? (WASSCE 2003 - NOV INT.SCIENCE Q3 b[i] Ref: STEM Dic. page 1086).

311. Define Hardness of water. (WASSCE 2003 - NOV INT.SCIENCE Q3 b[i] Ref: STEM Dic. page 605(Hard Water).

312. Draw and label PLANT CELL. (WASSCE 2003 - NOV INT. SCIENCE Q3 c[i] Ref: STEM Dic. page 223).

313. What is SOIL? (WASSCE 2003 - NOV INT.SCIENCE Q3 d[i] Ref: STEM Dic. page 1189).

314. List three KINGDOMS of living organisms and give one example of each (WASSCE 2003 - NOV INT.SCIENCE Q4 a Ref: STEM Dic. page 713).

315. What are PLASTICS? (WASSCE 2003 - NOV INT.SCIENCE Q4 c[i] Ref: STEM Dic. page 1001).

316. What is TEMPORARY HARDNESS OF WATER (WASSCE 2002 - JULY INT.SCIENCE Q1 b[i] Ref: STEM Dic. page 1275 (Temporary Hardness), page 1367 (water hardness) and page 605(Hard Water).

317. Draw and label a MOTOR NEURONE. (WASSCE 2002 - JULY INT.SCIENCE Q2 b[i] Ref: STEM Dic. page 850)

318. What is an ECHO. (WASSCE 2002 - JULY INT.SCIENCE Q3 d[i] Ref: STEM Dic. page 421).

319. What is meant by FEMALE GENITAL MUTILATION (WASSCE 2002 - NOV INT.SCIENCE Q1 a[i] Ref: STEM Dic. page 505 (Female Circumcision).

320. What are CONVECTIONAL CURRENTS (WASSCE 2002 - NOV INT.SCIENCE Q1 b[i] Ref: STEM Dic. page 306).

321. Name the components of STEEL (WASSCE 2002 - NOV INT. SCIENCE Q2 a[i]Ref: STEM Dic. page 1227).

322. Distinguish between PRIMARY COLOURS and SECONDARY of light (WASSCE 2002 - NOV INT.SCIENCE Q3 b[i] Ref: STEM Dic. page 1032 (Primary Colours) and page 1147 (secondary colours).

323. What is meant by MENOPAUSE (WASSCE 2002 - NOV INT. SCIENCE Q3 c[i] Ref: STEM Dic. page 810).

324. Distinguish between a DEHYDRATING AGENT and a DRYING AGENT (WASSCE 2002 - NOV INT.SCIENCE Q3 d[i] Ref: STEM Dic. page 357 (Dehydrating Agent) and page 412 (Drying Agent).

325. Name one natural sources of each of the following alkanoic acids (i) ETHANOIC ACID (WASSCE 2002 - NOV INT.SCIENCE Q4 a[i] Ref: STEM Dic. page 476). (ii) LACTIC ACID (WASSCE 2002 - NOV INT.SCIENCE Q4 a[ii] Ref: STEM Dic. page 720).

326. What are ENZYMES? (WASSCE 2001 - JULY INT.SCIENCE Q1 a[i] Ref: STEM Dic. page 463).

327. Define ISOTOPES. (WASSCE 2001 – JULY INT.SCIENCE Q1 b[i] Ref: STEM Dic. page 696).

328. State the PRINCIPLES OF FLOATATION - Flotation (WASSCE 2001 – JULY INT.SCIENCE Q3 b[i] Ref: STEM Dic. page 732 (Law of Flotation).

329. State the function of each of the following parts of mammalian circulatory system. (i) HEART (WASSCE 2001 – JULY INT.SC. Q4 b[i] Ref: STEM Dic. page 609). (ii) ARTERY (WASSCE 2001 – JULY INT.SC. Q4 b[i] Ref: STEM Dic. page 99). (iii) CAPILLARY (WASSCE 2001 – JULY INT.SCIENCE Q4 b[i]Ref: STEM Dic. page 204).

330. What is meant by VARIATION? (WASSCE 2001 - JULY INT. SCIENCE Q5 a[i] Ref: STEM Dic. page 1347).

331. What is BLOOD PLASMA? (WASSCE 2001 - NOV INT.SCIENCE Q2 b[i] Ref: STEM Dic. page 1000 (Plasma).

332. State one function each of the following blood cells: (i) ERYTHROCYTE (WASSCE 2001 - NOV INT.SCIENCE Q2 b[iii] Ref: STEM Dic. page 473). (ii) LEUCOCYTE (WASSCE 2001 - NOV INT.SCIENCE Q2 b[iii] Ref: STEM Dic. page 742). (iii) PLATELET (WASSCE 2001 - NOV INT.SCIENCE Q2 b[iii] Ref: STEM Dic. page 1002).

333. Distinguish between DENSITY and RELATIVE DENSITY of a substance (WASSCE 2001 - INT.SCIENCE Q1 c[i] Ref: STEM Dic. page 361 (Density) and page 1095 (Relative Density).

BIOLOGY QUESTIONS FROM THE WEST AFRICAN SENIOR SCHOOL CERTIFICATE EXAMINATIONS (W.A.S.S.CE.) AND STEM DICTIONARY REFERENCE PAGES

1. Name the two main parts of the NERVOUS SYSTEM: [WASSCE Biology June, 2019, Q. 1(a)] Ref: STEM Dictionary, page 227 (Central Nervous System) and page 967 (Peripheral Nervous system).

2. List two components each of the main parts of the nervous system named in (a) (i) CENTRAL NERVOUS SYSTEM. [WASSCE Biology June, 2019, Q. 1(a) Refs: STEM Dictionary, page 176 (Brain); page 1211 (Spinal cord). (ii)] PERIPHERAL NERVOUS SYSTEM. [WASSCE Biology June, 2019, Q. 1(a) (ii)] Refs: STEM Dictionary, page 1196 (Somatic nervous system) and page 115 (Automatic Nervous system).

3. What is a CONDITIONAL REFLEX? [WASSCE Biology June, 2019 Q.1(c)]. Ref: STEM Dictionary, page 293.

4. What is TRANSLOCATION in plants? [WASSCE BIOLOGY JUNE, 2019 Q. 2 (a) (i)] Ref: STEM Dictionary, page 1308.

5. Name the biological process that occurs before TRANSLOCATION takes place. [WASSCE Biology June, 2019 Q. 2(a) (ii)] Ref: STEM Dictionary, page 986 (Photosynthesis).

6. What is malnutrition? [WASSCE Biology June, 2019, Q. 2(b). (i)] Ref: STEM Dictionary, page 787.

7. State differences between a DNA and an RNA molecule. [WASSCE BIOLOGY JUNE, 2019, Q. 5(c)] Ref: STEM Dictionary, page 401(DNA) and page 1116 (RNA).

8. What is EXCRETION in living things? [WASSCE Biology November, 2019; Q1. (a)(i)] Ref: STEM Dictionary, page 485].

9. State the process by which single–celled aquatic organisms carry out excretion. [WASSCE Biology November, 2019. Q1. (a) (ii)] Ref: STEM Dictionary, page 486.

10. Make a labeled diagram of an AMOEBA. [WASSCE Biology November, 2019, Q. 1(b)] Ref: STEM Dictionary, page 63.

11. What is CHROMOSOME in genetics? [WASSCE Biology. November, 2019, Q. 4(a)] Ref: STEM Dictionary, page 249.

12. What is GENE POOL? [WASSCE Biology November, 2019. Q. 5(a) (i)] Ref: STEM Dictionary, page 562.

13. What is GENE FLOW? [WASSCE Biology November, 2019. Q. 5(a) (ii)] Ref: STEM Dictionary, page 562.

14. What is GENETIC DRIFT? [WASSCE Biology November, 2019, Q. 5 (a) (iii)] Ref: STEM Dictionary, page 565
15. State the importance of valves in the heart and veins of humans [WASSCE Biology November, 2019, Q. 5(b) (i)] Ref: STEM Dictionary, page 149 (Bicuspid) page 1314 (Tricuspid valve)
16. What is PINOCYTOSIS? [WASSCE Biology November, 2019. Q. 5 (c) (i)] Ref: STEM Dictionary, page 995.
17. State the application of BIOTECHNOLOGY to humans. [WASSCE Biology Nov., 2019. Q. 5(d)] Ref: STEM Dictionary, page 158(Biotechnology), page 673 (Insulin).
18. State the function of the following parts of the MICROSCOPE. [WASSCE Biology November, 2019. Q. 5(e)] i. IRIS DIAPHRAGM, Ref: STEM Dictionary: Diaphragm (page 374) and Iris (page 691). ii. OBJECTIVE LENS (page 902).
19. List examples of WATER POLLUTANTS. [WASSCE Biology November, 2019. Q. 5(f)] Ref: STEM Dictionary, page 1161 (Sewage), page 907 (Oil spillage).
20. What is meant by BIENNIAL PLANT? [WASSCE Biology November, 2019. Q. 6(a) (i)] Ref: STEM Dictionary, page 149.
21. What is meant by PERENNIAL PLANT? [WASSCE Biology November, 2019. Q. 6 (a) (ii)] Ref: STEM Dictionary, page 963.
22. Name the diseases of the liver. [WASSCE Biology November, 2019. Q. 6(g)] Ref: STEM Dictionary, page 254 (Cirrhosis), page 615 (Hepatitis)
23. State the characteristic features of members of CLASS MAMMALIA [WASSCE Biology June, 2018. Q. 1(a)] Ref: STEM Dictionary, page 788.
24. State the functions of the MAMMAL'S OVARY. [WASSCE Biology June, 2018. Q. 1(b) (i)] Ref: STEM Dictionary, Page page 928.
25. State the functions of the VAGINA. [WASSCE Biology June, 2018. Q. 1 (b) (ii)]. Ref: STEM Dictionary, page 1344.
26. State the functions of the UTERUS. [WASSCE Biology June, 2018. Q. 1 (b) (iii)]. Ref: STEM Dictionary, page 1341.
27. Make a labeled diagram of a TYPICAL FLOWER and its FLORAL PARTS [WASSCE Biology June, 2018. Q. 1 (c)] Ref: STEM Dictionary, page 523
28. What is DENTITION? [WASSCE Biology June, 2018. Q. 2 (a) (i)] Ref: STEM Dictionary, page 363
29. Describe the process of PHOTOSYNTHESIS. [WASSCE Biology June, 2018. Qn. 2 (b)]. Ref: STEM Dictionary, page 986.
30. What is POLLUTION? [WASSCE Biology June, 2018. Q. 3 (a) (i)]Ref: STEM Dictionary, page 1012.
31. What is TROPHIC LEVEL? [WASSCE Biology June, 2018. Q. 3(c)]Ref: STEM Dictionary, page 1318
32. State the role of MUSHROOM in a habitat. WASSCE Biology June, 2018. Q. (3) (c) (ii). Ref: STEM Dictionary, page 858.
33. State the causes of VARIATION in living things. [WASSCE Biology June, 2018.Q. 4 (a) (i)] Ref: STEM Dictionary, page 1347
34. Explain briefly HEALTH. [WASSCE Biology June, 2018.Q. 5 (a) (i)] Ref: STEM Dictionary, page 609
35. Explain briefly HYGIENE. (WASSCE Biology June, 2018. Q. 5(a) (ii) Ref: STEM Dictionary, page 642.
36. Explain briefly SANITATION. [WASSCE Biology June, 2018. Q. 5(a) (iii)Ref: STEM Dictionary, page 1135 (Sanitary).
37. Distinguish between: i. BILATERAL SYMMETRY and ii. RADIAL SYMMETRY. [WASSCE Biology June, 2018.Q. (b) (i)] Ref: STEM Dictionary, page 150 (Bilateral Symmetry) and page 1068 (Radial Symmetry).
38. State two ways in which the HUMAN HEART is adapted to its function.[WASSCE Biology June, 2018.Q. 5(e)] Ref: STEM Dictionary, page 609
39. State ways in which PROTEIN is important to animals. [WASSCE Biology June, 2018.Q. 5(f)]. Ref: STEM Dictionary, page 1044
40. What is CLASSIFICATION of living things? [WASSCE Biology June, 2017. Q. 1(a) (i)]. Ref: STEM Dictionary, page 257
41. State four characteristics of ENZYMES produced in the duodenum of humans. [WASSCE Bio. June, 2017. Q. 2(a)]. Ref: STEM Dic. page 463
42. List two DIGESTIVE ENZYMES produced in the duodenum of humans.[WASSCE Biology June, 2017. Q. 2(b) (i)]. Ref: STEM Dictionary, page 66 (Amylase), page 300 (Lipase), and page 1320 (Trypsin).
43. State one way in which CHLOROPHYLL is important in plants. [WASSCE Biology June, 2017. Q. 2(c) (i)]. Ref: STEM Dictionary, page 246.
44. Name one instrument used for collecting. i. Soil organisms from a soil sample. [WASSCE Biology June, 2017. Q. 3(a) (i)] Ref: STEM Dictionary, page 1321 (Tullgren Funnel or Berlese Funnel or Berlese Trap).
45. Tiny INSECTS from a leaf or stem. [WASSCE Biology June, 2017. Q. 3(a) (ii)]. Ref: STEM Dictionary, page 1018 (Pooter)
46. Describe briefly the external features of a FERN. [WASSCE Biology June, 2017. Q. 5(b)]. Ref: STEM Dictionary, page 506.
47. List five digestive organs of a rat that will be visible when the rat is cut open ventrally. [WASSCE Biology June, 2017. Q. 5(c)] Ref: STEM Dictionary, page 906 (oesophagus); page 1232 (stomach); page 413 (duodenum); page 941 (pancreas); page 760 (liver); and page 553 (gall bladder).
48. What is ANTENATAL CARE? [WASSCE Biology June, 2017. Q. 5(e) (i)] Ref: STEM Dictionary, page 80
49. Explain briefly the term RECOMBINANT DNA TECHNOLOGY. [WASSCE Biology June, 2017. Q. 5(f) (i)]. Ref: STEM Dic., page 1085.
50. What is PLACENTATION in flowering plants? [WASSCE Biology November, 2017. Q. 1(c) (i)] Ref: STEM Dictionary, page 998
51. Name four types of PLACENTATION in flowering plants. [WASSCE Biology November, 2017. Q. 1(c) (ii)] Ref: STEM Dictionary, Pages 941 (page 793); page 948(Parietal placentation) Axial placentation page 120 (Basal Placentation page 134).
52. Name two ENZYMES in the human alimentary canal that digest STARCH.[WASSCE Biology November, 2017. Q. 2(b) (i)]. Ref: STEM Dictionary, page 66 (Amylase) and page 1050 (ptyalin).
53. Name the part of the alimentary canal where the ENZYMES are most active. Ref: STEM Dictionary, page 413 (duodenum) and page 851 (mouth).

54. Explain briefly the following terms i. Community [WASSCE Biology November, 2017 Q. 3(a) (i)] Ref: STEM Dictionary, page 281 ii. Population. [WASSCE Biology November, 2017. Q. 3(a) (ii)]. Ref: STEM Dictionary, page 1018 iii. Ecological Niche. [WASSCE Biology November, 2017. Q. 3(a) (iii)] Ref: STEM Dictionary, page 422

55. State the effect of OVER GRAZING on a grassland vegetation [WASSCE Biology November, 2017. Q 3(c) (ii)]. Ref: STEM Dictionary, page 928

56. State three ways by which animals survive unfavourable conditions. [WASSCE Biology November, 2017. Q. 4(b)] Ref: STEM Dictionary, page 621 (Hibernation in winter or page 37 (aestivation in summer); page 828 (Migration); page 28 (Adaptation).

57. State evidences of Evolution. [WASSCE Biology November, 2017 Q. 4(d)] Ref: STEM Dictionary, page 1160 (SEROLOGY); and page 449 (EMBRYOLOGY).

58. What is a FLORAL FORMULAE? (WASSCE Biology November, 2017 Q. 5(i) Ref: STEM Dictionary, page 522

59. What is a protein? [WASSCE Biology November, 2017. Q. 5(d) (i)] Ref: STEM Dictionary, page 1044

60. Distinguish between TRANSCRIPTION and TRANSLATION as used in Protein synthesis. [WASSCE Biology November, 2017. Q. 5(d) (ii)] Ref: STEM Dictionary, page 1304 andpage 1307

61. What is BOD as used in a water industry? [WASSCE Biology November, 2017, Q. 5(e) (i)] Ref: STEM Dictionary, page 156 (Biological Oxygen Demand)

62. What is IRRITABILITY? [WASSCE Biology November, 2017, Q. 6(b) (i)] Ref: STEM Dictionary, page 692

63. What is TILLAGE as used in agriculture? [WASSCE Biology November, 2017, Q. 6(c) (i)] Ref: STEM Dictionary, page 1294

64. Explain briefly the following terms. i. Oviparity [WASSCE Biology November, 2017, Q. 6(e) (i)] Ref: STEM Dictionary, page 929. ii. Viviparity [WASSCE Biology November, 2017, Q. 6(e) (ii] Ref: STEM Dictionary, page 1360.

65. 67. Make a labeled diagram of a PLANT CELL. [WASSCE Biology June, 2016. Q. 1(b)]. Ref: STEM Dictionary, page 223.

66. What is a HABITAT? [WASSCE Biology June, 2016. Q. 3(a)]. Ref: STEM Dictionary, page 597.

67. Explain briefly the role of a DECOMPOSER in a habitat. [WASSC Biology June, 2016. Q. 3(a)(ii)]. Ref: STEM Dictionary, page 352

68. Explain the term AGGLUTINATION as used in blood transfusion. [WASSCE Biology June, 2016. Q. 4(a) (i)] Ref: STEM Dictionary, page 40

69. Explain briefly how the following structures of mammals are adapted for their function. i. VEINS: [WASSCE Biology November, 2016. Q. 1(c) (i)]Ref: STEM Dictionary, page 287. ii. ARTERIES: [WASSCE Biology November, 2016. Q. 1(C) (i)]. Ref: STEM Dictionary, page 99

70. Define ENZYMES. [WASSCE Biology November, 2016. Q. 2(a) (i)]Ref: STEM Dictionary, page 463.

71. Name three groups of MICROORGANISMS. [WASSCE Biology November, 2016. Q. 3(a) (i)] Ref: STEM Dictionary, page 1358(virus), page 125 (bacteria), page 1046 (protozoa)

72. State four ways by which MICROORGANISMS can be controlled. [WASSCE Biology November, 2016. Q. 3(a) (iii)]Ref: STEM Dictionary, page 82 (Antibiotics); page 654 (immunisation); page 1229 (sterilisation); page 293 (vaccine).

73. Name three sources of ENERGY WHICH DO NOT DEPEND ON FOSSIL FUEL. [WASSCE Biology November, 2016. Q. 5(d) (i)]Ref: STEM Dictionary, page 569 (Geothermal energy); page 893 (Nuclear Energy); page 1192 (solar)

74. Describe the structure of CARTILAGE in mammals. [WASSCE Biology November, 2016. Q. 5(e)]. Ref: STEM Dictionary, page 213.

75. Explain briefly SEXUAL REPRODUCTION IN RHIZOPUS. [WASSCE Biology June, 2015. Q. 1(a)]. Ref: STEM Dictionary, page 297(Conjugation)

76. A student suddenly stepped on a big snake, he cried for help and ran way. i. Name the HORMONE produced in the body for the reaction. [WASSCE Biology June, 2015. Q. 1(b) (i)]. Ref: STEM Dictionary, page 34 (Adrenaline). ii. In which part of the human body is the HORMONE PRODUCED. [WASSCE Biology June, 2015. Q. 1(b) (ii)]. Ref: STEM Dic., page 34 (Adrenaline).

77. List the ORGANELLES found in a plant cell that are responsible for excretion of METABOLIC WASTE. [WASSCE Biology June, 2015. Q. 1(c) (ii)]. Ref: STEM Dictionary, page 246 (chloroplast) and page 832 (Mitochondrion).

78. State three benefits of ROUGHAGE in the diet of humans. [WASSCE Biology June, 2015. Q. 2(d) (i)]. Ref: STEM Dictionary, page 1122

79. What is SEWAGE? [WASSCE Biology June, 2015. Q. 3(a) (i)] Ref: STEM Dictionary, page 1161.

80. Explain TERRACING in soil conservation method. [WASSCE Biology June, 2015. Q. (b)] Ref: STEM Dictionary, page 1278

81. Explain the term ADAPTATION. [WASSCE Biology June, 2015. Q. 3(c)]Ref: STEM Dictionary, page 28

82. Explain two ways by which the following organisms adapt to their habitats i. Hydrophytes. ii. Xerophytes. [WASSCE Biology June, 2015. Q. 3(d) (i), Q. 3(d) (ii)]. Ref: STEM Dictionary, page 641 and page 1389.

83. Explain the following: i. TEST CROSS. ii. MONOHYBRID CROSS [WASSCE Biology June, 2015. Q. 4(a)]. Ref: STEM Dictionary, page 1279 and page 843

84. What is MAGNIFICATION? [WASSCE Biology June, 2015. Q. 5(a) (i)]Ref: STEM Dictionary, page 784

85. State three main characteristics /features each of the members of the following phyla. ARTHROPODA. ii. NEMATODA. [WASSCE Biology June, 2015. Q. 5(b)] Ref: STEM Dictionary, page 99 and page 868.

86. What is CELLULAR RESPIRATION? [WASSCE Biology June, 2015. Q. 5(c) (i)]. Ref: STEM Dictionary, page 225

87. State two differences between DNA and RNA [WASSCE Biology June, 2015. Q. 5(d) (i)] Ref: STEM Dictionary, page 401 and page 1116

88. What is HOMEOSTASIS? [WASSCE Biology November, 2015. Q. 1(b)] Ref: STEM Dictionary, page 627

89. State functions of the LIVER. [WASSCE Biology November, 2015. Q. 2(c)] Ref: STEM Dictionary, page 760.

90. What is MUTATION? [WASSCE Biology November, 2015. Q. 4(a)] Ref: STEM Dictionary, page 858

91. List three symptoms of DOWNS SYNDROME. [WASSCE Biology November, 2015. Q. 4(b) (i)]. Ref: STEM Dictionary, page 408

92. Explain the term COMPARATIVE ANATOMY. [WASSCE Biology November, 2015. Q 4(d) (i)] Ref: STEM Dictionary, page 282.

93. State four differences between ORDER DIPTERA and ORDER LEPIDOPTERA. [WASSCE Biology November, 2015. Q. 5(a)]. Ref: STEM Dictionary, page 388 and page 741.

94. Explain briefly: i. ENDOCYTOSIS. ii. EXOCYTOSIS. [WASSCE Biology November, 2015. Q.5(d)]. Ref: STEM Dictionary, page 454 and page 487.

95. Define the following terms: i. ELIMINATION. ii. EXCRETION. iii. SECRETION. [WASSCE Biology July, 2014. Q. 2(a)]. Ref: STEM Dictionary, page 446 (Elimination); page 485 (Excretion), page 1149 (Secretion).

96. Explain briefly the following ecological terms: (i) BIOSPHERE (ii) HABITAT. [WASSCE Biology July, 2014. Q. 3(a)]. Ref: STEM Dictionary, page 157 (Biosphere) and page 597(Habitat).

97. Explain the term CO-DOMINANCE. (WASSCE Biology July, 2014. Q. 4(i) Ref: STEM Dictionary, page 270.

98. Name the biological phenomenon used to describe loss of water from plant when transpiration is impossible. [WASSCE Biology July, 2014. Q.5f) (i)] Ref: STEM Dictionary, page 595 (Guttation).

99. What is COVER CROPPING? [WASSCE Biology November, 2014. Q. 3 (c)]. Ref: STEM Dictionary, page 318

100. What is BALANCED DIET? (WASSCE Biology November, 2014. Qn 5(b) Ref: STEM Dictionary, page 127.

101. Explain briefly the Genetic terms i. TRANSCRIPTION. ii. TRANSLOCATION (WASSCE Biology November, 2014. Q. 5(f) Ref: STEM Dictionary, page 1304 (Transcription), page 1308 (Translocation).

102. What is an ENDOCRINE GLAND? [WASSCE Biology July, 2013. Q. 1 (a) (i)]. Ref: STEM Dictionary, page 453).

103. What is AUTONOMIC NERVOUS SYSTEM? [WASSCE Biology July, 2013. Q. 1 (b)(i)]. Ref: STEM Dictionary, page 116.

104. What is an ORGANELLE? [WASSCE Biology July, 2013. Q. 2(b)] Ref: STEM Dictionary, page 920.

105. Describe the structure and function of each of the following organelles: i. MITOCHONDRION. ii. ENDOPLASMIC RETICULUM. iii. CHLOROPLAST [WASSCE Biology July, 2013. Q. 2 (c)]. Ref: STEM Dictionary, page 832 (Mitochondrion), page 455 (Endoplasmic Reticulum) and page 246 (Chloroplast).

106. What is POLLINATION? [WASSCE Biology July, 2013. Q. 5 (a)]Ref: STEM Dictionary, page 1011.

107. What is EVOLUTION? [WASSCE Biology July, 2013. Q. 4 (a)] Ref: STEM Dictionary, page 483

108. 110. Explain how each of the following evidence supports the theory of evolution i. FOSSIL RECORDS. ii. EMBRYOLOGY. [WASSCE Biology July, 2013. Q. 4 (b)]. Ref: STEM Dictionary, page 534(Fossil Records) and page 449 (Embryology).

109. 111. What are ANIMAL HORMONES? [WASSCE Biology November, 2013. Q. 1 (a)]. Ref: STEM Dictionary, page 630

110. 112. What are AUTOTROPHS? [WASSCE Biology November, 2013. Q. 2 (a)(i)]. Ref: STEM Dictionary, page 117.

111. State two main types of CARBOHYDRATES giving one example of each type. [WASSCE Biology November, 2013. Q. 2 (c)]. Ref: STEM Dictionary, page 392 (Disaccharides) and page 845 (Monosaccharides).

112. State four structural features of GRASSHOPPERS which adapt them to their habitat. [WASSCE Biology November, 2013. Q. 4 (a) (i)]. Ref: STEM Dictionary, page 585.

113. List the distinguishing features of the following cells. i. PROKARYOTIC CELLS. ii. EUKARYOTIC CELLS. [WASSCE Biology November, 2013. Q. 1 (a)]. Ref: STEM Dictionary, page 1040 and page 479.

114. Describe EPIGEAL GERMINATION of a seed. [WASSCE Biology July, 2012. Q. 1 (a) (i)]. Ref: STEM Dictionary, page 300.

115. Define the following terms giving example each. i.CO DOMINANCE. ii. SEX-LINKED CHARACTER (SEX LINKAGE). iii. GENETIC ENGINEERING [WASSCE Biology July, 2012. Q. 4 (a)]. Ref: STEM Dictionary, page 270, page 1162 and page 565.

116. What is PLASMOLYSIS? [WASSCE Biology July, 2012. Q. 5 (c) (i)]. Ref: STEM Dictionary, page 1001.

117. Briefly describe how ORAL CONTRACEPTIVE work? [WASSCE Biology July, 2012. Q. 6 (a) (i)]. Ref: STEM Dictionary, page 917.

118. What is VARIATION? [WASSCE Biology November, 2012. Q. 3 (a) (i)]. Ref: STEM Dictionary, page 1347.

119. Explain the term NATURAL SELECTION. [WASSCE Biology NOVEMBER, 2012. Q. 3 (b) (ii)]. Ref: STEM Dictionary, page 865 .

120. What is ANTIGEN? [WASSCE Biology November, 2012. Q. 4 (b) (i)] Ref: STEM Dictionary, page 83.

121. List the characteristics of CHORDATA. [WASSCE Biology November, 2012. Q. 6 (a) (ii)]. Ref: STEM Dictionary, page 247.

122. Explain the following term. i. IMBIBITION. ii. TURGIDITY. iii. FLACCIDITY. iv. PLASMOLYSIS. [WASSCE Biology July, 2011. Q. 1 (a).] Ref: STEM Dictionary, page 653, page 1323, page 518 and page 1001

123. Define OSMOSIS. [WASSCE Biology July, 2011. Q. 1 (b) (i)] Ref: STEM Dictionary, page 925.

124. Explain the following term. i. DIPLOID. ii. POLYGENIC INHERITANCE. [WASSCE Biology July, 2011. Q. 4 (a)]. Ref: STEM Dictionary, page 388 and page 1013(POLYGENE)

125. Explain the difference between sex linkage and autosomal linkage. [WASSCE Biology July, 2011. Q. 4(b) (i)]. Ref: STEM Dictionary, page 1162 and page 117

126. What is TAXONOMY? [WASSCE Biology July, 2011. Q. 5 (a) (i)]. Ref: STEM Dictionary, page 1271.

127. Explain the term NUTRITION. [WASSCE Biology November, 2011. Q. 2 (a)]. Ref: STEM Dictionary, page 899.

128. Explain the following terms giving one example in each case: i. POPULATION ii. ECOSYSTEM [WASSCE Biology November, 2011. Q. 3 (a)]. Ref: STEM Dictionary, page 1018 and page 423.

129. What is COMPETITION? [WASSCE Biology November, 2011. Q. 3 (b) (i)] Ref: STEM Dictionary, page 283

130. What is POLLUTION? [WASSCE Biology November, 2011. Q. 4 (a)]. Ref: STEM Dictionary, page 1012

131. Explain the following terms: i. MONOECIOUS PLANT ii. PROTANDROUS FLOWER. iii. ZYGOMORPHIC FLOWER. [WASSCE Biology November, 2011. Q. 6] Ref: STEM Dictionary, page 388, page 1044(PROTANDRY) and page 1401).

132. What are lipids? [WASSCE Biology November, 2010. Q. 2 (a) (i)]. Ref: STEM Dictionary, page 755.

133. What is root pressure? [WASSCE Biology November, 2010. Q. 2 (c) (i)] Ref: STEM Dictionary, Page page 1119

134. What is an endangered species? [WASSCE Biology November, 2010. Q. 3 (a) (i)]. Ref: STEM Dictionary, page 453

135. What are PREDATORS? [WASSCE Biology November, 2010. Q. 3 (b) (i)] Ref: STEM Dictionary, page 1028 (PREDATION).

136. Explain the following terms: i. ALLELES. ii. PURE BREED. i PHENOTYPE. [WASSCE Biology November, 2010. Q. 4(a)]. Ref: STEM Dictionary, page 51, page 1053 and page 977.

137. Define the following terms: i. POPULATION DENSITY. ii. IMMIGRATION iii. BIOMASS. [WASSCE Biology November, 2010. Q. 6 (a)] Ref: STEM Dictionary, page 1019, page 653 and page 156.

138. DIAGRAM OF THE VERTICAL SECTION OF THE MAMMALIAN EYE. (WASSCE Biology June, 2009. Qn 1 (a)(i). REF: STEM DIC page page 495.

139. Describe the eye defect of SHORT SIGHTEDNESS. [WASSCE Biology June, 2009. Qn. 1 (b) (i)]. Ref: STEM Dictionary, page 1166.

140. Outline differences between AUTOTROPHIC and HOLOZOIC modes of nutrition. [WASSCE Biology June, 2009. Q. 2]. Ref: STEM Dictionary, page 117 and page 626.

141. Name the disease that results from deficiency in VITAMIN C. [WASSCE Biology June, 2009. Q. 2 (c) (i)]. Ref: STEM Dictionary, page 1145 (scurvy).

142. List symptoms of the DISEASE. [WASSCE Biology June, 2009. Q. 2 (c) (ii)]. Ref: STEM Dictionary, page 1145 (scurvy).

143. Explain the following terms. i. OVERCROWDING. ii. CONSERVATION. [WASSCE Biology June, 2009. Q. 3 (a) (i)]. Ref: STEM Dictionary, page 928 ii. CONSERVATION: [WASSCE Biology June, 2009. Q. 3 (a) (ii)]. Ref: STEM Dictionary, page 299.

144. Explain the following terms. i. ARTIFICIAL SELECTION. ii. INBREEDING [WASSCE Biology June, 2009. Q. 4 (a)]. Ref: STEM Dictionary, page 100 and page 657.

145. Explain briefly what will happen in blood transfusion if the blood of the two individuals are not compatible. [WASSCE Biology June, 2009. Q. 4 (c)]. Ref: STEM Dictionary, page 40 (Agglutination).

146. What are AMPHIBIANS? [WASSCE Biology June, 2009. Q. 5 (a) (i)] Ref: STEM Dictionary, page 64.

147. State how the features of the LUNG adapt to its functions. (WASSCE Biology November, 2009. Q 2 (a). REF: STEM DIC page 770).

148. State differences between AEROBIC RESPIRATION and ANAEROBIC RESPIRATION. (WASSCE Biology November, 2009. Q 2 (b). REF: STEM DIC page 36 and page 67.

149. What is GLYCOLYSIS? (WASSCE Biology November, 2009. Q 2 (c) (i) REF: STEM DIC page 578).

150. List some ATMOSPHERIC POLLUTANTS and their SOURCES (WASSCE Biology November, 2009. Q 4 (a). REF: STEM DIC page 108).

151. What is a balanced diet? (WASSCE Biology July, 2008. Qn 2 (a). REF: STEM DIC page 127).

152. Mention two ENZYMES that act on protein. (WASSCE Biology July, 2008. Qn 2 (d)(ii). REF: STEM DIC page 1098 (Rennin), page 961 (Pepsin), page 1320 (Trypsin)

153. What is MONOHYBRID CROSS? (WASSCE Biology July, 2008. Qn 3 (a)(i). REF: STEM DIC page 843).

154. List three external features common to all ARTHROPODS. (WASSCE Biology July, 2008. Qn 4 (a)(i). REF: STEM DIC page 99.

155. List three characteristic features of INSECTS. (WASSCE Biology July, 2008. Qn 4 (a)(ii). REF: STEM DIC page 671.

156. What are social insects? (WASSCE Biology July, 2008. Qn 4 (c)(i). REF: STEM DIC page 1184).

157. State vegetative differences between DICOTYLEDONOUS and MONOCOTYLEDONOUS plant. (WASSCE Biology July, 2008. Qn 5 (b). REF: STEM DIC page 376 and page 843.

158. What is DRUG? (WASSCE Biology July, 2008. Qn 6 (a)(i) . REF: STEM DIC page 411.

159. State four signs of EPILEPTIC FIT. (WASSCE Biology July, 2008. Qn 6 (b)(i). REF: STEM DIC page 517 (FIT).

160. Name three types of ASEXUAL REPRODUCTION in microorganism and give one example of each type. (WASSCE Biology November, 2008. Qn 1 (c)(ii) REF: STEM DIC page 151 (Binary/prokaryotic fission), page 263 (Clonal fragmentation), page 184 (Budding).

161. Explain the following terms. i. DIGESTION. ii. PERISTALSIS iii. TRANSPIRATION. (WASSCE Biology November, 2008. Qn 2 (a) REF: STEM DIC page 381, page 967, page 1309).

162. Make a diagram of the longitudinal section of a VILLUS and label fully. (WASSCE Biology November, 2008. Qn 2 (c). REF: STEM DIC page 1356

163. Explain the following: a. INDEPENDENT ASSORTMENT. b. LINKAGE c. SEX – LINKED CHARACTER. d. SICKLE CELL ANAEMIA. (WASSCE Biology November, 2008. Qn 3. REF: STEM DIC page 659, page 755, page 1162, page 1167).

164. What is variation? (WASSCE Biology November, 2008. Qn 4 (a)(i). REF: STEM DIC page 1347.

165. Name the two types of variation and give two examples of each. (WASSCE Biology November, 2008. Qn 4 (a)(ii). REF: STEM DIC page 1347).

166. Explain the following. i. CHROMOSOME. ii. MUTATION. (WASSCE Biology November, 2008. Qn 4 (b). REF: STEM DIC page 249 and page 858.

167. State three types of MUTATION. (WASSCE Biology November, 2008. Qn 4 (c). REF: STEM DIC page 858).

168. Make a drawing of a PARAMECIUM and label fully. (WASSCE Biology November, 2008. Qn 5 (a). REF: STEM DIC page 946).

169. Describe the structure of the MAMMALIAN EAR. (WASSCE Biology July, 2007. Qn 1 (a). REF: STEM DIC page 418)

170. Name the parts of the ear responsible for i. HEARING. ii. MAINTAINING BALANCE. (WASSCE Biology July, 2007. Qn 1 (b). REF: STEM DIC page 418

171. List the EXCRETORY ORGANS in humans and name one WASTE PRODUCT in each case. (WASSCE Biology July, 2007. Qn 5 (b). REF: STEM DIC page 710 (kidney), page 388 (skin), page 760 (liver), page 770 (lung).

172. Draw a longitudinal section of the mammalian heart and label it fully (WASSCE Biology November, 2007. Qn 1 (a). REF: STEM DIC page 609)

173. Explain the AUTOTROPHIC MODE of NURITION. (WASSCE Biology November, 2007. Qn 1 (a) REF: STEM DIC page 117.

174. Differentiate between micronutrients and macronutrients (WASSCE Biology November, 2007. Qn 2 (c)(i) REF: STEM DIC page 825 and page 777.

175. What is a WATER –LOGGED soil? (WASSCE Biology November, 2007. Qn 3 (b)(i) REF: STEM DIC. page 1369

176. State the first and second laws of MENDEL. (WASSCE Biology November, 2007. Qn 4 (a) REF: STEM DIC page 809).

177. State two importance of CLASSIFICATION (WASSCE Biology November, 2007. Qn 5 (a)(i) REF: STEM DIC page 257

178. Outline the difference between the HYDRA and RABBIT with respect to i. LEVEL OF ORGANISATION ii. BODY SYMMETRY. iii. NUMBER OF BODY LAYERS. iv MODE OF DIGESTION. V. HABITAT (WASSCE Biology November, 2007. Qn 6 (b) REF: STEM DIC page 636 and page 1067.

179. Explain the following terms and give one example of each. i. PROKARYOTES ii. EUKARYOTES (WASSCE Biology November, 2007. Qn 5 (c)(i) REF: STEM DIC page 1040 and page 479).

180. State the LEVEL OF ORGANIZATION in living organisms, giving one example in each case. (WASSCE Biology July, 2006. Qn 1 (a)(i). REF: STEM DIC page 223 (Cell), page 1295 (tissue), page 920 (organ), page 1263 (system).

181. Explain two of the levels of organization you have named in a(i). REF: STEM DIC page 223 (Cell), page 1295 (tissue), page 920 (organ), page 1263 (system).

182. Give two advantages of AMOEBA over LIZARD in relation to their levels to ORGANIZATION (WASSCE Biology July, 2006. Qn 1 (b)(i) REF: STEM DIC page 63 and page 760).

183. Mention the two types of germination of seeds and give one example each. (WASSCE Biology July, 2006. Qn 2(b)(ii). REF: STEM DIC page 300 (EPIGEAL) and page 646 (HYPOGEAL).

184. Define the term ADAPTATION (WASSCE Biology July, 2006. Qn 3(a) REF: STEM DIC page 28.

185. Describe the effects of the following water pollutants on aquatic life i. SEWAGE ii. OIL SPILLAGE iii. FERTILIZER (WASSCE Biology July, 2006. Qn 3 (c)(i) REF: STEM DIC page 1161, page 907 and page 293.

186. Name the excretory organs found in i. INSECTS ii. NEMATODES. (WASSCE Biology July, 2006. Qn 4 (b) REF: STEM DIC page 671, page 868)

187. State the similarities between a skin and a leaf [WASSCE Biology July, 2006. Q.4 (c)] Ref: STEM Dictionary, page 388 and page 736.

188. List five characteristics of each of these classes i. OSTEICHTHYES ii. MAMMALIA (WASSCE Biology July, 2006. Qn 5 (b) REF: STEM DIC page 925 and page 788).

189. Define First AID (WASSCE Biology July, 2006. Qn 6 (a)(i) REF: STEM DIC page 515).

190. What is ARTIFICIAL RESPIRATION (WASSCE Biology July, 2006. Qn 6 (c) REF: STEM DIC page 100).

191. Explain briefly how the process of plasmolysis occur in a living cell. (WASSCE Biology November, 2006. Qn 1 (a)(i) REF: STEM DIC page 1001).

192. Describe how AMOEBA regulates water in its body (WASSCE Biology November, 2006. Qn 2 (a) REF: STEM DIC page 63).

193. Give a brief account of the following processes in a mammal. i. FORMATION OF UREA. ii. ELIMINATION OF UREA. (WASSCE Biology November, 2006. Qn 2 (b) REF: STEM DIC page 1338).

194. State four functions of the mammalian skeleton. (WASSCE Biology Nov. 2006. Qn 3 (a)(i) REF: STEM DIC page 1178).

195. Draw and label a large diagram to show the structure of a typical CERVICAL VERTEBRA. (WASSCE Biology November, 2006. Qn 3 (a)(ii) REF: STEM DIC page 230).

196. What is meant by PHOTOTROPISM? (WASSCE Biology November, 2006. Qn 4 (a) REF: STEM DIC page 987).

197. Explain how AUXINS influence phototropic movements in plants. (WASSCE Biology November, 2006. Qn 4 (c) REF: STEM DIC page 117.

198. What is meant by POPULATION DENSITY? (WASSCE Biology November, 2006. Qn 5 (a)(i) REF: STEM DIC page 1019.

199. What is meant by PERSONAL HYGIENE? (WASSCE Biology November, 2006. Qn 6 (a)(i) REF: STEM DIC page 642).

200. Distinguish between transpiration and guttation. (WASSCE Biology July, 2005. Qn 1 (a)(i) REF: STEM DIC page 1309 and page 595).

201. What is; i. CELL. ii. AN ORGANELLE (WASSCE Biology July, 2005. Qn 1 (a)(i) REF: STEM DIC page 223 and page 920).

202. Describe the structure of the following ORGANELLES. i. NUCLEUS ii. MITOCHONDRION. iii. VACUOLE. iv. CHLOROPLAST. (WASSCE Biology July, 2005. Qn 1 (a)(i) REF: STEM DIC page 896, page 832, page 1343, page 246).

203. Define each of the following terms, i. BIOMAS. ii. PRODUCER iii. PRIMARY CONSUMER iv. FOOD CHAIN (WASSCE Biology July, 2005. Qn 5 (a) REF: STEM DIC page 156, page 1038, page 1032, page 528)

204. Compare meiosis with mitosis in living organisms REF: STEM DIC page 807 and page 833).

205. Distinguish between infectious and non-infectious diseases. (WASSCE Biology July, 2005. Qn 6 (b) REF: STEM DIC page 665).

206. Explain the following terms i. DIFFUSION ii. OSMOSIS iii. ACTIVE TRANSPORT (WASSCE Biology November, 2005. Qn 1 (a) REF: STEM DIC page 380, page 925, page 26).

207. What is meant by growth? (WASSCE Biology November, 2005. Qn 3 (a) REF: STEM DIC page 592).

208. Describe the significant features of the i. INTERPHASE STAGE OF THE NUCLEUS ii. METAPHASE STAGE IN MITOSIS (WASSCE Biology November, 2005. Qn 3 (b)(i) (ii) REF: STEM DIC page 681, page 817).

209. What is the significance of mitosis (WASSCE Biology November, 2005. Qn 3 (d) REF: STEM DIC page 833.

210. What is HOMEOSTASIS? (WASSCE Biology November, 2005. Qn 5 (a)(i) REF: STEM DIC page 627).

211. ORGANS in the mammal that are associated with homeostasis. (WASSCE Biology November, 2005. Qn 5 (a)(i) REF: STEM DIC page 710 (kidney), page 388 (skin), page 760 (liver).

212. Describe: i. The vertebral column of humans ii. A typical MAMMALIAN VERTEBRA (WASSCE Biology November, 2005. Qn 7(a), REF: STEM DIC page 1289 (thoracic vertebrae), page 230(cervical vertebrae), page 121(axis vertebrae).

213. Draw and label a diagram of the DIGESTIVE SYSTEM OF HUMANS. (WASSCE Biology November, 2005. Qn 6 (a) REF: STEM DIC page 48).

214. What are the main components of a EUKARYOTIC CELL. (WASSCE Biology July, 2004. Qn 1 (a)(i) REF: STEM DIC page 479).

215. State the type of life cycle exhibited by each of the following organisms i. TERMITE ii. BUTTERFLY iii. MOSS. (WASSCE Biology July, 2004. Qn 2 (a) REF: STEM DIC page 1277, page 188 and page 849).

216. What is meant by each of the following terms? i. TEST-CROSS ii. CROSSING OVER. iii. LINKAGE. [(WASSCE Biology July, 2004. Qn 4 (a) REF: STEM DIC page 1279 (test-cross), page 324(crossing over), page 755 (linkage)].

217. How can it be proved experimentally that plants lose water during TRANSPIRATION? (WASSCE Biology July, 2004. Qn 5(a). REF: STEM DIC page 1309

218. What is the importance of transpiration to plant? (WASSCE Biology July, 2004. Qn 5(b). REF: STEM DIC page 1309.

219. Name the main components of a DNA molecule. (WASSCE Biology July, 2004. Qn 6(a). REF: STEM DIC. page 401.

220. Describe the role of DNA in protein synthesis. (WASSCE Biology July, 2004. Qn 6(b). REF: STEM DIC page 401 (DNA) .

221. What is an AUXIN? (WASSCE Biology July, 2004. Qn 7(b)(i). REF: STEM DIC page 117.

222. By what EXTERNAL FEATURES can a fruit be distinguished from a seed? (WASSCE Biology Nov., 2004. Qn 1(a). Ref: Stem Dic. page 543 (fruits) and page 1150(seed).

223. What is CHROMOSOME? (WASSCE Biology November, 2004. Qn 2(a) REF: STEM DIC page 249.

224. Describe four causes of GENETIC VARIATION: i. GENE MUTATION ii. CHROMOSOME MUTATION. iii. INDEPENDENT ASSORTMENT iv. SEGREGATION. (WASSCE Biology November, 2004. Qn 2(b) REF: STEM DIC page 562 (Gene mutation), page 250 (chromosome mutation), page 659(independent assortment), page 1151 (segregation).

225. What is TRANSPIRATION? (WASSCE Biology November, 2004. Qn 3(a)(i) REF: STEM DIC page 1309.

226. State the possible sites of transpiration in plants and indicate what happens at each of the sites. (WASSCE Biology November, 2004. Qn. 3(ii). REF: STEM DIC page 333 (cuticle), page 1232 (Stomata), page 741 (Lenticel)

227. Name four human diseases caused by each of the following PATHOGENS: i. BACTERIA. ii. VIRUSES. (WASSCE Biology November, 2004)? REF: STEM DIC page 1280 (Tetanus), page 1325 (Typhoid), page 1321 (Tuberculosis), page 247 (Cholera), and page 242 (Chicken pox), page 803 (Measles), page 666 (Influenza)

228. What is NUTRITION? (WASSCE Biology November, 2004. Qn 6(a) REF: STEM DIC page 899).

229. What is the importance of PHOTOSYNTHESIS? (WASSCE Biology November, 2004. Qn 6(c). REF: STEM DIC page 986.

230. Describe each of following processes in tissue respiration in living cells: i. GLYCOLYSIS. ii. THE KREBS CYCLE. (WASSCE Biology July, 2003. Qn 1(a) REF: STEM DIC page 578 and page 716).

231. State four differences between aerobic respiration and anaerobic respiration (WASSCE Biology July, 2003. Qn 1(b) REF: STEM DIC page 36 and page 67.

232. Distinguish between a FLOWER and a FRUIT. (WASSCE Biology July, 2003. Qn 2(a) REF: STEM DIC page 523 and page 543).

233. Explain the following terms i. CONSERVATION. ii. WILDLIFE. (WASSCE Biology July, 2003. Qn. 7(a) REF: STEM DIC page 299and page 300.

234. State the normal functions of the stem. (WASSCE Biology November, 2003. Qn. 1(a). REF: STEM DIC page 1227.

235. Describe stem modification which save different purposes giving one suitable example in each case. (WASSCE Biology November, 2003. Qn. 1(b). REF: STEM DIC page 1125 (Runner), page 1109(Rhizome).

236. Define the following terms. i. GENE. ii. CHIASMA. iii. MUTATION. (WASSCE Biology Nov., 2003. Qn. 2(a) REF: STEM DIC page 562 (Gene), page 242 (Chiasma/ chiasmata), page 858 (mutation).

237. Describe the activities that take place in the cell at the interphase of mitosis (WASSCE Biology November, 2003. Qn. 2(b) . REF: STEM DIC. page 681(interphase).

238. Compare mitosis and meiosis. (WASSCE Biology November, 2003. Qn. 2(c). REF: STEM DIC page 807 and page 833.

239. Describe the functional UNIT of the KIDNEY. (WASSCE Biology Nov., 2003. Qn. 3(a) REF: STEM DIC page 710).

240. State one function of each of the parts of the structure you have described in Qn. 3(a). (WASSCE Biology November, 2003. Qn. 3(b) REF: STEM DIC page 175 (Bowman's Capsule), page 766 (Loop of Henle), page 274 (Collecting Duct), page 576 (Glomerulus).

241. Define the following terms. i. BIOSPHERE. (WASSCE Biology November, 2003. Qn. 4(a) REF: STEM DIC page 157. ii. ECOSYSTEM (WASSCE Biology November, 2003. Qn. 4(a)(ii) REF: STEM DIC page 423). iii. ENVIRONMENT (WASSCE Biology November, 2003. Qn. 4(a)(iii). REF. STEM DIC page 461. iv. COMMUNITY. (WASSCE Biology November, 2003. Qn. 4(a)(iv). REF: STEM DIC page 281. v. HABITAT. (WASSCE Biology November, 2003. Qn. 4(a)(ii) REF: STEM DIC page 597.

242. Explain why it is necessary to classify living organisms. (WASSCE Biology November, 2003. Qn. 5(a). REF: STEM DIC page 257 (classification).

243. What are the main characteristic features of the phylum chordate. (WASSCE Biology November, 2003. Qn. 5(b). REF: STEM DIC page 247).

244. With a named example in each case, distinguish between the two main classes of fish. (WASSCE Biology November, 2003. Qn. 5(c). REF: STEM DIC page 247 (Chondrichthyes), page 925 (osteichthyes).

245. Explain the term WILDLIFE. (WASSCE Biology November, 2003. Qn. 6(a)(ii). REF: STEM DIC page 300.

246. Explain the following Phenomenon: i. ROOT PRESSURE (WASSCE Biology November, 2003. Qn. 7(a)(i) REF: STEM DIC. page 1119. ii. ACTIVE TRANSPORT (WASSCE Biology November, 2003. Qn. 7(a)(iii). REF: STEM DIC page 26).

247. Give a summary of the functions of the different parts of the human digestive system. (WASSCE Biology July, 2002. Qn. 1(a) REF: STEM DIC. page 851 (Mouth), page 974 (Pharynx), page 906 (oesophagus), page 1232 (stomach),

page 413 (Duodenum), page 553 (Gall Bladder), page 1182(small intestine), page 728 (Large intestine), page 1086 (Rectum), page 86 (Anus).

248. Explain the term co-ordination in mammals. (WASSCE Biology July, 2002. Qn. 4(a). REF: STEM DIC page 308).

249. Name the systems that bring about co-ordination in the mammals (WASSCE Biology July, 2002. Qn. 4(b). REF: STEM DIC page 454(Endocrine system), page 871 (Nervous system).

250. State the two main parts of the nervous system of mammals. (WASSCE Biology July, 2002. Qn. 4(c). REF: STEM DIC. page 227(Central Nervous System), page 967 (Peripheral Nervous system).

251. With the aid of a diagram, describe the structure of the motor neuron of a mammal. (WASSCE Biology July, 2002. Qn. 4(d). REF: STEM DIC. page 850.

252. Explain the following terms: i. GENOTYPES: (WASSCE Biology July, 2002. Qn. 5(a)(i) REF: STEM DI. page 566. ii. GENE: (WASSCE Biology July, 2002. Qn. 5(a)(ii). REF: STEM DIC page 562). iii. PHENOTYPES (WASSCE Biology July, 2002. Qn. 5(a)(iii). REF:STEM DIC page 977

253. List genetic factors that cause variation. (WASSCE Biology July, 2002. Qn. 5(b)(ii). REF: STEM DIC. page 466 (Epistasis), page 270 (Codominance), page 250 (Chromosome mutation), page 562(Gene Mutation), page 1151(Segregation), page 659 (Independent Assortment).

254. Name the cellular components of the HUMAN BLOOD. (WASSCE Biology July, 2002. Qn. 6(a)(i). REF: STEM DIC. page 66 (Erythrocytes), page 742 (Leucocytes), page 1002 (Platelets).

255. State the site of formation of each cellular component of the blood (WASSCE Biology July, 2002. Qn. 6(a)(ii). REF: STEM DIC. page 170 (Bone marrow-erythrocytes and leucocytes, Platelet), page 772 (lymphocyte – lymph node).

256. Explain the following terms i. Monoecious plant (WASSCE Biology July, 2002. Qn. 7(a)(i) REF: STEM DIC. page 388. ii.Protandrous flower (WASSCE Biology July, 2002. Qn. 7(a)(ii). REF: STEM DIC.page 1044. iii. Zygomorphic flower. (WASSCE Biology July, 2002. Qn. 7(a)(iii). REF: STEM DIC. page 1401.

257. Describe the parts of a complete flower. (WASSCE Biology July, 2002. Qn. 7(b). REF: STEM DIC. page 199 (calyx), page 312 (corolla), page 70 (Androecium), page 596 (Gynoecium).

258. Compare the features of entomophilous and anemophilous. (WASSCE Biology July, 2002. Qn. 7(c). REF: STEM DIC. page 388 (Entomophilous), page 71 (Anemophilous).

259. Explain the following terms: i. Receptor (WASSCE Biology November, 2002. Qn. 1(a)(i) REF: STEM DIC. page 1084. ii. Effector (WASSCE Biology November, 2002. Qn. 1(a)(ii) REF: STEM DIC. page 427. iii. Neurone (WASSCE Biology November, 2002. Qn. 1(a)(iii) REF: STEM DIC. page 874. iv. Synapse (WASSCE Biology November, 2002. Qn. 1(a)(iv) REF: STEM DIC. page 1260.

260. Distinguish between Voluntary action and Involuntary action (WASSCE Biology November, 2002. Qn. 1(b) REF: STEM DIC. page 1363 and page 686.

261. Compare NERVOUS AND ENDOCRINE co-ordination in a mammal (WASSCE Biology November, 2002. Qn. 1(c). REF: STEM DIC. page 871 and page 454

262. What is a DISEASE? (WASSCE Biology November, 2002. Qn. 3(a) REF: STEM DIC page 393.

263. Distinguish between the two main groups of diseases (WASSCE Biology November, 2002. Qn. 3(b) REF: STEM DIC. page 665 (infectious disease) and page 886 (Non-contagious)

264. For the disease schistosomiasis, state the following: i. the causative organism. (WASSCE Biology November, 2002. Qn. 3(c). REF: STEM DIC. page 1141.

265. What is meant by IMMUNITY. (WASSCE Biology November, 2002. Qn. 3(d). REF: STEM DIC. page 654.

266. Name the two main types of IMMUNITY. (WASSCE Biology November, 2002. Qn. 3(d)(ii). REF: STEM DIC. page 654.

267. Distinguish between biennials and perennials. (WASSCE Biology November, 2002. Qn. 4(a). REF: STEM DIC. page 149 and page 963.

268. Explain briefly the following terms. i. Asexual Reproduction. (WASSCE Biology November, 2002. Qn. 4(a)(i)REF: STEM DIC. page 101. ii. Vegetative Reproduction (WASSCE Biology November, 2002. Qn. 4(a)(ii). REF: STEM DIC. page 1350.

269. Define the term Ecology. (WASSCE Biology November, 2002. Qn. 5(a) REF: STEM DIC. page 423.

270. Define the following i. ARTERY (WASSCE Biology November, 2002. Qn. 6(a)(i) REF: STEM DIC. page 99. ii. CAPILLARY. (WASSCE Biology November, 2002. Qn. 6(a)(ii) REF: STEM DIC . page 203. iii. VEIN (WASSCE Biology November, 2002. Qn. 6(a)(iii) REF: STEM DIC. page 287.

271. Give a labeled diagram of the mammalian heart. (WASSCE Biology November, 2002. Qn. 6(b)(ii). REF: STEM DIC. page 609.

272. Explain each of the following terms: i. Isotonic Solution, (WASSCE Biology July, 2001. Qn. 1(a)(i) Ref: STEM Dictionary, page 696. ii. HYPOTONIC SOLUTION (WASSCE Biology July, 2001. Qn. 1(a)(ii) REF: STEM DIC, page 647. iii. HYPERTONIC SOLUTION (WASSCE Biology July, 2001. Qn. 1(a)(iii) REF: STEM DIC. page 645.

273. Name four organelles present in a typical animal cell. (WASSCE Biology July, 2001. Qn. 1(c)(i) REF: STEM DIC. page 1343 (vacuole), page 832 (Mitochondrion), page 1343 (Vacuole), page 246 (Chloroplast).

274. With the aid of a diagram, describe the structure of EUGLENA. (WASSCE Biology July, 2001. Qn. 2(a) (i) REF: STEM DIC, page 479).

275. Describe CONJUGATION in SPIROGYRA. (WASSCE Biology July, 2001. Qn. 2(a)(i). REF: STEM DIC. page 297).

276. Explain each of the terms: i. Food chain (WASSCE Biology July, 2001. Qn. 2(a)(i) REF: STEM DIC. page 528. ii. Food Web (WASSCE Biology July, 2001 Qn. 2(a)(ii) REF: STEM DIC. page 529). iii. PYRAMID OF BIOMASS (WASSCE Biology July, 2001. Qn. 2(a)(iii) REF: STEM DIC. page 1054 iv. DECOMPOSERS (WASSCE Biology July, 2001. Qn. 3(a)(iii) REF: STEM DIC. page 352). v. PYRAMID OF NUMBERS (WASSCE Biology July, 2001. Qn. 3(a)(ii) REF: STEM DIC. page 117).

277. State five characteristic features of reptiles. (WASSCE Biology July, 2001. Qn. 4(a)(i) REF: STEM DIC. page 1100).

278. Distinguish between AMPHIBIANS and REPTILES [(WASSCE Biology July, 2001. Qn. 4(a)(ii). REF: STEM DIC. page 64 (Amphibia) and page 1100 (Reptile).

279. Describe the structure of the following parts of the mammalian eye i. SCLEROTIC LAYER (WASSCE Biology July, 2001. Qn. 6(a)(i) REF: STEM DIC. page 1143. ii. CHOROID (WASSCE Biology July, 2001. Qn. 6(a)(ii) REF: STEM DIC. page 247. iii. RETINA (WASSCE Biology July, 2001. Qn. 6(a)(iii) REF: STEM DIC page 1106).

280. What is SHORT SIGHT? (WASSCE Biology July, 2001. Qn. 6(b)(i) REF: STEM DC. page 1166 (shortsighted).

281. Explain the term EVOLUTION (WASSCE Biology July, 2001. Qn. 7(a)REF: STEM DIC. page 483.

282. Distinguish between Convergent Evolution and Divergent Evolution (WASSCE Biology July, 2001. Qn. 7(b) REF: STEM DIC. page 306 and page 399.

283. Describe the two girdles found in the mammal skeleton. (WASSCE Biology November, 2001. Qn. 1(a)(i) REF: STEM DIC. page 959 (Pelvic girdle), page 121 (Pectoral girdle).

284. Explain briefly the following terms: i. ECDYSIS (WASSCE Biology November, 2001. Qn. 2(a)(i) REF: STEM DIC. page 420). ii. COMPLETE METAMORPHOSIS (WASSCE Biology November, 2001. Qn. 2(a)(ii) REF: STEM DIC. page 284). iii. INCOMPLETE METAMORPHOSIS (WASSCE Biology November, 2001. Qn. 2(a)(iii) REF: STEM DIC. page 658).

285. Distinguish between TISSUE RESPIRATION and GASEOUS EXCHANGE (WASSCE Biology November, 2001. Qn. 2(a) REF: STEM DIC. page 1296 (TISSUE RESPIRATION), page 558 (GASEOUS EXCHANGE).

286. What are the main stages in GLYCOLYSIS? (WASSCE Biology November, 2001. Qn. 3(b)(i) REF: STEM DIC. page 578).

287. What is the importance of the KREBS CYCLE? (WASSCE Biology November, 2001. Qn. 3(b)(ii) REF: STEM DIC. page 716).

288. List three atmospheric pollutants and i. State their sources ii. their effects on human life (WASSCE Biology November, 2001. Qn. 4(a)(i)(ii) REF: STEM DIC. page 1247 (SO_2), page 206, (CO_2), page 48 (NO, Nitrogen oxides).

289. Describe the girdles found in the mammalian skeleton (WASSCE Biology July, 2000. Qn. 1(a)(i) REF: STEM DIC. page 959 (Pelvic girdle), page 121 (Pectoral girdle).

290. Explain briefly the following terms i. COMPLETE METAMORPHOSIS (WASSCE Biology July, 2000. Qn. 2(a)(ii) REF: STEM DIC. page 284.

291. (ii) INCOMPLETE METAMORPHOSIS. (WASSCE Biology July, 2000. Qn. 2(a)(ii) REF: STEM DIC. page 658).

292. Name the causative organism of MALARIA (WASSCE Biology July, 2000. Qn. 4(a)(i). REF: STEM DIC. page 786).

293. Distinguish between continuous variation and discontinuous variation, citing two examples of each. (WASSCE Biology July, 2000. Qn. 5(a)(ii) REF: STEM DIC. page 304 (Continuous) and page 393 (Discontinuous variation). REF: STEM DIC. page 1151 (Segregation), page 858 (Mutation), page 117 (Recombination).

294. Name two CARNIVOROUS PLANTS. [WASSCE Biology July, 2000. Qn. 7(b)(i) REF: STEM DIC. page 211.

CHEMISTRY QUESTIONS FROM THE WEST AFRICAN SENIOR SCHOOL CERTIFICATE EXAMINATIONS (W.A.S.S.CE.) AND STEM DICTIONARY REFERENCE PAGES

1. Differentiate between ELECTRON AFFINITY and ELECTRO NEGATIVITY. (WASSCE –Chemistry JUNE 2019 Q1.b Ref: STEM Dic. page 438 & page 441).

2. Define FARADAY'S CONSTANT. (WASSCE –Chemistry JUNE 2019 Q1.c[i] Ref: STEM Dic. page 501).

3. State the differences between a DEHYDRATING AGENT and DRYING AGENT. (WASSCE –Chemistry JUNE 2019 Q1.g Ref: STEM Dic. page 357 & page 412).

4. Distinguish between a PRIMARY CELL and SECONDARY CELL (WASSCE –Chemistry JUNE 2019 Q2.c[i] Ref: STEM Dic. page 1031 & page 1147)

5. Define SATURATION SOLUTION. (WASSCE –Chemistry JUNE 2019 Q3. Ref: STEM Dic. page 1137).

6. State three CONSTITUENT of CEMENT (WASSCE –Chemistry JUNE 2019 Q5.d Ref: STEM Dic. page 226).

7. What is a CATALYST? (WASSCE –Chemistry NOV 2019 Q2.a[i] Ref: STEM Dic. page 216).

8. State one advantage of SYNTHETIC POLYMERS over NATURAL POLYMERS. (WASSCE –Chemistry NOV 2019 Q1.b Ref: STEM Dic. page 1014).

9. Name one metal that is extracted by ELECTROLYSIS. (WASSCE –Chemistry NOV 2019 Q2.c[i] Ref: STEM Dic. page 436 (Electrolysis – Metal - Aluminum). Others include reactive metals such as sodium, potassium and calcium.

10. Distinguish between a HYGROSCOPIC COMPOUND and a DELIQUESCENT COMPOUND. (WASSCE – Chemistry NOV 2019 Q3.b[i] Ref: STEM Dic. page 642 (hygroscopic compound) & page 358 (deliquescent).

11. Explain briefly what is meant by the term SOLUBILITY PRODUCT (WASSCE –Chemistry NOV 2019 Q4.a[i] Ref: STEM Dic. Page page 48).

12. Define each of the following terms: i. BRONSTED-LOWRY ACID (WASSCE –Chemistry NOV 2019 Q4.b[i] Ref: STEM Dic. page 181) ii. LEWIS ACID: (WASSCE –Chemistry NOV 2019 Q4.a [ii] Ref: STEM Dic. page 744).

13. What is POLAR COVALENT BOND? (WASSCE –Chemistry NOV 2019 Q4.d[i] Ref: STEM Dic. page 1009).

14. What is a FUNCTIONAL GROUP (WASSCE –Chemistry JUNE 2018 Q1.a Ref: STEM Dic. page 546)

15. Define each of the following terms: ALIPHATIC COMPOUND (WASSCE – Chemistry JUNE 2018 Q1. b[i] Ref: STEM Dic. page 49)

16. AROMATIC COMPOUND (WASSCE –Chemistry JUNE 2018 Q1.b [ii].a Ref: STEM Dic. page 97).

17. Outline the principles involved in PARTITION CHROMATOGRAPHY (WASSCE –Chemistry JUNE 2018 Q1.d[i] Ref: STEM Dic. page 952).

18. State two functions of a SALT-BRIGDE in an electrochemical cell (WASSCE –Chemistry JUNE 2018 Q1.h Ref: STEM Dic. page 1133).

19. Explain briefly the term END-POINT as used in titration (WASSCE –Chemistry JUNE 2018 Q3.e Ref: STEM Dic. page 457).
20. Define the term HYDRATION ENERGY. (WASSCE –Chemistry JUNE 2018 Q5.c[i] Ref: STEM Dic. page 636).
21. What is GEOMETRIC ISOMERISM? (WASSCE –Chemistry JUNE 2018 Q5.d[i] Ref: STEM Dic. page 568)
22. What is an ACID – BASE INDICATOR? (WASSCE –Chemistry JUNE 2017 Q1.a[i] Ref: STEM Dic. page 19).
23. Give one example of ACID-BASE INDICATOR? (WASSCE –Chemistry JUNE 2017 Q5. a [ii] Ref: STEM Dic. page 19).
24. Define the term MOLE. (WASSCE –Chemistry JUNE 2017 Q1.e Ref: STEM Dic. page 838).
25. State the PERIODIC LAW. (WASSCE –Chemistry JUNE 2017 Q2.a [i] Ref: STEM Dic. page 965).
26. Define RELATIVE ATOMIC MASS. (WASSCE –Chemistry JUNE 2017 Q2.b Ref: STEM Dic. page 1094).
27. Define HYBRIDIZATION. (WASSCE –Chemistry JUNE 2017 Q4.a [i] Ref: STEM Dic. page 636).
28. State FARADAYS FIRST LAW OF ELECTROLYSIS. (WASSCE –Chemistry JUNE 2017 Q4.b [i] Ref: STEM Dic. page 501).
29. Explain briefly the term HOMOLYTIC FISSION and give an example (WASSCE –Chemistry JUNE 2017 Q4.e Ref: STEM Dic. page 628).
30. What is the nature of the following radiations: i. ALPHA: (WASSCE –Chemistry JUNE 2017 Q5. a [i] Ref: STEM Dic. page 54) ii. BETA: (WASSCE –Chemistry JUNE 2017 Q5.a [ii] Ref: STEM Dic. page 147) iii. GAMMA: (WASSCE –Chemistry JUNE 2017 Q5.a [iii] Ref: STEM Dic. page 555).
31. What is STANDARD HYDROGEN ELECTRODE? (WASSCE –Chemistry NOV 2017 Q1. a Ref: STEM Dic. page 1221).
32. What is CHEMICAL CHANGE? (WASSCE –Chemistry NOV 2017 Q1.d[i] Ref: STEM Dic. page 238).
33. What are ISOTOPES? (WASSCE –Chemistry NOV 2017 Q1.e Ref: STEM Dic. page 696).
34. Define the term pH. (WASSCE –Chemistry NOV 2017 Q1.g Ref: STEM Dic. page 973).
35. State two postulates of the KINETIC THEORY OF GASES. (WASSCE –Chemistry NOV 2017 Q4.b[i] Ref: STEM Dic. page 713).
36. What are NUCLEONS? (WASSCE –Chemistry JUNE 2016 Q1.a Ref: STEM Dic. page 896).
37. State GRAHAM'S LAW OF DIFFUSION? (WASSCE –Chemistry JUNE 2016 Q1.b Ref: STEM Dic. page 583).
38. Define STRUCTURAL ISOMERISM. (WASSCE –Chemistry JUNE 2016 Q1.a[i] Ref: STEM Dic. page 1240).
39. Define the term SOLUBILITY. (WASSCE –Chemistry JUNE 2016 Q4.a Ref: STEM Dic. page 1195).
40. Define NUCLEAR FUSION: (WASSCE –Chemistry JUNE 2016 Q4.d[i] Ref: STEM Dic. page 894).
41. What is a BUFFER SOLUTION? (WASSCE –Chemistry JUNE 2016 Q5.a[i] Ref: STEM Dic. page 184).
42. State LE CHATELIER'S PRINCIPLE as it applies to chemical equation. (WASSCE –Chemistry NOV 2016 Q1.d[i] Ref: STEM Dic. page 734)
43. What is DYNAMIC EQUILIBRIM? (WASSCE –Chemistry NOV 2016 Q1.e Ref: STEM Dic. page 415).
44. State the AUFBAU PRINCIPLE. (WASSCE –Chemistry NOV 2016 Q1. [ii] Ref: STEM Dic. page 113).
45. What are VAN DER WAALS FORCES. (WASSCE –Chemistry NOV 2016 Q2.a [i] Ref: STEM Dic. page 1345).
46. Explain briefly each of the following separation techniques: i. CRYSTALLIZATION: (WASSCE –Chemistry NOV 2016 Q3.a [i] Ref: STEM Dic. page 328) ii. EVAPORATION: (WASSCE –Chemistry NOV 2016 Q3.a [ii] Ref: STEM Dic. page 482). iii. FILTRATION: (WASSCE –Chemistry NOV 2016 Q3.a [iii] Ref: STEM Dic. page 513).
47. Define the following terms: i. ATOMIC NUMBER: (WASSCE – Chemistry NOV 2016 Q3.e[i] Ref: STEM Dic. page 110). ii. MASS NUMBER: (WASSCE –Chemistry NOV 2016 Q3.e [ii] Ref: STEM Dic. page 796).
48. Define ESTERIFICATION. (WASSCE –Chemistry JUNE 2015 Q1.a Ref: STEM Dic. page 475 (Ester).
49. State CHARLE'S LAW. (WASSCE –Chemistry JUNE 2015 Q1.e [i] Ref: STEM Dic. page 236).
50. Explain briefly the term ATOMIC ORBITAL. (WASSCE – Chemistry JUNE 2015 Q2.bi [i] Ref: STEM Dic. page 110).
51. Explain briefly an EQUILIBRIM REACTION. (WASSCE – Chemistry JUNE 2015 Q5.c [i] Ref: STEM Dic. page 469).
52. Define BRONSTED-LOWRY BASE. (WASSCE –Chemistry JUNE 2015 Q4.b [i] Ref: STEM Dic. page 181).
53. Define the term HYDROCARBONS. (WASSCE –Chemistry NOV 2015 Q1.a Ref: STEM Dic. page 637).
54. Define OXIDATION in terms of ELECTRON TRANSFER. (WASSCE –Chemistry NOV 2015 Q1.c Ref: STEM Dic. page 930).
55. State DALTON'S LAW OF PARTIAL PRESSURES. (WASSCE –Chemistry NOV 2015 Q4. Ref: STEM Dic. page 343).
56. What are ISOMERS. (WASSCE –Chemistry NOV 2015 QI.[i] Ref: STEM Dic. page 694).
57. What are GALVANIC CELLS? (WASSCE –Chemistry NOV 2015 Q1.j Ref: STEM Dic. page 554).
58. Define the term ELECTROLYSIS. (WASSCE –Chemistry NOV 2015 Q4.a Ref: STEM Dic. page 436).
59. Define Rate of Reaction. (WASSCE –Chemistry NOV 2015 Q4.d[i] Ref: STEM Dic. page 1079).
60. Define the term HYDROLYSIS. (WASSCE –Chemistry NOV 2015 Q5.a[i] Ref: STEM Dic. page 640).
61. Define LEWIS ACID. (WASSCE –Chemistry NOV 2015 Q5.c[i] Ref: STEM Dic. page 744).
62. Define IONIC BOND. (WASSCE –Chemistry JUNE 2014 Q1.a[i] Ref: STEM Dic. page 689).
63. State two differences between HYDRATION and HYDROLYSIS (WASSCE –Chemistry JUNE 2014 Q4.a Ref: STEM Dic. page 636 & page 640)
64. Explain briefly the term TITRATION (WASSCE –Chemistry JUNE 2014 Q4. b[i] Ref: STEM Dic. page 1296).
65. What is STANDARD SOLUTION? (WASSCE –Chemistry JUNE 2014 Q4. d[i] Ref: STEM Dic. page 1222).
66. Define the term, FAT. (WASSCE –Chemistry JUNE 2014 Q5.b[i] Ref: STEM Dic. page 502).

67. State HUNDS RULE OF MAXIMUM MULTIPLICITY (WASSCE –Chemistry NOV 2014 Q1.b Ref: STEM Dic. page 634).

68. Define each of the following terms i. BIOTECHNOLOGY (WASSCE –Chemistry NOV 2014 Q1.j[i] Ref: STEM Dic. page 158). ii. FERMENTATION (WASSCE –Chemistry 2014 Q1.j [ii] Ref: STEM Dic. page 506).

69. What are CATHODE RAYS? (WASSCE –Chemistry NOV 2014 Q2.b[i] Ref: STEM Dic. page 218).

70. Define each of the following terms: i. HEAT OF NEUTRALISATION (WASSCE –Chemistry NOV 2014 Q5.a[i] Ref: STEM Dic. page 610). ii. EXOTHERMIC REACTION (WASSCE –Chemistry NOV 2014 Q5.b[i] Ref: STEM Dic. page 488)

71. What is BASICITY of an acid? (WASSCE –Chemistry NOV 2014 Q5.b[i] Ref: STEM Dic. page 136).

72. Explain briefly each of the following terms: i. HOMOLYTIC FISSION: (WASSCE –Chemistry JUNE 2013 Q1.d[i] Ref: STEM Dic. page 628) ii. HETEROLYTIC FISSION (WASSCE –Chemistry JUNE 2013 Q1.d [ii] Ref: STEM Dic. page 121). iii. FREE RADICALS (WASSCE –Chemistry JUNE 2013 Q1.d [i] Ref: STEM Dic. page 539).

73. Define each of the following terms: i. CLOSED SYSTEM (WASSCE –Chemistry JUNE 2013 Q2.a [i] Ref: STEM Dic.page 264). ii. ENDOTHERMIC REACTION (WASSCE –Chemistry JUNE 2013 Q2.a [ii] Ref: STEM Dic. page 456). iii. HEAT OF NEUTRALISATION: (WASSCE –Chemistry JUNE 2013 Q2.a [iii] Ref: STEM Dic. page 610).

74. What is an amphoteric oxide? (WASSCE –Chemistry JUNE 2013 Q3.a [i] Ref: STEM Dic. page 65).

75. Explain briefly each of the following terms: i. POLYMER: (WASSCE –Chemistry JUNE 2013 Q4.a [i] Ref: STEM Dic. page 1014). ii. POLYMERIZATION: (WASSCE –Chemistry JUNE 2013 Q4.a [i] Ref: STEM Dic. page 1015).

76. Define each of the following terms i. RATE OF REACTIONS (WASSCE –Chemistry JUNE 2013 Q5.a [i] Ref: STEM Dic. page 1079). ii. RATE CONSTANT (WASSCE –Chemistry JUNE 2013 Q5.a [i] Ref: STEM Dic. page 1078). iii. RATE – DETERMINE STEP (WASSCE –Chemistry JUNE 2013 Q5.a [i] Ref: STEM Dic. page 1079).

77. State DALTONS LAW OF PARTIAL PRESSURES (WASSCE –Chemistry JUNE 2013 Q6.c [i] Ref: STEM Dic. page 343).

78. Name the components of the following alloys i. BRONZE (WASSCE –Chemistry NOV 2013 Q2.c [i] Ref: STEM Dic. page 181). ii. DURALUMIN (WASSCE –Chemistry NOV 2013 Q2.c [ii] Ref: STEM Dic. page 413). iii. STEEL (WASSCE –Chemistry NOV 2013 Q2.c [iii] Ref: STEM Dic. page 1227).

79. Explain briefly each of the following terms as applied in the petroleum industry i. CRACKING (WASSCE –Chemistry NOV 2013 Q3.a [i] Ref: STEM Dic. page 319). ii. REFORMING (WASSCE –Chemistry NOV 2013 Q3.a [ii] Ref: STEM Dic. page 1091). iii. OCTANE NUMBER (WASSCE –Chemistry NOV 2013 Q3.c [iii] Ref: STEM Dic. page 904).

80. Define each of the following terms: i. EFFLORESCENCE (WASSCE –Chemistry NOV 2013 Q6.a [i] Ref: STEM Dic. page 427). ii. DELIQUESCENCE: (WASSCE –Chemistry NOV 2013 Q6.a [ii] Ref: STEM Dic. page 358).

81. What is an ACID ANHYDRIDE (WASSCE – Chemistry NOV 2013 Q6.c [i] Ref: STEM Dic. page 18).

82. Define ELECTRONEGATIVITY (WASSCE –Chemistry JUNE 2012 Q1.a [i] Ref: STEM Dic. page 441).

83. Define ISOTOPE (WASSCE –Chemistry JUNE 2012 Q1.c [i] Ref: STEM Dic. page 696).

84. Define the term SOLUBILITY. (WASSCE – Chemistry JUNE 2012 Q2.a [i] Ref: STEM Dic. page 1195).

85. What is an ACID-BASE INDICATOR? (WASSCE –Chemistry 2 JUNE 012 Q2. b [i] Ref: STEM Dic. page 19).

86. Define HYBRIDIZATION (WASSCE – Chemistry JUNE 2012 Q3.b [i] Ref: STEM Dic. page 636 (Hybridisation).

87. Describe briefly how each of the following bonds are formed i. DATIVE BOND (WASSCE –Chemistry JUNE 2012 Q4.a [i] Ref: STEM Dic. page 346). ii. METALLIC BOND (WASSCE –Chemistry JUNE 2012 Q4.a [i] Ref: STEM Dic. page 816).

88. Explain briefly the term CHEMICAL EQUILIBRIM (WASSCE –Chemistry NOV 2012 Q2. a [i] Ref: STEM Dic. page 239).

89. What is an electrolyte (WASSCE –Chemistry NOV 2012 Q4.a [i] Ref: STEM Dic. page 436).

90. Explain briefly each of the following terms: i. ORDER OF REACTION (WASSCE –Chemistry NOV 2012 Q5.a [i] Ref: STEM Dic. page 919). ii. RATE LAW (WASSCE –Chemistry NOV 2012 Q5.a [ii] Ref: STEM Dic. page 1078 (Rate Equation). iii. ACTIVATION ENERGY (WASSCE –Chemistry NOV 2012 Q5.a [iii] Ref: STEM Dic. page 24).

91. Define the term SAPONIFICATION (WASSCE –Chemistry NOV 2012 Q6.a [i] Ref: STEM Dic. page 1136).

92. Explain the following terms: i. AVOGADRO'S CONSTANT (WASSCE –Chemistry JUNE 2011 Q2.a [i] Ref: STEM Dic. page 119). ii. THE MOLE (WASSCE –Chemistry JUNE2011 Q2.a [ii] Ref: STEM Dic. page 838). iii. ISOPTOPES (WASSCE –Chemistry JUNE 2011 Q2.a [iii] Ref: STEM Dic. page 696).

93. Define each of the following and indicate one use of each i. NUCLEAR FISSION (WASSCE –Chemistry JUNE 2011 Q1.a [i] Ref: STEM Dic. page 893. ii. NUCLEAR FUSION (WASSCE –Chemistry JUNE 2011 Q1.a [i] Ref: STEM Dic. page 894). iii. HALF – LIFE (WASSCE –Chemistry JUNE 2011 Q1.a [iii] Ref: STEM Dic. page 600).

94. What do you understand by ORDER OF A REACTION? (WASSCE –Chemistry JUNE 2011 Q5.a [i] Ref: STEM Dic. page 919).

95. What is a POLLUTANT? (WASSCE –Chemistry NOV 2011 Q2.a [i] Ref: STEM Dic. page 1012).

96. What is CRACKING? (WASSCE –Chemistry NOV 2011 Q2.e Ref: STEM Dic. page 319).

97. What is meant by RATE OF A CHEMICAL REACTION? (WASSCE –Chemistry NOV 2011 Q3. a [i] Ref: STEM Dic. page 1079).

98. What are CARBOHYDRATES? (WASSCE –Chemistry NOV 2011 Q4.a [i] Ref: STEM Dic. page 205).

99. What is a TRANSITION ELEMENT? (WASSCE –Chemistry NOV 2011 Q6.a [i] Ref: STEM Dic. page 1307).

100. Define the term mass number. (WASSCE – Chemistry NOV. 2010 Q1.a [i] Ref: STEM Dic. page 796).

101. Define the term NATURAL RADIOACTIVITY. (WASSCE – Chemistry NOV 2010 Q1.c [i] Ref: STEM Dic. page 1072).

102. Define the term BOND DISSOCIATION ENERGY. (WASSCE –Chemistry NOV 2010 Q6. a [i] Ref: STEM Dic. page 169).

103. Define LATTICE ENERGY (WASSCE – Chemistry NOV 2010 Q6.c [i] Ref: STEM Dic. page 731).

104. Define the term: i. RADIOACTITVITY (WASSCE –Chemistry JUNE 2009 Q1.a [i] Ref: STEM Dic. page 1072). ii. HALF – LIFE (WASSCE –Chemistry JUNE 2009 Q1.a [i] Ref: STEM Dic. page 600).

105. Define FERMENTATION (WASSCE –Chemistry JUNE 2009 Q2.a [i] Ref: STEM Dic. page 506).

106. What are POLYMERS? (WASSCE –Chemistry JUNE 2009 Q2.b [i] Ref: STEM Dic. page 1014).

107. What is WATER OF CRYSTALLIZATION? (WASSCE –Chemistry JUNE 2009 Q4. a [i] Ref: STEM Dic. page 328).

108. Define each of the following terms: i. POLLUTION (WASSCE –Chemistry JUNE 2009 Q4.c [i] Ref: STEM Dic. page 1012). ii. POLLUTANT (WASSCE –Chemistry JUNE 2009 Q4.c [i] Ref: STEM Dic. page 1012).

109. Define: i BRONSTED-LOWRY ACID. (WASSCE –Chemistry JUNE 2009 Q5.a [i] Ref: STEM Dic. page 181) ii. LEWIS ACID (WASSCE –Chemistry JUNE 2009 Q5. a [i] Ref: STEM Dic. page 744).

110. What is meant by CONJUGATE ACID-BASE PAIR (WASSCE –Chemistry JUNE 2009 Q5. a [ii] Ref: STEM Dic. page 297).

111. What is LATTICE ENERGY (WASSCE –Chemistry JUNE 2009 Q5.c [i] Ref: STEM Dic. page 731).

112. What is ISOMERISM? (WASSCE –Chemistry NOV 2009 Q1.a [i] Ref: STEM Dic. page 695).

113. What is GALVANIC CELL? (WASSCE –Chemistry NOV 2009 Q3.a [i] Ref: STEM Dic. page 554).

114. Define each of the following terms: i. AN ION (WASSCE – Chemistry NOV 2009 Q4.a [i] Ref: STEM Dic. page 687) ii. ATOMICITY (WASSCE – Chemistry NOV 2009 Q4.a [iii] Ref: STEM Dic. page 111).

115. What is BOND ENERGY? (WASSCE –Chemistry NOV 2009 Q5.b [i] Ref: STEM Dic. page 169).

116. Define the term RADIOACTIVITY? (WASSCE –Chemistry JUNE 2008 Q1. a [i] Ref: STEM Dic. page 1072).

117. What are CARBOHYDRATES? (WASSCE –Chemistry JUNE 2008 Q2.a [i] Ref: STEM Dic. page 205).

118. Define COVALENT BOND (WASSCE –Chemistry JUNE 2008 Q1.a [i] Ref: STEM Dic. page 318).

119. State GRAHAM'S LAW OF DIFFUSION (WASSCE –Chemistry JUNE 2008 Q3. a [i] Ref: STEM Dic. page 583).

120. Define the terms: i. ACTIVATION COMPLEX (WASSCE – Chemistry JUNE 2008 Q4.c [i] Ref: STEM Dic. page 24 (Activated Complex). ii. ELECTRON AFFINITY (WASSCE –Chemistry JUNE 2008 Q4.c [i] Ref: STEM Dic. page 438).

121. Define the term RATE OF REACTION (WASSCE –Chemistry JUNE 2008 Q5.a Ref: STEM Dic. page 1079).

122. Define the term HYBRIDIZATION (WASSCE –Chemistry JUNE 2008 Q6.a [i] Ref: STEM Dic. page 636).

123. What is meant by a FUNCTIONAL GROUP? (WASSCE – Chemistry NOV 2008 Q1. a Ref: STEM Dic. page 546).

124. What is meant by each of the following terms i. STANDARD ELECTRODE POTENTIAL (WASSCE –Chemistry NOV 2008 Q2.a Ref: STEM Dic. page 1220). ii. STANDARD HYDROGEN ELECTRODE (WASSCE –Chemistry NOV 2008 Q2. a [i] Ref: STEM Dic. page 1221).

125. Define the term ELECTRON AFFINITY (WASSCE –Chemistry NOV 2008 Q1.a[i] Ref: STEM Dic. page 438).

126. Define the term HYBRIDIZATION (WASSCE – Chemistry NOV 2008 Q5.b[i] Ref: STEM Dic. page 636).

127. What are TRANSITION ELEMENTS (WASSCE –Chemistry NOV 2008 Q6. a Ref: STEM Dic. page 1307).

128. Define in terms of electron transfer: i. OXIDATION (WASSCE –Chemistry JUNE 2007 Q1.a [ii] Ref: STEM Dic. page 930). ii. REDUCTION (WASSCE –Chemistry JUNE 2007 Q1.a [ii] Ref: STEM Dic. page 1089).

129. Define the following terms: i. SATURATION SOLUTION (WASSCE –Chemistry JUNE 2007 Q2.a [i] Ref: STEM Dic. Page page 121 (Saturation). ii. STANDARDIZED SOLUTION (WASSCE – Chemistry JUNE 2007 Q2.a [ii] Ref: STEM Dic. page 1222 (Standard Solution).

130. What are ISOTOPES? (WASSCE –Chemistry JUNE 2007 Q3.a [i] Ref: STEM Dic. page 696).

131. Define the term ISOMERISM (WASSCE –Chemistry JUNE 2007 Q4.a [i] Ref: STEM Dic. page 695).

132. What is a: i. BRONSTED–LOWRY ACID (WASSCE –Chemistry JUNE 2007 Q5.a [i] Ref: STEM Dic. page 181). ii. BRONSTED – LOWRY BASE (WASSCE –Chemistry JUNE 2007 Q5.a [i] Ref: STEM Dic. page 181).

133. State HESS' LAW (WASSCE –Chemistry JUNE 2007 Q5.b [i] Ref: STEM Dic. page 617).

134. Define the term BUFFER SOLUTION (WASSCE –Chemistry JUNE 2007 Q6. a [i] Ref: STEM Dic. page 184).

135. Define the term IONIZATION ENERGY (WASSCE –Chemistry JUNE 2007 Q6. d [i] Ref: STEM Dic. page 689).

136. Define the term HYDROCARBON (WASSCE –Chemistry NOV 2007 Q1.a [i] Ref: STEM Dic. page 637).

137. What is meant by the following terms: i. SAPONIFICATION (WASSCE –Chemistry NOV 2007 Q1.d [i] Ref: STEM Dic. page 1136). ii. ESTERIFICATION (WASSCE –Chemistry NOV 2007 Q1.a [i] Ref: STEM Dic. page 475).

138. What is meant by each of the following terms: i. ACID-BASE INDICATOR (WASSCE –Chemistry NOV 2007 Q4.a [i] Ref: STEM Dic. page 19). ii. pH (WASSCE –Chemistry NOV 2007 Q4.a [ii] Ref: STEM Dic. page 973).

139. Explain the term STANDARD ELECTRODE POTENTIAL (WASSCE –Chemistry NOV 2007 Q6.a [i] Ref: STEM Dic. page 1220).

140. Define the term SOLUBILITY (WASSCE –Chemistry JUNE 2006 Q2.d [i] Ref: STEM Dic. page 1195).

141. Define STRUCTURAL ISOMERISM (WASSCE –Chemistry NOV 2006 Q3. a Ref: STEM Dic. page 1240).

142. Define each of the following terms: i. ISOTOPES: (WASSCE –Chemistry NOV 2006 Q1.a [i] Ref: STEM Dic. page 696). ii. LATTICE ENERGY: (WASSCE –Chemistry NOV 2006 Q1.a [ii] Ref: STEM Dic. page 731).

143. Define the term ACTIVATION ENERGY: (WASSCE –Chemistry NOV 2006 Q2. a [i] Ref: STEM Dic. page 24).

144. Define each of the following terms i. ORDER OF REACTION (WASSCE –Chemistry NOV 2006 Q3.a [i] Ref: STEM Dic. page 919). ii. RATE LAW: (WASSCE –Chemistry NOV 2006 Q3.a [ii] Ref: STEM Dic. page 1078).

145. Define the terms: i. CONJUGATE ACID (WASSCE –Chemistry NOV 2006 Q5.a [i] Ref: STEM Dic. page 297). ii. CONJUGATE BASE (WASSCE –Chemistry NOV 2006 Q5.a [ii] Ref: STEM Dic. page 297).

146. Explain each of the following terms: i. ELECTRONEGATIVITY (WASSCE –Chemistry NOV 2006 Q4.a [i] Ref: STEM Dic. page 441). ii. ELECTRONCHEMICAL SERIES (WASSCE –Chemistry NOV 2006 Q4. a [ii] Ref: STEM Dic. page 435).

147. What is BIOTECHNOLOGY (WASSCE –Chemistry NOV 2006 Q1.c [i] Ref: STEM Dic. page 158).

148. What is: REDOX REACTION (WASSCE –Chemistry NOV 2005 Q1.a [i] Ref: STEM Dic. page 1088).

149. SOLUBILITY (WASSCE –Chemistry NOV 2005 Q1. a [ii] Ref: STEM Dic. page 1195).

150. WATER OF CRYSTALLIZATION. (WASSCE –Chemistry NOV 2005 Q1. a [iii] Ref: STEM Dic. page 328).

151. State three postulates of the COLLISION THEORY of REACTION RATES (WASSCE –Chemistry NOV 2005 Q2.a [i] Ref: STEM Dic. page 275).

152. Define acid and bases according to the i. BRONSTED LOWRY THEORY (WASSCE –Chemistry NOV 2005 Q4.a [i] Ref: STEM Dic. page 181). ii. LEWIS THEORY (WASSCE –Chemistry NOV 2005 Q4.a [ii] Ref: STEM Dic. page 744).

153. Define the following terms: i. CATALYTIC CRACKING: (WASSCE –Chemistry NOV 2005 Q2. a [i] Ref: STE.M Dic. page 319 (cracking). ii. POLYMERIZATION: (WASSCE –Chemistry NOV 2005 Q1.a [i] Ref: STEM Dic. page 1015). iii. SAPONIFICATION (WASSCE –Chemistry NOV 2005 Q2.a [i] Ref: STEM Dic. page 1136).

154. Explain the following terms using electron transfer: i. OXIDATION (WASSCE –Chemistry NOV 2005 Q4.a [i] Ref: STEM Dic. page 930). ii. REDUCTION (WASSCE –Chemistry NOV 2005 Q4.a [ii] Ref: STEM Dic. page 1089).

155. What is meant by the HALF-LIFE OF A RADIOISOTOPE? (WASSCE –Chemistry NOV 2005 Q7.a Ref: STEM Dic. page 600.

156. What is HALF-LIFE OF RADIOACTIVE SUBSTANCE (WASSCE –Chemistry JULY 2004 Q1.d [i] Ref: STEM Dic. Page 564).

157. What is LEWIS ACID (WASSCE –Chemistry JULY 2004 Q1. a [i] Ref: STEM Dic. page 744)

158. What is the difference between HYDRATION and HYDROLYSIS? (WASSCE –Chemistry JULY 2004 Q5.a [i] Ref: STEM Dic. page 636 & page 640).

159. State one use of AMMONIA (WASSCE –Chemistry JULY 2004 Q6.c [ii] Ref: STEM Dic. page 62).

160. Define the following term: i. NUCLEONS (WASSCE –Chemistry NOV 2004 Q1.a [i] Ref: STEM Dic. page 896). ii. NUCLIDE: (WASSCE –Chemistry NOV 2004 Q1.a [ii] Ref: STEM Dic. page 897). iii. MOLE (WASSCE –Chemistry NOV 2004 Q1.a [iii] Ref: STEM Dic. page 838).

161. What is FIRST IONIZATION ENERGY of an atom (WASSCE –Chemistry NOV 2004 Q2.a [i] Ref: STEM Dic. page 515).

162. Explain the following terms and give one example of each: i. OXIDIZING AGENT (WASSCE –Chemistry NOV 2004 Q13.a [i] Ref: STEM Dic. page 931). ii. REDUCING AGENT (WASSCE –Chemistry NOV 2004 Q3.a [ii] Ref: STEM Dic. page 1089).

163. State DALTON'S LAW OF PARTIAL PRESSURES (WASSCE –Chemistry NOV 2004 Q4. a [i] Ref: STEM Dic. page 343).

164. What is a TRANSITION ELEMENT (WASSCE –Chemistry NOV 2004 Q1.a [i] Ref: STEM Dic. page 1307).

165. WHAT IS AN EQUILIBRIM REACTION (WASSCE –Chemistry JULY 2003 Q4.b [i] Ref: STEM Dic. page 469).

166. Define each of the following terms: i. LATTICE ENERGY (WASSCE –Chemistry JULY 2003 Q6.b [i] Ref: STEM Dic. page 731). ii. SATURATED SOLUTION (WASSCE –Chemistry JULY 2003 Q4.b [ii] Ref: STEM Dic. page 121 (Saturation).

167. State three differences between PI-BOND and SIGMA BOND (WASSCE –Chemistry JULY 2003 Q7. b [i] Ref: STEM Dic. page 992(PI-Bond) and page 1168 (sigma bond).

168. State the following principles/rules using 0 to illustrate in each case: i. AUFBAU'S PRINCIPLE (WASSCE –Chemistry NOV 2003 Q1.a [i] Ref: STEM Dic. page 113). ii. PAULI'S EXCULSION PRINCIPLE (WASSCE –Chemistry NOV 2003 Q1.a [ii] Ref: STEM Dic. page 955). iii. HUND'S RULES OF MAXIMUM MULTIPLICITY (WASSCE –Chemistry NOV 2003 Q1.a [iii] Ref: STEM Dic. page 634).

169. What is meant by the following terms: i. SPECIFIC RATE CONSTANT (WASSCE –Chemistry NOV 2003 Q3.a [i] Ref: STEM Dic. page 1204). ii. EFFECTIVE COLLISION (WASSCE –Chemistry NOV 2003 Q3.a [ii] Ref: STEM Dic. page 426). iii. ACTIVATION ENERGY (WASSCE –Chemistry NOV 2003 Q3.a [iii] Ref: STEM Dic. page 24).iv. HYBRIDIZATION (WASSCE –Chemistry NOV 2003 Q3.b [iv] Ref: STEM Dic. page 636).

170. What is BIOTECHNOLOGY (WASSCE –Chemistry NOV 2003 Q6.c [i] Ref: STEM Dic. page 158).

171. State the FIRST LAW OF ELECTROLYSIS (WASSCE –Chemistry NOV 2003 Q6.b [i] Ref: STEM Dic. page 501).

172. Define each of the following terms: i. IONIZATION ENERGY (WASSCE –Chemistry JULY 2002 Q1.a [i] Ref: STEM Dic. page 689).ii. ELECTRON AFFINITY (WASSCE –Chemistry JULY 2002 Q1.a [ii] Ref: STEM Dic. page 438). iii. ELECTRONEGATIVITY (WASSCE –Chemistry JULY 2002 Q1.a [iii] Ref: STEM Dic. page 441).

173. Define each of the following terms: i. MOLE (WASSCE –Chemistry NOV 2002 Q2.a [i] Ref: STEM Dic. page 838) ii. STANDARD SOLUTION (WASSCE –Chemistry NOV 2002 Q2.a [ii] Ref: STEM Dic. page 1222).

174. Distinguish between: i. ELECROLYTIC CELLS AND ELECROCHEMICAL CELLS (WASSCE –Chemistry NOV 2002 Q5.a [i] Ref: STEM Dic. page 437 & page 435). ii. PRIMARY CELLS AND SECONDARY CELLS (WASSCE –Chemistry NOV 2002 Q5. a [ii] Ref: STEM Dic. page 1031 & page 1147).

175. What is a TRANSITION ELEMENT? (WASSCE –Chemistry NOV 2002 Q6. a [i] Ref: STEM Dic. page 1307).

176. Name three particles emitted by RADIOACTIVITY SUBSTANCES (WASSCE –Chemistry JULY 2001 Q1.a [i] Ref: STEM Dic. page 1072).

177. What is the HALF-LIFE of a radioactive substance (WASSCE –Chemistry JULY 2001 Q1. a [iii] Ref: STEM Dic. Page 564).

178. 183. Define each of the following terms: i. STANDARD SOLUTION (WASSCE –Chemistry JULY 2001 Q2.b [i] Ref: STEM Dic. page 1222). ii. SATURATED SOLUTION (WASSCE –Chemistry JULY 2001 Q2.b [ii] Ref: STEM Dic. page 1222).

iii. STANDARD ENTHALPY OF FORMATION (WASSCE – Chemistry JULY 2001 Q2.b [iii] Ref: STEM Dic. page 1221).

179. Explain the following term as applied in chemical kinetics: RATE OF REACTION (WASSCE –Chemistry JULY 2001 Q3.a [i] Ref: STEM Dic. page 1079).

180. What is: i. AN AMPHOTERIC SUBSTANCE (WASSCE – Chemistry JULY 2001 Q5.a [i] Ref: STEM Dic. page 65). ii. A DELIQUESCENT SUBSTANCE (WASSCE –Chemistry JULY 2001 Q5.a [ii] Ref: STEM Dic. page 358(Deliquescent).

181. What is BIOTECHNOLOGY (WASSCE –Chemistry JULY 2001 Q6.a [i] Ref: STEM Dic. page 158).

182. What is POLLUTION? (WASSCE –Chemistry JULY 2001 Q6.c [i] Ref: STEM Dic. page 1012).

183. State two postulates of the KINETIC THEORY OF GASES (WASSCE –Chemistry NOV 2001 Q2.a Ref: STEM Dic. page 713).

184. State GRAHAM'S LAW OF DIFFUSION (WASSCE –Chemistry NOV 2001 Q2.c [i] Ref: STEM Dic. page 583).

185. What is: i. BRONSTED-LOWRY ACID: (WASSCE –Chemistry NOV 2001 Q3.a [i] Ref: STEM Dic. page 181) ii. BRONSTED-LOWRY BASE (WASSCE –Chemistry NOV 2001 Q3.a [ii] Ref: STEM Dic. page 181).

186. State the difference between STRONG ACIDS & WEAK ACIDS (WASSCE –Chemistry NOV 2001 Q3.c [i] Ref: STEM Dic. page 1238 & page 1371).

187. What is REDOX REACTION (WASSCE –Chemistry NOV 2001 Q5.a [i] Ref: STEM Dic. page 1088).

188. What is a HYDROCARBON (WASSCE –Chemistry NOV 2001 Q6. a [i] Ref: STEM Dic. page 637).

189. What is: i. POLYMERIZATION (WASSCE –Chemistry NOV 2001 Q7. a [i] Ref: STEM Dic. page 1015). ii. ADDITION POLYMER: (WASSCE –Chemistry NOV 2001 Q7. a [ii] Ref: STEM Dic. page 30). iii. CONDENSATION POLYMER (WASSCE –Chemistry NOV 2001 Q7. a [iii] Ref: STEM Dic. page 292)

190. What is: i. ELECTRON AFFINITY (WASSCE –Chemistry JULY 2000 Q1.b [i] Ref: STEM Dic. page 438). ii. ELECRONEGATIVITY (WASSCE –Chemistry JULY 2000 Q1.b [ii] Ref: STEM Dic. page 441).

191. State and illustrate HUND'S RULE OF MAXIMUM MULTIPLICITY (WASSCE –Chemistry JULY 2000 Q1.e Ref: STEM Dic. page 634 (Hund's Rule).

192. State CHARLES' LAW (WASSCE –Chemistry JULY 2000 Q2.b [i] Ref: STEM Dic. page 236).

193. Explain ISOTOPIC ABUNDANCE WASSCE –Chemistry JULY 2000 Q3.a [i] Ref: STEM Dic. page 696).

194. What is HALF – LIFE of a nuclide (WASSCE –Chemistry JULY 2000 Q3.b [i] Ref: STEM Dic. page 600).

195. Explain the term CONJUGATE ACID BASE PAIR and illustrate your answer with an example. (WASSCE –Chemistry JULY 2000 Q4.a Ref: STEM Dic. page 297).

196. Distinguish between HYDRATION and HYDROLYSIS (WASSCE –Chemistry JULY 2000 Q5. a Ref: STEM Dic. page 636 & page 640).

PHYSICS QUESTIONS FROM THE WEST AFRICAN SENIOR SCHOOL CERTIFICATE EXAMINATIONS (W.A.S.S.CE.) AND STEM DICTIONARY REFERENCE PAGES

1. Define THERMAL CONDUCTIVITY. (WASSCE 2019-NOV physics Q2 [a] [b] Ref: STEM Dic. page 1283).

2. State the conditions fulfilled by a body executing SIMPLE HARMONIC MOTION. (WASSCE 2019 – NOV physics Q8 [a] Ref: STEM Dic page page 1173).

3. Explain the concept of RELATIVE VELOCITY. (WASSCE 2019-NOV physics Q8. C [i] Ref: STEM Dic. page 1096).

4. State the principle of CONSERVATION OF ENERGY. (WASSCE 2019-NOV physics Q9. [9] Ref: STEM DIC. page 299).

5. Define CRITICAL ANGLE (WASSCE 209-NOV physics Q10. D[i] Ref: STEM Dic. page 321).

6. What is ELECTRIC FIELD? (WASSCE 2019-NOV physics Q11. [a] Ref: STEM Dic. page 432).

7. State two differences between (i) GRAVITATIONAL POTENTIAL and (ii) GRAVITATIONAL FIELD STRENGTH. (WASSCE 2019-NOV. Physics Q11[b]. Ref: STEM Dic. page 587 and page 587.

8. Define NATURAL RADIOACTIVITY. (WASSCE 2019-NOV Physics Q12. a[i] Ref: STEM Dic. page 865).

9. List 4 components of the NUCLEAR REACTOR? (WASSCE 2019-NOV physics Q12. b [i] [iii] Ref: STEM Dic page 894).

10. Define STRAIN (WASSCE -physics June 2018 Q1.[a] Ref: STEM Dic page 1234).

11. What is an INTRINSIC SEMI-CONDUCTOR. (WASSCE -physics June 2018 Q4 [a], 2017 Q6. Ref: STEM Dic page 684)

12. What does the acronym LASER stand for? (WASSCE -physics June 2018 Q7. [a] Ref: STEM Dic. page 729).

13. What is a LASER? (WASSCE -physics June 2018 Q7.[b] Ref: STEM Dic page 729).

14. Define UNIFORM ACCELERATION. (WASSCE -physics June 2018 Q8.[a].Ref: STEM Dic page 1331).

15. Explain the statement: The SPECIFIC LATENT HEAT OF VAPORIZATION OF MERCURY is 2.72 105 JKg-1K-1. (WASSCE -physics June 2018 Q9. C Ref: STEM Dic page 1204 (Specific Latent Heat of Vaporization) and page 810 (Mercury).

16. Define DIFFRACTION. (WASSCE 2019- physics June Q10 [a]. Ref: STEM Dic page 379).

17. Define a) i) RESISTANCE (WASSCE -physics June 2018 Q11. [a] I, June 2015. Ref: STEM Dic page 1101).

18. Define a) ii) IMPEDANCE. (WASSCE -physics June 2018 Q11[a] (ii) Ref: STEM Dic page 655).

19. Define BINDING ENERGY in an atom. (WASSCE -physics June 2018 Q12[a] Ref: STEM Dic page 152).

20. State two factors on which SURFACE TENSION depends. (WASSCE -physics June 2017 Q5. [a]. Ref: STEM Dic page 1254).

21. Define STABLE EQUILIBRIUM as applied to a rigid body. (WASSCE -physics June 2017 Q9. Ref: SYETEM Dic. page 1219).

22. What is a WAVE FRONT? (WASSCE -physics June 2017 Q10[a] Ref: STEM Dic page 1370).

23. Draw and label a diagram of an ASTRONOMICAL TELESCOPE in normal adjustment. (WASSCE -physics June 2017 Q10. C[i] Ref: STEM Dic. page 105)

24. What is a DIELECTRIC? (WASSCE -physics June 2017 Q11.A[i]. Ref: STEM Dic. page 377).

25. What is NUCLEAR FISSION? (WASSCE -physics June 2017 Q12 a[i] Ref: STEM Dic. Ppage 893).

26. State the function of each of the following materials in a NUCLEAR REACTOR (a) graphite (b) boron rods (WASSCE -physics June 2017 Q12.a [i]Ref: STEM Dic. page 894).

27. State three uses of LASERS. (WASSCE –physics NOV 2017 Q2. Ref: STEM Dic. page 729).

28. State three magnetic elements. (WASSCE –physics NOV 2017 Q3Ref: STEM Dic. page 780).

29. Identify the physical quantities of an electrochemical cell. (WASSCE –physics NOV 2017 Q7. Ref: STEM Dic. page 435 (electrochemical cell).

30. What is a FORCE? (WASSCE –physics NOV 2017 Q8; Ref: STEM Dic. page 530).

31. State two practical applications of poor CONDUCTOR of heat. (WASSCE –physics NOV 2017 Q9. [a]. Ref: STEM Dic. page 293)

32. Explain the term CLOSE SYSTEM in relation to the definition of energy. (WASSCE –physics NOV 2017 Q9 C. Ref: STEM Dic. page 264)

33. Define ELECTRIC LINE OF FORCE. (WASSCE –physics NOV 2017 Q11. Ref: STEM Dis page 432).

34. Write an equation for WHEATSTONE BRIDGE at balance point. (WASSCE –physics NOV 2017 Q11 C Ref: STEM Dic. page 1376).

35. Explain the term RESISTIVITY (WASSCE –physics NOV 2017 Q11.d Ref: STEM Dic page 206).

36. Define HALF-LIFE (WASSCE –physics NOV 2017 Q12.[a] 15A[i] Ref: STEM Dic page 600).

37. What is NUCLEAR CHAIN REACTION? (WASSCE –physics NOV 2017 Q12. [b] Ref: STEM Dic page 231 (Chain Reaction).

38. What is DOPING? (WASSCE –physics JUNE 2016 Q6.a,b; Ref: STEM Dic page 404).

39. Draw the diagram to illustrate a cathode ray tube and identity the COMPONENTS OF CATHODE RAY TUBE. (WASSCE –physics JUNE 2016 Q7; Ref: STEM Dic page 218).

40. Explain the term NET FORCE (WASSCE –physics JUNE 2016 Q8.[a] Ref: STEM Dic. page 872).

41. Define the PRINCIPLE OF CONSERVATION OF LINEAR MOMENTUM. (WASSCE –physics JUNE 2016 Q8.[b] Ref: STEM Dic. page 1035).

42. Explain THERMAL EQUILIBRIM. (WASSCE –physics JUNE 2016 Q9.a[i] Ref: STEM Dic. page 1283).

43. List two uses of HYDRAULIC PRESS. (WASSCE –physics JUNE 2016 Q9.b. Ref: STEM Dic page 637).

44. Define CRITICAL ANGLE. (WASSCE –physics JUNE 2016 Q10.a Ref: STEM Dic. page 321).

45. How are ANTI-NODE created in a STATIONARY WAVE? (WASSCE –physics JUNE 2016 Q10.b. Ref: STEM Dic. page 84(Anti-node) and page 1225

46. Explain the following terms: i. MASS DEFECT ii. BINDING ENERGY OF A NUCLEUS. (WASSCE –physics JUNE 2016 Q12.A[i],[ii].Ref: STEM Dic. page 795 (Mass Defect) and page 152 (Binding Energy).

47. State one use of SATELLITES (WASSCE – physics NOV 2016 Q4.a Ref: STEM Dic. page 1137).

48. State two uses of ROCKETS. (WASSCE –physics NOV 2016 Q4.b. Ref: STEM Dic. page 1117).

49. Define TERMINAL VELOCITY. (WASSCE –physics NOV 2016 Q8.a Ref: STEM Dic. page 1277)

50. Define LINEAR MOMENTUM. (WASSCE –physics NOV 2016 Q8.b. Ref: STEM Dic. page 753).

51. Define/State Boyle's LAW (WASSCE –physics NOV 2016 Q9.a[i] Ref: STEM Dic. page 175).

52. State the Assumptions of the KINETIC THEORY of gas necessary for a real gas to be considered as ideal (page 713).

53. Explain the term RENEWABLE ENERGY. (WASSCE –physics NOV 2016 Q9. b, 2015 June Q9.b. Ref: STEM Dic. page 1098).

54. The SIX'S MAXIMUM AND MINIMUM THERMOMETER (WASSCE –physics NOV 2016 Q9.c Ref: STEM Dic. page 1177).

55. Define the FARAD. WASSCE –physics NOV 2016 Q11.C[i] Ref: STEM Dic. page 501).

56. Define DECAY CONSTANT. (WASSCE –physics NOV 2016 Q12.a[i] Ref: STEM Dic. page 350).

57. State the function and material used for each of the under listed components in a nuclear fission reactor. i. MODERATOR ii. CONTROL RODS iii. COOLANT (WASSCE –physics NOV 2016 Q12. B [i, ii, iii] Ref: STEM Dic. page 305 (Control Rod), page 308 (Coolant) and page 836 (Moderator).

58. What is a PROJECTILE? (WASSCE –physics JUNE 2015 Q1.a Ref: STEM Dic. page 1040).

59. What is DOPING (WASSCE –physics JUNE 2015 Q5(a) Ref: STEM Dic. page 404).

60. Write the DE-BROGLIE EQUATION. (WASSCE –physics JUNE 2015 Q7.a Ref: STEM Dic. page 347).

61. Define the term UNIFORM ACCELERATION. (WASSCE –physics JUNE 2015 Q8.a [i] Ref: STEM Dic. page 1331).

62. Define Specific LATENT HEAT OF VAPORIZATION. (WASSCE –physics JUNE 2015 Q9(a). Ref: STEM Dic. page 1204).

63. What is RESONANCE? (WASSCE –physics JUNE 2015 Q10 (a). Ref: STEM Dic. page 1103).

64. Differentiate between LOUDNESS (page 767) and INTENSITY (page 675) of Sound.

65. Define the COULOMB. (WASSCE –physics JUNE 2015 Q11.a[i] Ref: STEM Dic. page 316).

66. Define NUCLEON NUMBER. (WASSCE –physics JUNE 2015 Q12.a Ref: STEM Dic. page 206).

67. Explain the following terms. (a) TENSILE STRESS. (WASSCE –physics NOV 2015 Q5(a) Ref: STEM Dic. page 1236 (Stress)

68. b. YOUNG'S MODULUS (WASSCE –physics NOV 2015 Q5(b) Ref: STEM Dic. page 1393).

69. Define FORCE (WASSCE –physics NOV 2015 Q8.b [i] Ref: STEM Dic. page 530).

70. List the three main features of a CLINICAL THERMOMETER. (WASSCE –physics NOV 2015 Q9.b. Ref: STEM Dic. page 261)

71. Define the ROOT MEAN SQUARE VALUE OF an alternating current. (WASSCE –physics NOV 2015 Q11.a Ref: STEM Dic. page 1119)

72. Explain the term TRANSMUTATION as it relates to radioactivity. (WASSCE –physics NOV 2015 Q12.a[i] Ref: STEM Dic. page 1309)

73. State NEWTON'S SECOND LAW OF MOTION. (WASSCE – physics JUNE 2014 Q8. (c) Ref: STEM Dic. page 877).

74. Explain the term INERTIA. (WASSCE –physics JUNE 2014 Q8.d Ref: STEM Dic. page 664).

75. Define HEAT CAPACITY and state its unit. (WASSCE – physics JUNE 2014 Q9.a Ref: STEM Dic. page 609).

76. Explain RESONANCE as applied to sound. (WASSCE –physics JUNE 2014 Q10.b Ref: STEM Dic. page 97).

77. Define ELECTROMOTIVE FORCE. (WASSCE –physics JUNE 2014 Q11.a Ref: STEM Dic. page 438).

78. State the PRINCIPLE OF OPERATION of POTENTIOMETER. (WASSCE –physics JUNE 2014 Q11.a [ii] Ref: STEM Dic. page 1026 (Potentiometer).

79. On what physical principle does the FIBER OPTIC CABLE work? (WASSCE –physics NOV 2014 Q4.a Ref: STEM Dic. page 1301 (Total Internal Reflection).

80. List two applications of FIBER OPTICS. (WASSCE –physics NOV 2014 Q4.b Ref: STEM Dic. page 509).

81. What is FREE FALL? (WASSCE –physics NOV 2014 Q8.a Ref: STEM Dic. page 539).

82. Define ESCAPE VELOCITY. (WASSCE –physics NOV 2014 Q8.c Ref: STEM Dic. page 474).

83. Define HEAT CAPACITY (WASSCE –physics NOV 2014 Q9. a Ref: STEM Dic. page 609)

84. Define RADIOACTIVITY. (WASSCE –physics NOV 2014 Q12.c, 2008 Q5. a[i] Ref: STEM Dic. page 1072).

85. What is PARKING ORBIT for a Satellite? (WASSCE –physics JUNE 2013 Q3.a Ref: STEM Dic. page 949).

86. What is meant by DIMENSION OF A PHYSICAL QUANTITY? (WASSCE –physics JUNE 2013 Q4. a Ref: STEM Dic. page 385).

87. Define HEAT CAPACITY. (WASSCE –physics JUNE 2013 Q12.a Ref: STEM Dic. page 609.)

88. Explain DIFFRACTION. (WASSCE –physics JUNE 2013 Q13.a[i]. Ref: STEM Dic. page 379).

89. In relation to an a.c circuit explain: INDUCTIVE REACTANCE (WASSCE –physics JUNE 2013 Q14.a[i] Ref: STEM Dic. page 663).

90. In relation to an a.c circuit, explain IMPEDANCE. (WASSCE –physics JUNE 2013 Q14.a [ii] Ref: STEM Dic. page 655).

91. Explain EDDY CURRENT. (WASSCE –physics JUNE 2013 Q14.b [iii] Ref: STEM Dic. page 425).

92. Define EXCITEMENT POTENTIAL. (WASSCE –physics JUNE 2013 Q15.a Ref: STEM Dic. page 485).

93. State one similarity and two differences between a SEMI CONDUCTOR DIODE and TRANSISTOR. (WASSCE –physics NOV 2013 Q7. Ref: STEM Dic. page 1156 (Semi Conductor), page 386 (Diode) and page 1306 (Transistor).

94. State KIRCHHOFF'S LAW OF electrical networks. (WASSCE –physics NOV 2013 Q9. a, 2009 Q8a,b Ref: STEM Dic. page 713).

95. Give the meaning of the term NETWORK as used in connection with KIRCHHOFF'S laws. (WASSCE –physics NOV 2013 Q9.b Ref: STEM Dic. page 872).

96. Explain the term (i) INERTIA (ii) INERTIAL MASS (WASSCE –physics NOV 2013 Q12.a Ref: STEM Dic. page 664 and page 66).

97. Define PRINCIPAL FOCUS for a diverging lens (WASSCE –physics NOV 2013 Q13.a[i] Ref: STEM Dic. page 1034).

98. Define PRINCIPAL AXIS for a diverging lens. (WASSCE – physics NOV 2013 Q13.a[i] Ref: STEM Dic. page 1034).

99. Explain the term EDDY CURRENTS and state two devices in which the currents are applied. (WASSCE –physics NOV 2013 Q14.b Ref: STEM Dic. page 425).

100. What are BEATS? (WASSCE –physics JUNE 2012 Q2. a, 2008 Q3.a Ref: STEM Dic. page 140).

101. What is a BLACK BODY? (WASSCE –physics JUNE 2012 Q3. a Ref: STEM Dic. page 161).

102. Define ELASTIC LIMIT. (WASSCE –physics JUNE 2012 Q11.a[i] Ref: STEM Dic. page 430).

103. Define TENSILE STRESS. WASSCE –physics JUNE 2012 Q11.b[i] (WASSCE –physics JUNE 2012 Q11.a[i]b Ref: STEM Dic. page 1236 (Stress).

104. What is DIFFRACTION? (WASSCE –physics JUNE 2012 Q13.a[i] Ref: STEM Dic. page 379).

105. Distinguish between a POSITIVELY CHARGED object and NEGATIVELY CHARGED object. (WASSCE –physics JUNE 2012 Q14.a Ref: STEM Dic. page 867 (Negatively Charged) and page 1021 (Positive charged).

106. Define HALF-LIFE. (WASSCE –physics JUNE 2012 Q15.a, 2008 – June Q15.a [I,ii] Ref: STEM Dic. page 600).

107. List 3 situations in which DOPPLER EFFECT is observed. (WASSCE –physics NOV 2012 Q5. 2008-NOV Q4 Ref: STEM Dic. page 404).

108. What is CENTRIPETAL FORCE? (WASSCE – physics NOV 2012 Q11.b[i] Ref: STEM Dic. page 228).

109. Define VELOCITY RATIO. (WASSCE –physics NOV 2012 Q12.d[i] Ref: STEM Dic. page 1351).

110. What is a DOPING? (WASSCE –physics JUNE 2011 Q8.a Ref: STEM Dic. page 404).

111. Define the term BEAT FREQUENCY. (WASSCE –physics JUNE 2011 Q5.a Q10.A. Ref: STEM Dic. page 140).

112. List 3 facts about ACCELERATION OF FREE FALL due to gravity. (WASSCE –physics JUNE 2011 Q11.a Ref: STEM Dic. page 12).

113. Define DECAY CONSTANT. (WASSCE –physics JUNE 2011 Q5.a[i], 2008 JUNE a[i] Ref: STEM Dic. page 350).

114. What is an IDEAL GAS? (WASSCE –physics NOV 2011 Q2.a, 2010 Q5.a Ref: STEM Dic. page 650).

115. What is a COOLING CURVE? (WASSCE –physics NOV 2011 Q3.a Ref: STEM Dic. page 308).

116. Distinguish between metals, insulators and semi-conductors in terms of the BAND THEORY. (WASSCE –physics NOV 2011 Q9. Ref: STEM Dic. page 130).

117. State the PRINCIPLE OF CONSERVATION OF LINEAR MOMENTUM. (WASSCE –physics NOV 2011 Q11. Ref: STEM Dic. page 299 (Conservation of Linear Momentum).

118. State the LAWS OF REFLECTION. (WASSCE –physics NOV 2011 Q13.Aa Ref: STEM Dic. page 733).

119. State two differences between ELECTRIC POTENTIAL and ELECTRIC FIELD INTENSITY. (WASSCE –physics NOV 2011 Q14.b Ref: STEM Dic. page 433 (Electric Potential), page 206 (Electric Field Intensity).

120. Define NUCLEAR BINDING ENERGY. (WASSCE –physics NOV 2011 Q15.b[i] Ref: STEM Dic. page 893).

121. Define a JOULE. (WASSCE –physics NOV 2010 Q12.a Ref: STEM Dic. page 702).

122. What is the difference between a THERMOMETER and a THERMOSTAT? (WASSCE –physics NOV 2010 Q12.a Ref: STEM Dic. page 1286(Thermometer), page 1287 (Thermostat).

123. Explain the term RESONANCE as used in sound. (WASSCE –physics NOV 2010 Q13.a Ref: STEM Dic. page 1103).

124. What is a LIGHTING CONDUCTOR? (WASSCE –physics NOV 2010 Q14.a[i] Ref: STEM Dic. page 748).

125. Define i. PHOTOELECTRONS ii. STOPPING POTENTIAL. (WASSCE –physics NOV 2010 Q15.a [i, ii] Ref: STEM Dic. Page i. page 984 (Photoelectrons), ii. page 1233 (Stopping Potential).

126. List 3 characteristics of GAMMA RAYS. (WASSCE –physics NOV 2010 Q15.b Ref: STEM Dic. page 555).

127. How are CATHODE RAYS produced? i. State 2 properties of CATHODE RAYS. (WASSCE –physics JUNE 2009 Q1.a, b Ref: STEM Dic. page 218).

128. What is meant by the DIMENSION OF A PHYSICAL QUANTITY? (WASSCE –physics JUNE 2009 Q5. a, 2008 NOV Q1.a Ref: STEM Dic. page 385).

129. What is an ACCEPTOR IMPURITY? (WASSCE –physics JUNE 2009 Q6.a Ref: STEM Dic. page 13).

130. What is MOTION? (WASSCE –physics JUNE 2009 Q11.a Ref: STEM Dic. page 849).

131. Explain the following terms. i. PRINCIPAL FOCUS ii. OPTICAL CENTRE iii. FOCAL LENGTH. (WASSCE –physics JUNE 2009 Q13.b Ref: STEM Dic. Page i. page 1034 (Principal Focus), ii page 915 (Optical Centre) and iii page 526 (Focal Length).

132. Define BINDING ENERGY of a nucleus. (WASSCE –physics JUNE 2009 Q15.a, 2015 Q9 c. Ref: STEM Dic. page 152).

133. Explain CHAIN REACTION. (WASSCE –physics JUNE 2009 Q15.c[i] Ref: STEM Dic. page 231).

134. Explain the phenomenon of DOPPLER EFFECT for a source of sound moving towards a stationary observe. (WASSCE –physics NOV 2009 Q5. Ref: STEM Dic. page 404).

135. State NEWTON'S i. FIRST LAW OF MOTION. ii. SECOND LAW OF MOTION. (WASSCE –physics NOV 2009 Q11.a[i], 2008 Q11.a Ref: STEM Dic. page 877).

136. The principle of CONSERVATION OF LINEAR MOMENTUM. (WASSCE –physics NOV 2009 Q11.a [ii] Ref: STEM Dic. page 299).

137. Explain LAND BREEZE and SEA BREEZE. (WASSCE –physics JUNE 2008 Q12.a [ii] Ref: STEM Dic. page 726 (Land Breeze), page 1145(Sea Breeze).

138. State the LAWS OF REFRACTION OF LIGHT. (WASSCE –physics JUNE 2008 Q13.a Ref: STEM Dic. page 733).

139. Explain what is meant by ELECTROMOTIVE FORCE. (WASSCE –physics JUNE 2008 Q14.a[i] Ref: STEM Dic. page 438).

140. Explain what is meant by INTERNAL RESISTANCE OF A CHEMICAL CELL. (WASSCE –physics JUNE 2008 Q14.a [ii] Ref: STEM Dic. page 679 (Internal Resistance)

141. Define i. VELOCITY ii. ACCELERATION (WASSCE –physics JUNE 2008 Q11.a[i] Ref: STEM Dic. page 11 (Acceleration) and page 1351 (Velocity).

142. Define SPECIFIC HEAT CAPACITY of a substance. (WASSCE –physics JUNE 2008 Q12.a, JUNE 2007 Q3.a Ref: STEM Dic. page 66).

143. State SNELL'S LAW. (WASSCE –physics NOV 2008 Q13.a[i] Ref: STEM Dic. page 1183).

144. State OHM'S LAW. (WASSCE –physics NOV 2008 Q14.a[i] Ref: STEM Dic. page 907).

145. Define i. WORK FUNCTION ii. THRESHOLD FREQUENCY (WASSCE –physics NOV 2008 Q15.a [i, ii] Ref: STEM Dic. page 1384 (Work Function), ii. page 1291 (Threshold).

146. What is a HEAT ENGINE? (WASSCE –physics JUNE 2007 Q2.a Ref: STEM Dic. page 610).

147. State Newton's LAW OF COOLING. (WASSCE –physics JUNE 2007 Q3.b Ref: STEM Dic. page 877).

148. What is DOPPLER EFFECT? (WASSCE –physics JUNE 2007 Q6.a Ref: STEM Dic. page 404).

149. State in word's KIRCHHOFF'S LAW of electrical work. (WASSCE –physics JUNE 2007 Q7, 2005 Q8.a Ref: STEM Dic. page 713).

150. Distinguish between TEMPERATURE and HEAT. (WASSCE –physics JUNE 2007 Q12.a[i] Ref: STEM Dic. page 1274(TEMPERATURE), ii page 609(HEAT).

151. What is meant by RESONANCE? (WASSCE –physics JUNE 2007 Q13.a[i], 2005 Q8.c Ref: STEM Dic. page 97).

152. What is STATIONARY WAVE? (WASSCE –physics 2006 Q13.a[i] Ref: STEM Dic. page 1225).

153. Define AMPLITUDE (WASSCE – physics 2006 Q13.a[ii]. Ref: STEM Dic. page 66).

154. Explain the terms i. REACTANCE and IMPEDANCE in an alternating current circuit. (WASSCE –physics 2006 Q14.b Ref: STEM Dic. page 655 (Impedance), ii.page 1081 (Reactance).

155. Explain i. GROUND STATE ENERGY. ii. EXCITATION ENERGY. iii. BINDING ENERGY. WASSCE –physics 2006 Q15.b Ref: STEM Dic. page 590 (Ground State Energy), page 485 (Excitation Energy), page 152 (Binding Energy).

156. What is a DERIVED UNIT? (WASSCE –physics 2005 Q1.a[i] Ref: STEM Dic. page 366).

157. Define ANGULAR ACCELERATION. (WASSCE –physics 2005 Q1.b[i] Ref: STEM Dic. page 73).

158. What is WEIGHT? (WASSCE –physics 2005 Q2.b[i] Ref: STEM Dic. page 1374).

159. Define an ELASTIC COLLISION. (WASSCE –physics 2005 Q2.c[i] Ref: STEM Dic. page 430).

160. State the LAW of CONSERVATION OF ENERGY. (WASSCE –physics 2005 Q3. a[i]. Ref: STEM Dic. page 732).

161. Define DEW POINT. (WASSCE –physics 2005 Q3.b[i]. Ref: STEM Dic. page 371).

162. Define LATENT HEAT OF FUSION of a substance. (WASSCE –physics 2005 Q3.c[i] Ref: STEM Dic. page 730).

163. Define RELATIVE HUMIDITY. (WASSCE –physics 2005 Q4.a[i] Ref: STEM Dic. page 1095).

164. Explain DIFFRACTION with reference to a wave. (WASSCE –physics 2005 Q5. a[i] Ref: STEM Dic. page 379).

165. State Coulombs law (WASSCE –physics 2005 Q7.a[i] Ref: STEM Dic. page 316).

166. What is NATURAL RADIOACITIVITY? (WASSCE –physics 2005 Q9.a Ref: STEM Dic. page 1071 (RADIOACTIVITY).

167. What is INTRINSIC SEMI CONDUCTOR? (WASSCE –physics 2005 Q10.a[i] Ref: STEM Dic. page 684).

168. Define SPECIFIC LATENT HEAT and state its unit. (WASSCE –physics 2005. Q4. a Ref: STEM Dic. page 1204).

169. Define RELATIVE DENSITY (WASSCE –physics 2004 Q1.a Ref: STEM Dic. page 1095).

170. Define the following terms as used with a machine i. MECHANICAL ADVANTAGE ii. EFFICIENCY (WASSCE –physics 2004 Q3.a Ref: STEM Dic. Page (i) page 804 (Mechanical Advantage, ii page 427 (Efficiency).

171. Draw ray diagrams to illustrate each of the following high phenomena i. REFLECTION ii. REFRACTION (WASSCE –physics 2004 Q5.a Ref: STEM Dic. page 1090 (Reflection), page 1091 (Refraction).

172. Define ELECTRICAL POTENTIAL (WASSCE –physics 2004 Q7. a[i] Ref: STEM Dic. page 433).

173. Define MAGNETIC FLUX DENSITY and state it unit. (WASSCE –physics 2004 Q8. a Ref: STEM Dic. page 781).

174. Define NUCLEAR FUSION. (WASSCE –physics 2004 Q9.a[i]. Ref: STEM Dic. page 894).

175. What is NUCLEAR BINDING ENERGY? (WASSCE –physics 2004 Q9.a [ii Ref: STEM Dic. page 893).

176. What are LINE SPECTRA? (WASSCE –physics 2004 Q9.c[i] Ref: STEM Dic. page 752).

177. What is THERMIONIC EMISSION (WASSCE –physics 2004 Q10.a Ref: STEM Dic. page 1284).

178. What is a CONTACT FORCE? (WASSCE –physics 2003 Q2.a[i] Ref: STEM Dic. page 302)

179. Define LINEAR EXPANSIVITY of a substance. (WASSCE –physics 2003 Q3.a[i] Ref: STEM Dic. page 488 (Expansivity).

180. What is a PRIMARY CELL? (WASSCE –physics 2003 Q7. a[i] Ref: STEM Dic. page 1031).

181. What is a photon? (WASSCE –physics 2003 Q10.a[i].Ref: STEM Dic. page 985).

182. What is QUANTIZATION OF ELECTRIC CHARGE? (WASSCE –physics 2003 Q9.c[i] Ref: STEM Dic. page 1060).

183. What is a SIMPLE PENDULUM? (WASSCE –physics 2003 Q2.c[i] Ref: STEM Dic. page 959).

184. Explain RELATIVE VELOCITY. (WASSCE –physics 2002 Q1.a[i] Ref: STEM Dic. page 1096).

185. State the LAW OF FLOATATION. (WASSCE –physics 2002 Q1.b[i] Ref: STEM Dic. page 732).

186. What is INERTIA? (WASSCE –physics 2002 Q1.c[i]. Ref: STEM Dic. page 664).

187. Define GRAVITATIONAL POTENTIAL. (WASSCE –physics 2002 Q2.a[i] Ref: STEM Dic. page 587).

188. Define EFFORT. (WASSCE –physics 2002 Q3.a[i]. Ref: STEM Dic. page 427).

189. Define temperature on the KELVIN SCALE. (WASSCE –physics 2002 Q3. a[i] Ref: STEM Dic. page 707).

190. State two differences between LIGHT WAVES and SOUND WAVES. (WASSCE –physics 2002 Q5.a[i]. Ref: STEM Dic. page 746 (Light) and page 1198 (Sound).

191. What is FORBIDDEN BAND? (WASSCE –physics 2002 Q9.a[i]. Ref: STEM Dic. page 97).

192. State NEWTON'S LAW OF UNIVERSAL GRAVITATION. (WASSCE –physics 2001 Q1.a Ref: STEM Dic. page 877.

193. Define REFRACTIVE INDEX. [WASSCE –physics 2001 Q6.a (i). Ref: STEM Dic. page 1091].

194. Define MAGNETIC FIELD. [WASSCE –physics 2001 Q8.a(i) Ref: STEM Dic. page 875].

195. Define ACCELERATION. (WASSCE –physics 2000 Q1.a Ref: STEM Dic. page 11).

196. Define VELOCITY RATIO as applied to a machine. (WASSCE –physics 2000 Q3.a Ref: STEM Dic. page 1351).

197. Define ACCOMODATION as applied to human eye. (WASSCE –physics 2000 Q6.a Ref: STEM Dic. page 14).

198. With the aid of diagrams explain TOTAL INTERNAL REFLECTION OF LIGHT. (WASSCE –physics 2000 Q6.(b) Ref: STEM Dic. page 1301 (Total Internal Reflection).

199. Distinguish between (i) ALPHA (ii) BETA and (iii) GAMMA radiations under the heading nature, speed and ionizing power. (WASSCE –physics 2000 Q9. a. Ref: STEM Dic. page 54 (Alpha), ii.page 147 (Beta) iii. page 555 (Gamma).

CONCLUSION

With technology now at an advanced stage, classrooms no longer need teachers to always stand in front of pupils, lecturing them, or students, reading long notes and text books continuously before gaining a little knowledge. These could accumulate easily, by merely playing such self-tuition Games as the Osmosis and Initiation or adopting any of the modern-self-tuition approaches. Also, working with syllabi designed for particular classes and age groups seriously limit the ability of all students to explore their full potentials, because such syllabi are designed with the average student in mind; and all students, including the most brilliant are forced to fly at that level - or below them in most classrooms, due to other constraints, such as extra-curricula activities, teacher/student absenteeism, lack of certain vital resources, etc.

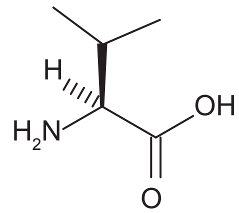
Gamification and most of the new approaches to learning, seamlessly offer knowledge, far above the syllabi at the level of students, textbooks and even teachers in most cases. All students, including the write-offs are the gainers, not just the few exoteric ones that formal teaching and learning or luck produces. It takes only just one individual, brilliant or otherwise, to totally transform the world. Let us give all students/pupils chances that all of them could grab, not just those chances that only a few exoteric or lucky ones could grasp.

The 6-in-1, Quick, Daily Reference

SCIENCE, TECHNOLOGY AND MATHEMATICS DICTIONARY

A

2-AMINO-3-METHYLBUTANOIC ACID (VALINE): One of the essential amino acids with the chemical formula $C_5H_{11}NO_2$, density 1.316 g/cm^3, melting point of 298 °C (571 K), which decomposes. It is found in most proteins required for normal growth in animals. It cannot be synthesised by the body and human dietary sources include cheese, fish, poultry, peanuts, sesame seeds and lentils. It is produced industrially by hydrolysis of proteins or by chemical synthesis. Together with two other amino acids, isoleucine and leucine, they promote normal growth, repair tissues, regulate blood sugar, and provide the body with energy. It helps to stimulate the central nervous system, and is helpful in the treatment of liver and gallbladder diseases.



Chemical Structure of Valine

2-AMINOPROPANOIC ACID (ALANINE): Any of the two non-essential α-amino acids with the chemical formula $C_3H_7NO_2$, molecular mass 89.09 gmol^{-1}, density 1.424 g/cm^3 and melting point 258 °C (531 K). Alpha-alanine is one of the non-essential amino acids, found in most proteins and particularly abundant in fibroin, the protein in silk. Beta-alanine is a naturally occurring amino acid not found in proteins. It is an important constituent of the vitamin pantothenic acid and is used in its synthesis, as well as in biochemical research, electroplating and organic synthesis. It is also used as a dietary supplement.

4-ACETAMIDOPHENOL [ACETAMINOPHEN; N-(4-HYDROXYPHENYL)ETHANAMIDE; N-(4-HYDROXYPHENYL)ACETAMIDE; PARACETAMOL]: A synthetic analgesic drug similar to aspirin having the chemical formula $C_8H_9NO_2$, molecular mass 151.163 g/mol, density 1.263 g/cm^3, melting point 169 °C and boiling point of 420 °C. It is effective for relieving pain such as headache, muscle pain and lower back pain; and also used in reducing fever. It has weak anti-inflammatory properties. An overdose can cause severe, often irreversible or even fatal, liver and kidney damage.

a: In mathematics, it is a symbol for atto-, which is used in the Système international (S.I.) units for fractions of the physical units. It is used to denote the sub-multiple 10^{-18} or $1/10^{18}$ in the measurement of quantities. For example, 6.02×10^{-23} becomes 0.0000602 atto.

A: 1. A prefix meaning no, lacking or without. For example, **abiotic** means non-living or non-biological; **acarpous** means without carpel or pistil; **anomalous** means not normal, as in anomalous expansion of water; **asexual** means not involving gametes, as in asexual reproduction; **asymmetry** means without symmetry; and **asymptomatic** means without symptoms. 2. In chemistry, it is the symbol for mass number, the number of protons and neutrons in the nucleus of an atom. 3. In mathematics, it represents the number 10 in the hexadecimal notation. 4. In physics, it is the symbol for ampere, a unit of electrical current.

A BOMB (A-BOMB; ATOM BOMB; ATOMIC BOMB; NUCLEAR BOMB): A bomb of extreme destructive power fuelled by the fission of uranium or plutonium. Only two atomic bombs have ever been used in war. They were both dropped on Japanese cities, Hiroshima and Nagasaki at the end of the Second World War (World War II) by American planes. The bomb used against Hiroshima had a yield of 10 kilotons and used trinitrotoluene (TNT) charges to blow two subcritical masses of ^{235}U together. The bomb used against Nagasaki had a yield of 20 kilotons and was a ball of ^{239}Pu crushed together by trinitrotoluene (TNT).

A FORTIORI: Latin for "from the stronger", used to express a conclusion for which there is a stronger evidence than for a previously accepted one. For example, since the number 7 is prime, it is a fortiori not divisible by 3.

A HORIZON (HORIZON A): One of the main classifications of soil layer, which is a weathered, usually dark-coloured, and found at the uppermost part of soil. It is made up of some small mineral materials, organic matter and is also home to some organisms such as earthworms, fungi, and bacteria; plants grow in this layer. Most soils have three main soil horizons or layers which are Horizon A, Horizon B, and Horizon C. Horizon O is also sometimes found in some soils.

A POSTERIORI: Relating to or derived by reasoning from observed facts rather than through an understanding of how certain things work.

A PRIORI: 1. Proceeding from a known or assumed cause to a necessarily related effect; deductive. 2. Derived by or designating the process of reasoning without reference to particular facts or experience. 3. Evidence based on hypothesis or theory rather than experiment.

A PRIORI PROBABILITY: Relating to or derived by reasoning from self-evident propositions or using logic or reason to form a conclusion or opinion about something. It is used to distinguish the ways in which values for probabilities can be obtained.

A PROGRAMMING LANGUAGE (APL): A general-purpose, third-generation (3GL) programming language that allows certain data manipulations to be expressed with a special non-ASCII (American Code for Information Interchange) set of symbols, resulting in programs that are shorter than would be possible using most other languages. APL's notation allows matrix manipulation as well as recursion functions to be built into simple expressions rather than requiring multiple language statements.

AAA (AUTHENTICATION, AUTHORIZATION AND ACCOUNTING): A system of management and security protocols for controlling a client's activities on a secure network which enables the network server (NAS) to identify users who want to access the secure network (**Authentication**); to grant access to the secure network resources (**Authorization**); and to collect data regarding things such as what network resources are accessed, time spent in the network, the amount of data transferred and to enforce policies (**Accounting**).

***a*-:** 1. In mathematics, it is a prefix used to denote "not", for example, an asymmetrical figure is one that is not symmetrical. 2. An abbreviation used for arc, anti- or argument that indicates an inverse of a function, for example, alog for antilogarithm; atan for arctangent; arcsin and arccoth. They are usually indicated with -1 as a superscript, for example, in trigonometry, arcsin, which is the inverse of sine function is written as $\sin^{-1}$; arccos, which is the inverse of the cosine function is written as $\cos^{-1}$; and arccoth, which is the inverse of hyperbolic cotangent is written as $\coth^{-1}$.

A-BAND (A BAND; ANISOTROPIC BAND): The dark section of a striated muscle (skeletal muscle) fibre that contains both myosin (thick) and actin (thin) filaments with a narrow and lighter region of myosin filaments called H band, seen bisecting the A-band. The "A" denotes "anisotropic" because in a polarising light microscope, the dark bands are birefringent. It also contains the M line, which connects the central portion of each thick filament; and the zone of overlap, in which thin filaments are situated between the thick